VIBRIDGE: AN IOT-BASED VIBRATION AND STRUCTURAL CONDITION MONITORING SYSTEM FOR SMALL BRIDGES

A Capstone Proposal
Presented to the Faculty of the
College of Computer Studies, University of Cebu

In Partial Fulfillment of the Requirement
for the degree of Bachelor of Science in Information Technology

By

Paul Vincent Lepatan
Maverick Angelo Beltran
Lordjan Carombana
Christian Rosales

Adviser Name
Mr. Joemarie C. Amparo

April 2026

ACKNOWLEDGEMENT

First and foremost, the researchers express their heartfelt gratitude to God Almighty for granting them the wisdom, strength, and perseverance to complete this research. His guidance has been a constant source of inspiration throughout this journey.

The researchers extend their sincere appreciation to their research adviser, Mr. Joemarie C. Amparo, for his unwavering support, insightful guidance, and encouragement. His expertise and dedication have been instrumental in shaping the direction and success of this study.

The researchers are deeply grateful to the professionals and personnel involved in bridge monitoring and maintenance, particularly those from the Department of Public Works and Highways (DPWH), for their valuable insights, guidance, and willingness to share their experiences. Their contributions helped ground this study in real-world engineering practices and challenges.

Moreover, the researchers extend their heartfelt thanks to their parents, friends, and relatives for their unwavering love, encouragement, and support. Their presence has been a constant source of motivation, enabling the researchers to overcome challenges and complete this project successfully.

Finally, the researchers would like to express their deepest gratitude to all participants who contributed to this study. Their time, cooperation, and valuable insights played a significant role in the completion and success of this research.

The Researchers

DEDICATION

This research study is dedicated to the hardworking and dedicated professionals in the field of infrastructure and public service, especially those committed to ensuring the safety, reliability, and sustainability of bridge structures. It is also dedicated to the research adviser who guided the direction of this work, and to the families and loved ones of the researchers, whose unwavering support remained constant throughout this journey. This study is further dedicated to the engineers and personnel involved in bridge monitoring and maintenance, particularly those working in government agencies, in the hope that this project will contribute to improving current practices and enhancing public safety. The researchers also dedicate this work to future innovators and developers who aspire to advance technology in structural health monitoring. Lastly, this study is dedicated to the Almighty God, the source of strength, wisdom, and perseverance behind this endeavor.

TABLE OF CONTENTS

		Page No.
TITLE PAGE		I
ACKNOWLEDGEMENT		II
DEDICATION		III
TABLE OF CONTENTS		IV
LIST OF TABLES		VI
LIST OF FIGURES		VII
CHAPTER I	INTRODUCTION	
	Rationale of the Study	2
	Objectives of the Study	4
	Scope and Limitations	4
	Significance of the Study	5
	Flow of the Study	5
	Definition of Terms	6
CHAPTER II	RELATED LITERATURES AND STUDIES	
	Theoretical Background	8
	Related Literature	10
	Related Studies	11
	Comparative Matrix	12
CHAPTER III	RESEARCH METHODOLOGY	
	Software Engineering Methodology	18
	Planning/Concept - Initiation Phase	27
	Business Model Canvas	27
	Program Workflow	29

Validation Board 32
Gantt Chart 34
Functional Decomposition Diagram 36
Architectural Layout 40
Block Diagram 43
Analysis - Design Phase 43
Use Case Diagram 43
User Interface Design 45
Storyboard 115
Database Design 119
Entity Relationship Diagram 119
Data Dictionary 123
Network Design 137
Network Model 138
Network Topology 139
Development/Construction/Build Phase 139
Technology Stack Diagram 140
Software Specification 141
Hardware Specification 142
Program Specification 144
List of Modules 145

REFERENCES 149
APPENDICES 159
TEAM COMPOSITION 172

LIST OF TABLES

Table No.	Table Name	Page No.
1	Comparative Matrix	13
2	Comparative Matrix Summary	15
3	Agile Sprint Plan	20
4	Business Model Canvas	28
5	Validation Board	32
6	Gantt Chart	34
7	Users	123
8	Bridges	124
9	Bridge Assignments	125
10	Sensor Nodes	126
11	Vibration Readings	127
12	Monitoring Configurations	129
13	Inspection Tasks	130
14	Inspection Records	131
15	Anomaly Cases	132
16	Anomaly Feedback	133
17	Sensor Issue Reports	134
18	Notifications	135
19	Reports	136
20	Technology Stack	140
21	Software Specification	141
22	Hadware Specification	143
23	List of Modules	145

LIST OF FIGURES

Figure No.	Figure Name	Page No.
1	Flow of the Study	5
2	Agile Software Development Cycle	18
3	Program Workflow (DPWH Administrator)	30
4	Program Workflow (DPWH Engineer)	31
5	Functional Decomposition Diagram (ViBridge Mobile Application)	37
6	Functional Decomposition Diagram (ViBridge Web Application)	38
7	Side View	40
8	Under Side/Bottom View	41
9	Protection Box	42
10	Block Diagram	43
11	Use Case Diagram of ViBridge	44
12	Engineer Login Page	45
13	Dashboard Screen Page	46
14	Assigned Bridges Screen Page	47
15	Select Bridges Page	48
16	Bridge Profile Page	49
17	View Bridge Location Page	50
18	Bridge Monitoring Page	51
19	Select Monitoring Page	53
20	Select Sensor Page	54
21	View Real Time Vibration Page	55
22	View Baseline and Threshold	56
23	View Monitoring History Page	57
24	Notification Page	58
25	System Detected Anomaly Alert Page	59
26	Inspection Task Page	60
27	Sensor Issue Notice Page	61
28	Anomaly Feedback Request Page	62

29	Anomaly Cases Page	63
30	System Anomalies Page	64
31	Anomaly Details Page	65
32	Review Vibration Data Page	66
33	Add Feedback Anomaly Page	67
34	Record Anomaly Detected Page	68
35	Select Inspected Bridge Page	69
36	Inspection Findings Page	70
37	Submit Anomaly Record Page	71
38	Report Sensor Issues Page	72
39	Select Sensor Node Page	73
40	Submit Sensor Issue Report Page	74
41	Manage Profile Page	75
42	View Profile Page	76
43	Update Profile Page	77
44	Web Administration Login Page	78
45	Web Administration Dashboard Page	79
46	Web Administration Review Anomaly Alerts Page	80
47	Administration Review Bridge Records Management Page	81
48	Administration Bridge Profile Page	82
49	Administration Update Bridge Details Page	83
50	Administration Register New Records Page	84
51	Administration Installed Sensors Page	85
52	Administration Sensors Details Page	86
53	Administration Edit Sensors Page	87
54	Administration Link New Sensor Page	88
55	Administration Monitor Bridge Condition Page	89
56	Administration Select Bridge Page	90
57	Administration Select Sensor Node Page	91
58	Administration Real Time Data Vibration Page	92
59	Administration Baseline and Threshold Page	93

60	Administration Monitoring History Page	94
61	Administration Manage Sensor Nodes Page	95
62	Administration View Sensor Node Profile Page	96
63	Administration Flag Sensor Node for Maintenance Page	97
64	Administration Manage Monitoring Configuration Page	98
65	Administration View Baseline and Threshold Page	99
66	Administration Update Configuration Page	100
67	Administration Update Sensor Node Details Page	101
68	Administration Review Active Anomaly Alerts Page	102
69	Administration View Alert Details Page	103
70	Administration Update Alert Status Page	104
71	Administration Update Review Remarks Page	105
72	Administration Generate Anomaly Reports Page	106
73	Administration View Bridge Location Page	107
74	Administration View Notifications Page	108
75	Administration Manage Engineer Accounts Page	109
76	Administration Add Engineer Account Page	110
77	Administration Update Engineer Account Page	111
78	Administration Deactivate Engineer Account Page	112
79	Administration Manage Admin Account Page	113
80	Administration View Admin Profile Page	114
81	Administration Update Admin Information Page	115
82	Storyboard of Mobile Monitoring Application	116
83	Storyboard of Web Monitoring Application	118
84	Entity Relationship Diagram	120
85	Network Models	138
86	Network Topology	139

CHAPTER I
INTRODUCTION

Bridges are critical transport assets because they support mobility, access to services, and the movement of goods across communities. Their safety and serviceability are therefore essential to public access, economic activity, and infrastructure reliability. However, bridge performance changes over time due to traffic loads, environmental exposure, seismic activity, and long-term material aging. Recent bridge-monitoring literature emphasizes that maintaining bridge safety requires more than periodic record keeping. Reviews by Deng et al. (2023) and Kang et al. (2025) explain that modern bridge structural health monitoring increasingly relies on continuous sensing and data interpretation to support safer and more timely maintenance decisions.

Despite this importance, bridge monitoring still commonly relies on periodic visual inspection. Although visual inspection remains an important part of bridge assessment, it cannot provide continuous structural information between scheduled evaluations. Gaviña et al. (2021) explained that visual inspections are highly variable, lack resolution, and may fail to detect damage unless it is already visible. Similarly, De La Torre et al. (2020) noted that visual inspection provides insufficient data about the actual condition of bridge structures. As a result, hidden deterioration, abnormal vibration behavior, and early structural changes may remain unnoticed until they become more serious or difficult to manage.

This concern becomes more significant for small bridges that are exposed to repeated traffic effects, deterioration mechanisms, typhoons, high-salinity environmental conditions, and seismic hazards. Ibrahim et al. (2024) explained that bridge components are affected by multiple deterioration factors, while Montes et al. (2020) linked weak load-limit monitoring to deterioration and collapse. Montes et al. (2020) also reported that the 2013 Bohol earthquake damaged 41 bridges and 18 roads. These findings highlight that after major catastrophic events, visual inspection alone may not be sufficient to ensure structural safety, because some forms of damage may not be immediately visible yet can still affect the bridge's actual condition and performance.

To respond to this need, this study proposes "ViBridge," an IoT-based bridge vibration and structural condition monitoring system designed to support real-time bridge assessment. The proposed system uses ADXL355 triaxial MEMS accelerometers installed on bridge components, ESP32-based wireless sensing nodes for data transmission, and Android mobile and web-based monitoring platforms for viewing and organizing collected readings. The system is intended to

support continuous vibration data collection, baseline and threshold checking, Random Forest-assisted anomaly classification, alert generation, and map-based identification of bridge and sensor locations. Recent studies show that wireless sensing, low-cost IoT hardware, and connected monitoring platforms can support practical bridge monitoring when data are collected continuously and transmitted for analysis (Sonbul & Rashid, 2023; Bhatta & Dang, 2024; Hidalgo-Fort et al., 2024).

Through this proposed system, “ViBridge” aims to provide DPWH bridge monitoring and maintenance personnel, particularly DPWH administrators and engineers, with timely monitoring information that can support maintenance and safety-related decisions for small bridges. Rather than replacing professional inspection, the system is designed to complement existing inspection practices by providing continuous vibration readings, anomaly alerts, and organized monitoring records that can help engineers identify when and where further assessment may be needed.

The proposed system processes vibration readings and compares current behavior against baseline and threshold conditions to help determine whether bridge response remains normal or may indicate a possible anomaly. It also uses a Random Forest algorithm to support clearer interpretation of unusual vibration patterns. Recent studies have shown that machine-learning-based structural health diagnosis can improve pattern recognition in monitoring data, while Random Forest models can support bridge-condition classification (Bao et al., 2025; Sunchikala et al., 2025). Through these functions, "ViBridge" is intended to support earlier anomaly detection, clearer condition reporting, and more responsive bridge-monitoring decisions.

Rationale of the Study

Bridge monitoring remains a significant concern because bridges must continue operating safely even as their structural condition changes over time. In many local engineering practices, bridge assessment still depends mainly on periodic visual inspection. While inspection is necessary, it is limited because it cannot continuously observe the bridge’s actual response during regular use. This creates a gap between scheduled inspections, were early signs of deterioration, abnormal vibration behavior, or hidden structural changes may remain undetected.

This study is important because small bridges in Cebu and similar local contexts are exposed to several risk factors, including traffic loading, aging materials, typhoons, seismic activity, and high-salinity environmental conditions. These conditions may affect bridge performance even before visible damage appears. Because of this, relying only on surface-based inspection may not provide enough timely information for engineers who need to assess bridge safety and plan maintenance actions. A monitoring approach that observes vibration behavior while

the bridge is in service can provide additional data that may help support more informed engineering decisions.

The study is also justified by the limitations of existing bridge monitoring applications. Some existing systems provide isolated functions such as smartphone-based vibration measurement, structural data viewing, or inspection documentation. However, these functions are often not combined into one accessible platform that supports continuous sensing, threshold-based checking, machine-learning-based anomaly classification, alert generation, and map-based location identification. For engineers handling multiple bridges, the lack of a centralized monitoring platform can make it difficult to quickly view vibration readings, identify possible anomalies, locate affected bridge areas, and organize inspection records.

ViBridge addresses this gap by proposing an IoT-based monitoring system that combines hardware sensing, data transmission, anomaly detection, and monitoring platforms in one system. The use of ADXL355 triaxial MEMS accelerometers and ESP32 wireless sensing nodes allows the system to collect and transmit bridge vibration data in real time. The use of baseline and threshold checking, supported by Random Forest classification, provides a method for identifying possible structural anomalies based on changes in vibration behavior. Meanwhile, the Android and web-based platforms allow users to view vibration data, receive alerts, locate monitored bridges and sensor nodes, and manage monitoring records in a centralized environment.

This is the gap that makes the development of “ViBridge” important. While recent studies already support wireless sensing, connected platforms, and intelligent diagnosis, these capabilities are often discussed separately in the literature or applied in settings that are broader than the small-bridge context intended in this study. By combining ADXL355 triaxial MEMS accelerometer sensing, ESP32-based wireless data transmission, baseline and threshold checking, Random Forest based analysis for anomaly classification, and alert generation in one system, “ViBridge” is intended to support earlier detection of unusual bridge behavior, clearer monitoring decisions, and more timely reporting of possible structural concerns (Abdrabo, 2024; Bao et al., 2025; Sunchikala et al., 2025). The expected impact of the system is to help monitor small bridges more responsively by providing timely alerts and organized vibration data that can guide inspection priorities. Through this, the system may support earlier maintenance action, improve bridge safety awareness, reduce reliance on delayed manual observation, and help lessen the risk of unnoticed structural issues affecting commuters, local communities, and transport services.

Objectives of the Study

This study aims to develop “ViBridge”, an IoT-based bridge vibration and structural condition monitoring system through an Android mobile application and web-based monitoring platform that supports real-time monitoring, vibration analysis, and timely alert generation for bridge condition assessment.

Specifically, this study pursues the following objectives:

1. To gather data and understand the current practices and challenges in the inspection, vibration monitoring, and maintenance of small bridges, with focus on how DPWH engineers manage bridge maintenance activities.
2. To develop an IoT-based bridge monitoring system that uses the ADXL355 triaxial MEMS accelerometer and wireless data transmission to collect, transmit, and display real-time bridge vibration data.
3. To apply baseline vibration patterns, threshold-based checking, and a Random Forest algorithm in interpreting sensor data, classifying vibration patterns, and detecting possible structural anomalies.
4. To implement an in-app notification and email alert mechanism, together with a dynamic map-based interface dashboard, for reporting detected anomalies and identifying the location of affected bridges and sensor nodes.

Scope and Limitations of the Study

This study focuses on the development of ViBridge, an IoT-based bridge vibration and structural condition monitoring system for selected small bridges. The system covers the use of an ADXL355 triaxial MEMS accelerometer, ESP32-based wireless data transmission, Android mobile application, and web-based monitoring platform to collect, transmit, display, and monitor bridge vibration data in real time.

The study also includes baseline and threshold checking, Random Forest-based anomaly classification, in-app and email alert generation, and a map-based interface for identifying monitored bridges and affected sensor nodes. These features are intended to help authorized users monitor bridge conditions, detect possible anomalies, and support timely reporting.

The study is limited to the selected bridge or bridges used during pilot implementation. The system depends on proper sensor placement, calibration, power supply, and internet connection. It is also limited to Android and web platforms and does not support iOS devices. The system does not replace professional engineering inspection, provide official structural certification, perform automatic repair actions, or connect directly to emergency dispatch services.

Significance of the Study

The study holds significant importance for various stakeholders involved in bridge monitoring. The following are the key areas where the study's findings can have a positive impact:

Department of Public Works and Highways (DPWH). The primary beneficiaries of the study are DPWH offices and personnel responsible for bridge monitoring, maintenance, and management. Through ViBridge, DPWH administrators and engineers may access real-time vibration data, anomaly alerts, baseline and threshold information, bridge records, and sensor-node locations. These features may help support timely maintenance decisions, improve monitoring of selected small bridges, and provide organized records for bridge condition assessment.

Commuters and Pedestrians. The general public in Cebu will benefit from enhanced safety and reliability in their daily transportation. With continuous monitoring, the risk of sudden structural failure is minimized, ensuring that the bridges used by commuters, students, and local businesses are structurally sound and secure for transit.

Local Economy and Small Businesses. Although not direct users of "ViBridge", local businesses and logistics providers will benefit from the prevention of unexpected bridge closures. By shifting to proactive maintenance, the platform helps ensure the uninterrupted flow of goods and services across Cebu's secondary road networks, supporting economic stability.

Future Researchers. Researchers and developers in the field of civil engineering and the Internet of Things (IoT) will benefit from the methodologies and vibration datasets developed in this study. It will encourage further innovation in the design of affordable, AI-powered monitoring systems for varied types of infrastructure in high-salinity and seismically active regions.

Flow of the Study

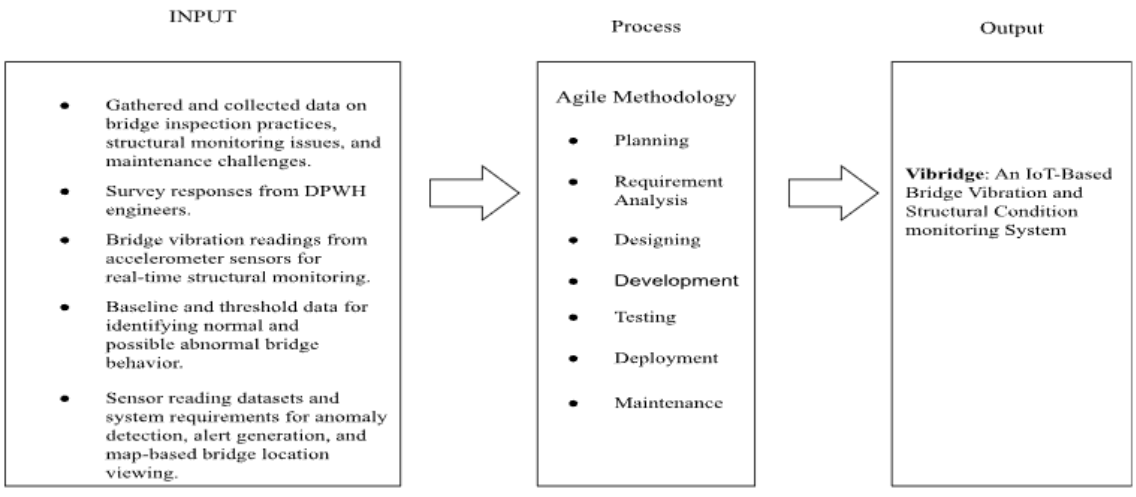


Figure 1: **Flow of the Study**

Figure 1 shows the flow of the “ViBridge” study. The process begins with gathering and collecting data on current bridge inspection practices, structural monitoring issues, and maintenance challenges related to bridge condition assessment. It also includes survey or consultation responses from DPWH engineering personnel to understand the needs and challenges in bridge monitoring. In addition, the study collects bridge vibration readings from the ADXL355 triaxial MEMS accelerometer, together with baseline data, threshold-based checking requirements, sensor reading datasets, and the hardware and software requirements needed for the system. These inputs serve as the foundation for developing the proposed bridge monitoring platform.

The next stage includes system development, system testing, and dataset analysis. In this stage, the collected readings are organized, analyzed, and prepared to support the system’s monitoring functions, including vibration analysis, baseline and threshold-based checking, Random Forest-based anomaly classification, alert generation, and map-based monitoring support. This process helps prepare the system for real-time bridge condition monitoring.

The study follows the Agile Methodology, which includes planning, requirement analysis, designing, development, testing, deployment, and maintenance. This approach allows the system to be improved step by step based on testing results, system performance, and user feedback. It also supports continuous refinement of both the hardware and software components of the platform.

The final output of the study is “ViBridge,” an IoT-based bridge vibration and structural condition monitoring system. The system is designed to support real-time monitoring, baseline and threshold-based assessment, Random Forest-based analysis, in-app alert generation, email alerts, and map-based location viewing through Android and web platforms. Through this process, the study aims to produce a functional system that can support more responsive and data-driven bridge monitoring.

Definition of Terms

This section presents the operational definition of terms used in the study for clearer understanding of the concepts discussed.

AI-Assisted Analysis. It refers to the use of artificial intelligence in helping the system interpret sensor readings and identify possible abnormal bridge vibration patterns.

Anomaly Detection. It refers to the process of identifying unusual vibration behavior that may indicate possible structural problems in the bridge.

Baseline Data. It refers to the normal vibration readings collected from the bridge, which are used as the reference for comparison in monitoring bridge condition.

Random Forest. It refers to the AI-based method used in the system to classify sensor data and help detect abnormal vibration patterns or possible structural anomalies.

Dynamic Map Feature. It refers to the map-based interface of the system that shows the location of monitored bridges and installed sensor nodes and helps users identify the affected bridge when an anomaly is detected.

Oscillation Period. It refers to the time it takes for one complete cycle of bridge movement or vibration, which is used as one of the indicators in bridge condition analysis.

Real-Time Monitoring. It refers to the continuous collection and display of bridge vibration data as they are recorded by the system.

Structural Condition Monitoring System. It refers to the proposed system, ViBridge, which is designed to monitor bridge vibration and structural condition using sensors, wireless transmission, analysis, and alerts.

Threshold Value. It refers to the set limit used by the system to determine whether the observed bridge vibration remains normal or may already indicate unusual structural behavior.

Wireless Data Transmission. It refers to the sending of sensor readings from the bridge-mounted devices to the Android mobile application and web-based monitoring platform without the use of wired connections.

CHAPTER II

REVIEW OF RELATED LITERATURE AND STUDIES

This chapter presents the literature and studies that support the development of ViBridge: An IoT-Based Bridge Vibration and Structural Condition Monitoring System for Small Bridges. It discusses concepts and existing works related to bridge structural health monitoring, vibration-based sensing, IoT-based data transmission, sensor readings, baseline and threshold-based checking, anomaly detection, and machine-learning-supported bridge assessment.

Theoretical Background

This study is anchored on Structural Health Monitoring (SHM) theory, which explains that the condition of a bridge can be assessed through continuous or periodic measurement of its structural response. Recent reviews describe SHM as a data-driven monitoring approach that supports safety evaluation, early diagnosis, and maintenance planning by observing changes in bridge behavior over time (Deng et al., 2023; Yang et al., 2024). In relation to ViBridge, this theory supports the continuous collection of vibration data instead of relying only on occasional visual inspection.

The study is also supported by vibration-based monitoring theory. In bridge applications, vibration is important because changes in acceleration, frequency, and other response patterns may reflect changes in structural condition. Recent open-access reviews explain that vibration-based methods remain central in bridge SHM because they help reveal how a structure behaves under actual service loads and environmental conditions (Sonbul & Rashid, 2023; Kang et al., 2025). In ViBridge, this supports the use of bridge vibration readings as the primary basis for observing structural response.

Another theoretical support is MEMS accelerometer sensing theory, which explains that small sensor devices can measure acceleration and movement along different axes. In structural monitoring, accelerometers are used to detect vibration responses that may occur when a bridge is affected by traffic loads, environmental conditions, or possible structural changes. The ADXL355 triaxial MEMS accelerometer is relevant to this study because it can acquire three-axis acceleration readings and send the measured data to a microcontroller for processing and transmission. Muttillo et al. (2020) used the ADXL355 as the main accelerometer in an IoT-based SHM system, while Hasani et al. (2024) developed a wireless bridge monitoring system using MEMS accelerometers for operational modal analysis. In ViBridge, this theory supports the use of the ADXL355 sensor to collect vibration readings along the X, Y, and Z axes from the monitored bridge.

Another theoretical support is the Internet of Things (IoT), which enables sensors, embedded devices, and communication modules to collect and transmit structural data through connected networks. Recent literature explains that IoT-based SHM makes monitoring more practical by linking sensing devices with wireless communication, edge processing, and cloud or platform-based access (Bhatta & Dang, 2024; Hidalgo-Fort et al., 2024). This supports ViBridge because the proposed system depends on connected sensor nodes, ESP32-based wireless data transmission, and accessible web and Android mobile monitoring interfaces.

The study is further supported by signal processing theory. Raw acceleration signals are not immediately ready for interpretation because they often need cleaning, organization, and preparation before useful patterns can be identified. Recent studies show that bridge monitoring systems commonly process vibration signals to obtain clearer indicators from measured bridge movement (Bao et al., 2025; Sunchikala et al., 2025). In ViBridge, this theory supports the conversion of raw sensor readings into usable monitoring data for baseline comparison, threshold checking, anomaly detection, and alert generation.

Another supporting concept is threshold-based and anomaly-aware monitoring. Bridge monitoring systems commonly compare current sensor values against baseline behavior, calibrated ranges, or learned normal patterns so that unusual responses can be flagged early. Recent open access studies emphasize the importance of calibration, threshold definition, and anomaly detection methods in improving the reliability of bridge monitoring decisions (Yang et al., 2024; Abdrabo, 2024). This supports ViBridge because the system is designed to issue alerts when measured vibration behavior departs from expected conditions.

Lastly, the study is supported by Random Forest classification theory. Random Forest is a machine-learning approach that combines several classification outputs to produce a more reliable result. In structural health monitoring, this type of model can help classify vibration patterns and support the identification of possible structural anomalies. Recent bridge-focused studies show the relevance of Random Forest in bridge condition assessment. Yaghoubzadehfard et al. (2024) reported that Random Forest was effective in identifying bridge damage location and severity, while other recent studies also support the use of machine-learning models for classifying damage-related vibration patterns in bridge monitoring (Bao et al., 2025; Sunchikala et al., 2025). In ViBridge, this theory supports the AI-assisted component of the system, where vibration data are classified to help identify whether a bridge condition is normal or possibly anomalous.

Taken together, these theories support the design and operation of ViBridge. SHM theory supports continuous bridge condition assessment, vibration-based monitoring theory supports the use of bridge motion data, MEMS accelerometer sensing theory supports the use of ADXL355 triaxial acceleration readings, IoT theory supports connected sensing and wireless transmission, signal processing theory supports the preparation of raw vibration readings into useful monitoring data, threshold-based monitoring supports alert generation, and Random Forest classification theory supports clearer classification of abnormal bridge behavior.

Related Literature

Recent literature shows that bridge SHM has increasingly shifted toward integrated systems that combine sensing, communication, data processing, and condition assessment. Reviews by Deng et al. (2023), Bhatta and Dang (2024), and Kang et al. (2025) explain that modern bridge monitoring no longer focuses on sensors alone; instead, it connects sensing, transmission, preprocessing, and decision support into one monitoring workflow. This direction is consistent with the purpose of ViBridge.

Literature also shows that wireless sensor networks and ADXL355 triaxial MEMS accelerometer-based measurement remain practical for bridge applications. Sonbul and Rashid (2023) found that wireless platforms are widely used in bridge SHM because they reduce installation constraints and support dynamic response monitoring. Likewise, Kang et al. (2025) noted that bridge SHM systems commonly capture vibration, strain, and environmental responses in real time. These findings support the use of ADXL355-based sensor nodes in ViBridge.

Another strong theme in recent literature is the growing role of IoT and edge-enabled monitoring. Bhatta and Dang (2024) described IoT as a major enabler of real-time SHM because it connects sensors, communication modules, storage, and analytics into a unified monitoring environment. Hidalgo-Fort et al. (2024) further demonstrated that low-cost and low-power edge-computing designs can support wireless bridge monitoring while reducing dependence on heavier infrastructure. This literature supports the practical implementation of ViBridge using sensor nodes and lightweight networked devices.

Recent literature also emphasizes that bridge monitoring systems depend on proper data processing. Bao et al. (2025) explained that intelligent SHM depends on data cleaning and reliable diagnosis because raw monitoring data are often incomplete, noisy, or inconsistent. Sunchikala et al. (2025) likewise showed that processed vibration data can improve the classification of bridge conditions. This supports the need for data processing in ViBridge before alerting or classifying structural behavior.

Machine-learning and anomaly-detection methods are also becoming more important in recent bridge monitoring literature. Bao et al. (2025) reviewed machine-learning-based structural health diagnosis and highlighted its value in handling complex bridge data. Abdrabo (2024) showed that one-class classification can be applied to bridge SHM for anomaly detection when the goal is to recognize departures from normal behavior. These works support the AI-assisted analysis component of ViBridge, especially for identifying abnormal vibration patterns from monitored data.

Finally, recent literature suggests that monitoring and inspection are becoming more complementary. Ribeiro et al. (2025) reviewed remote bridge inspection methods and showed that digital and sensor-based approaches can improve observation, documentation, and decision support. While ViBridge is mainly a vibration-based monitoring system rather than a full inspection platform, this literature supports the idea that digital tools can help engineers make faster and better-informed assessment decisions.

Overall, the reviewed literature supports the major technological components of "ViBridge" continuous monitoring, ADXL355 triaxial MEMS accelerometer-based sensing, IoT communication, data processing, anomaly detection, and decision support. The reviewed works also show that recent bridge monitoring research favors systems that are practical, connected, and capable of supporting earlier response.

Related Studies

Recent bridge-related applications show that digital tools are already being used to support bridge monitoring, structural assessment, and inspection documentation. These applications provide functions such as vibration measurement, sensor-based monitoring, live structural data viewing, graphical reports, field inspection recording, and photo documentation. The following applications were reviewed to identify the features that are already available in existing systems and the functions that may still be limited or separated.

App4SHM is a smartphone application for structural health monitoring of bridges and other civil structures. The application uses the phone's internal accelerometer to measure structural acceleration. It also applies artificial intelligence techniques to detect possible damage almost in real time. App4SHM is intended for assessing the condition of bridges or other civil structures after a catastrophic event or when required by authorities and stakeholders.

Bridge Monitoring by 4 EMME is a mobile application developed for monitoring bridges and other structures. The application is designed to work with control units and sensors installed on the structure. It supports continuous monitoring, accurate data collection, graphical data

visualization, activity history, and detailed reports. Its public page states that it can collect data on temperature, vibration, humidity, and other important structural parameters.

Structural Health Monitoring by iLink Air is a BLE-based mobile application designed to connect with structural health monitoring acquisition systems. The application provides real-time monitoring of structural deflections and loads. It also supports live data analysis and mobile access to monitoring information, making it useful for engineers, technicians, students, and academic users who need to view structural data from compatible acquisition systems.

Bridge Inspection (BI) App is a field inspection application for bridge inspectors. The application allows inspectors to capture photographs, record findings, and synchronize photos and notes over Wi-Fi. It is designed to help inspectors organize bridge inspection activities in the field and improve the documentation process during inspection work.

BRIDGE by Woodlands North Inc. is a bridge inspection application available on Google Play. Its public description states that it provides complete bridge inspection and includes four inspection types: Basic, Engineering, Restoration, and New Bridge Installation. Each inspection type is designed based on regulatory authority requirements and industrial standards.

The reviewed applications show that existing bridge-related systems already provide useful features for structural monitoring and inspection. App4SHM supports smartphone-based vibration measurement and AI-assisted damage detection. Bridge Monitoring by 4 EMME supports sensor-based structural monitoring, graphical reports, and activity history. Structural Health Monitoring by iLink Air supports real-time monitoring of deflections and loads through BLE-connected acquisition systems. Bridge Inspection (BI) App and BRIDGE support bridge inspection documentation and field assessment.

However, these features are often separated across different applications. Some applications focus on vibration measurement, some focus on sensor data viewing, and others focus on inspection records. This creates an opportunity for a system that combines dedicated accelerometer-based vibration sensing, wireless data transmission, baseline and threshold-based checking, Random Forest-based anomaly classification, alert generation, inspection records, and map-based bridge location viewing in one platform for small bridges.

Comparative Matrix

This section presents a comparative matrix developed to organize and assess relevant studies and downloadable applications in relation to " ViBridge". It highlights shared and distinct features across systems, including bridge inspection support, vibration or structural monitoring, data recording, mobile access, alerts, and condition assessment functions. The matrix also identifies the strengths, limitations, platform details, support mechanisms, and monetization schemes of each

system. Through this comparison, the study is able to position “ViBridge” as an IoT-based bridge vibration and structural condition monitoring system that combines sensor-based data collection, connected monitoring platforms, and anomaly interpretation in a more focused solution for bridge safety and condition assessment.

Table 1.1

COMPARATIVE MATRIX

Related Studies	Features	Limitations	Platform Details	Support	Monetization Scheme
Name: “App4SHM “ URL: https://play.google.com/store/apps/details?id=pt.cofac.deisi.labs.app4shm Company name: Universidade Lusófona	Uses smartphone accelerometer for bridge/civil structure SHM; supports AI-assisted damage detection; intended for post-event or required structural assessment.	Uses phone sensor instead of dedicated bridge-mounted sensor nodes; best for measurement sessions rather than fixed continuous deployment; map-based bridge location viewing and inspection record handling are not clearly stated.	Mobile Application (IOS & Android)	Official website and app store pages provide access and information.	Publicly downloadable/research-oriented application; no clear public subscription plan stated.
Name:”Bridge Monitoring “ URL:https://play.google.com/store/apps/details?hl=en&id=com.quattroemme.monitoraggi_app&utm Company name:4 EMME	Supports bridge/structure monitoring through control units and sensors; collects vibration, temperature, humidity, and other structural data; provides graphs and reports.	Requires compatible control units and sensors; implementation may be costly; Random Forest-based anomaly classification and map-based affected bridge identification are not clearly stated.	Mobile application (Android)	Google Play listing and developer support channel.	Commercial/professional monitoring solution tied to hardware and services.

Table 1.2

COMPARATIVE MATRIX cont'd

Related Studies	Features	Limitations	Platform Details	Support	Monetization Scheme
Name: "Structural Health Monitoring " URL: https://play.google.com/store/apps/details?hl=en&id=com.app.structuralhealthmonitioring&utm Company name: iLink Air	Connects to SHM acquisition systems through BLE; supports real-time monitoring of deflections and loads; provides live data analysis.	Depends on compatible acquisition system; not clearly bridge-specific; does not clearly state automated alerts, Random Forest classification, or map-based bridge location viewing.	Mobile Application (Android)	Google Play developer support.	Companion monitoring application; pricing not clearly stated in public listing.
Name: "Bridge Inspection " URL: https://apps.apple.com/us/app/bridge-inspection-bi-app/id1423008941?utm Company name: E-management services LLC	Supports bridge inspection documentation; allows photo capture; records findings; syncs notes and photos.	Focuses on inspection records, not continuous vibration monitoring; no sensor-based vibration analysis; no AI anomaly classification or threshold-based checking stated.	Mobile Application (IOS)	App Store support/developer listing.	Professional inspection support application.
Name: "BRIDGE" URL: https://play.google.com/store/apps/details?hl=en&id=com.woodlandsnorth.bridge Company name: Woodlands North	Supports bridge inspection activities; provides Basic, Engineering, Restoration, and New Bridge Installation inspection types.	Focuses on inspection documentation rather than continuous vibration monitoring; does not clearly state IoT-based sensor monitoring, Random Forest-based anomaly classification, threshold-based checking, or automated alert generation.	Mobile application (Android)	Google Play app support and developer contact are provided.	Publicly downloadable application; no clear subscription plan is stated on the Google Play listing.

Tables 1.1 and 1.2 present on what ViBridge shares similarities with the related applications mentioned, particularly in its use of digital tools to support bridge monitoring and bridge condition assessment. Like existing systems such as App4SHM, Bridge Monitoring, and Structural Health Monitoring, ViBridge focuses on collecting and presenting structural data that can help users observe bridge conditions. These systems show that bridge-related applications can already support functions such as vibration monitoring, sensor-based data collection, real-time data viewing, and structural health assessment. Similar to Bridge Inspection and BRIDGE, ViBridge also recognizes the importance of organized bridge records that can support inspection and monitoring activities.

ViBridge also presents key differences in its approach. While many related applications focus only on one part of the monitoring process, such as smartphone-based vibration measurement, structural data viewing, or bridge inspection recording, ViBridge combines these functions into one system. It uses dedicated bridge-mounted sensors to collect vibration readings, applies baseline and threshold-based checking to identify possible abnormal behavior, and uses Random Forest-based anomaly classification to support the interpretation of sensor data. The system also includes notifications and alerts, graphical monitoring, inspection records, and map-based bridge location viewing.

ViBridge sets itself apart through its integrated monitoring approach for small bridges. Instead of relying only on visual inspection records or separate monitoring tools, the system provides real-time vibration data, possible anomaly detection, alert generation, and organized bridge information through Android and web platforms. Through these features, ViBridge is intended to support engineers and administrators in making faster, more informed, and more responsive bridge-monitoring decisions.

Table 2.1

COMPARATIVE MATRIX SUMMARY

Related Studies	**Sensor-Based Structural Monitoring**	**Vibration-Based Monitoring**	**Baseline and Threshold-Based Checking**	**AI-Assisted Anomaly Detection**	**Notifications and Alerts**	**Map-Based Bridge Location Viewing**	**Inspection Documentation**
Name/Title: App4SHM	✓	✓	X	✓	X	X	X

Table 2.2

COMPARATIVE MATRIX SUMMARY cont'd

Related Studies	Sensor-Based Structural Monitoring	Vibration-Based Monitoring	Baseline and Threshold-Based Checking	AI-Assisted Anomaly Detection	Notifications and Alerts	Map-Based Bridge Location Viewing	Inspection Documentation
Name/Title: Bridge Monitoring	✓	✓	X	X	X	X	X
Name/Title: Structural Health Monitoring	✓	X	X	X	X	X	X
Name/Title: Bridge Inspection	X	X	X	X	X	X	✓
Name/Title: BRIDGE	X	X	X	X	X	X	✓
Name/Title: "ViBridge": An IoT-Based Bridge Vibration and Structural Condition Monitoring System for Small Bridges	✓	✓	✓	✓	✓	✓	✓

Tables 2.1 and 2.2 present the comparative matrix summary of the proposed "ViBridge" system and selected related systems. The table compares the systems based on key features relevant to bridge vibration and structural condition monitoring, including sensor-based structural monitoring, vibration-based monitoring, baseline and threshold-based checking, AI-assisted anomaly detection, notifications and alerts, map-based bridge location viewing, and inspection documentation.

Based on the comparison, some existing systems already provide useful monitoring or inspection-related functions. App4SHM supports sensor-based and vibration-based monitoring using a smartphone accelerometer and also includes AI-assisted checking. Bridge Monitoring supports sensor-based and vibration-based monitoring through structural sensors and monitoring equipment. Structural Health Monitoring provides sensor-based structural monitoring, while Bridge Inspection focuses mainly on inspection documentation rather than continuous vibration monitoring. These systems show that current applications can support certain parts of bridge monitoring, data viewing, or field inspection activities.

However, the table also shows that most existing systems only focus on selected features. Some are mainly designed for vibration measurement, some are intended for structural data viewing, while others are focused on inspection documentation. In contrast, “ViBridge” combines sensor-based structural monitoring, vibration-based monitoring, baseline and threshold-based checking, AI-assisted anomaly detection, notifications and alerts, map-based bridge location viewing, and inspection documentation in one integrated platform.

Through this comparison, “ViBridge” is positioned as a more complete and responsive bridge vibration and structural condition monitoring system for small bridges. Its main advantage is the integration of dedicated sensor-node monitoring, baseline and threshold checking, anomaly detection, alert generation, location identification, and inspection-related documentation. This allows authorized users to monitor bridge conditions, identify affected sensor locations, and support more organized inspection and maintenance decisions.

CHAPTER III

RESEARCH METHODOLOGY

This chapter includes the procedures involved in creating the proposed system. Each section focuses on the analysis and technical details that support the proposal. It also presents the program workflow, diagrams, designs, future methodologies, resources, and tools for implementing "ViBridge": An IoT-Based Bridge Vibration and Structural Condition Monitoring System for Small Bridges. The system is intended to support DPWH bridge monitoring and maintenance personnel in monitoring bridge vibration data, identifying possible anomalies, and organizing bridge condition records. This chapter provides a clear understanding of the system's features, capabilities, risks, and technical framework to support more informed bridge monitoring and maintenance decisions.

Software Engineering Methodology

The ViBridge application employs the Agile Method as the main framework for the development of the system. The Agile method supports the concept of iterative development, collaboration, flexibility, and continuous improvement throughout the system lifecycle. Unlike traditional linear development models, Agile allows the researchers to refine system components progressively based on testing results, user feedback, and real-world monitoring conditions.

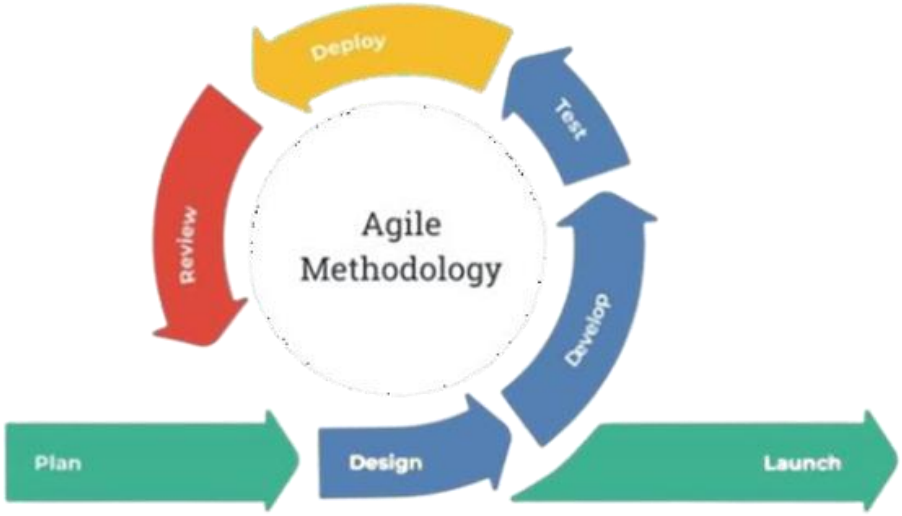


Figure 2: **Agile Software Development Cycle**

The researchers will utilize the iterative nature of Agile software development to guide the planning, design, development, testing, deployment, and review of the "ViBridge" system. This

methodology is appropriate for "ViBridge" because the system integrates multiple components, including IoT sensor data collection, real-time vibration monitoring, anomaly detection, cloud-based storage, alert notifications, and web and mobile applications. The flexibility of Agile allows the researchers to improve the system through repeated testing, feedback, and refinement during each phase. This reduces the risk of major revisions in the later stages of development and ensures that the system remains aligned with the needs of bridge monitoring personnel, particularly DPWH administrators and engineers.

Planning Phase. The researchers will identify the key structural parameters and indicators necessary for monitoring the condition of small bridges. This phase includes gathering information on vibration frequency, acceleration levels from the ADXL355 sensor, sensor reading patterns, and environmental disturbances that may affect bridge stability. The researchers will also determine the project scope, user needs, and major functional requirements of the system, including real-time monitoring, anomaly detection, cloud-based data storage, and automated alert notifications.

Design Phase. The researchers will design the overall architecture of the ViBridge system. This includes planning the placement of the ADXL355 triaxial MEMS accelerometer on bridge structures, designing the database schema for storing vibration and monitoring data, and creating the user interface layout for the web-based dashboard and Android mobile application. During this phase, the researchers will also define the baseline vibration patterns and safety threshold parameters to be used as the basis for structural alerts and anomaly detection.

Development Phase. The researchers will develop both the hardware and software components of the system. This includes interfacing the ADXL355 triaxial MEMS accelerometer with the ESP32 microcontroller, programming the firmware for data collection and transmission, and implementing the cloud-based database and backend services. The web dashboard and Android application will also be coded and integrated with the backend and monitoring system.

Testing Phase. The researchers will conduct iterative testing to evaluate the reliability, accuracy, and overall performance of the system. Sensor calibration will be carried out to verify the correctness of vibration measurements. The stability of data transmission and synchronization with the cloud database will also be examined. In addition, the anomaly detection model will be tested to assess its capability to distinguish normal structural vibrations from possible structural anomalies. Necessary revisions and adjustments will be made based on the results of the testing activities.

Deployment Phase. The researchers will deploy the ViBridge prototype on selected small bridges for pilot implementation. Sensor nodes will be installed and configured to enable real-time structural monitoring. The web dashboard and Android mobile application will then be activated so that authorized users, such as DPWH administrators and engineers, can access live vibration data, system alerts, and bridge monitoring information.

Review Phase. The researchers will continuously evaluate the performance of the deployed system using collected monitoring data and user feedback. Findings from field implementation and testing will be analyzed to refine sensor sensitivity, improve alert threshold settings, and enhance the overall functionality and usability of the system. This phase ensures that the system remains effective, accurate, and responsive to the monitoring needs of bridge management personnel.

Table 3.1

AGILE SPRINT PLAN

SDLC Phases	**1st Sprint**	**2nd Sprint**	**3rd Sprint**	**4th Sprint**	**5th Sprint**
	Establish the study foundation and research direction. Time Frame: 2 weeks	Establish the theoretical, literature, and comparative basis of the study. Time Frame: 2 weeks	Establish the design and technical planning foundation of the proposed system. Time Frame: 2 weeks	Conduct survey or consultation with engineers or related bridge personnel. Time Frame: 2 weeks	Finalize the functional and user interaction design of the system. Time Frame: 2 weeks
Plan	Mapped out the study foundation, research direction, and initial technical focus of the proposed capstone.	Outlined the coverage of Chapter II, including the theoretical background, related literature, related studies, and comparative matrix.	Set the scope for the Chapter III contents, including the business model, user workflows, UI direction, database choice, network structure, and IoT placement on the bridge.	Scheduled the survey or consultation process for engineers or personnel involved in bridge inspection and maintenance.	Identified the system functions, user roles, and interaction flow of the proposed system.

Table 3.2

AGILE SPRINT PLAN cont'd

Design	Sketched the initial structure of Chapter I and identified the possible IoT devices, algorithm, and key features to be used in the proposed system.	Arranged the structure and flow of the Chapter II sections based on the study objectives and system direction.	Formulated the initial concepts for the business model canvas, UI for users, database structure, network model, network topology, and architectural layout.	Crafted the questionnaire or interview guide to gather opinions about the proposed capstone manuscript and system concept.	Structured the program workflow, use case diagram, and functional decomposition diagram.
Develop	Compiled the early study direction, including the problem focus, proposed title scope, possible hardware components, possible anomaly detection approach, and major system features.	Developed the Chapter II content, including the supporting theories, recent literature, related studies, and comparative analysis.	Drafted the Chapter III planning outputs and supporting explanations for the proposed system design.	Assembled the survey or interview instrument and the validation board draft.	Produced the functional and interaction design outputs with their explanations.
Test	Verified whether the proposed devices, algorithm, and key features were aligned with the capstone title and study objectives.	Evaluated whether the selected references and discussions were relevant, updated, and supportive of ViBridge.	Validated whether the planned methodology and design outputs were aligned with the study objectives and proposed system features.	Confirmed whether the questions were clear, relevant, and aligned with the objectives and features of ViBridge.	Verified whether the workflows and diagrams correctly represent the system processes and user responsibilities.

Table 3.3

AGILE SPRINT PLAN cont'd

Deploy	N/A for manuscript phase.	N/A for manuscript phase.	N/A for manuscript phase.	Conducted the survey or consultation and collected responses from engineers or related personnel.	N/A for manuscript phase.
Review	Assessed the initial study foundation and research direction for refinement.	Reexamined the completeness and consistency of Chapter II.	Examined the feasibility and alignment of the Chapter III planning outputs.	Analyzed and summarized the gathered responses for manuscript support and system refinement.	Checked the completeness and clarity of the functional design outputs.
SDLC Phases	**6th Sprint** Build the technical data structure and documentation outputs. Time Frame: 2 weeks	**7th Sprint** Finalize the network, architecture, and technology design of the system. Time Frame: 2 weeks	**8th Sprint** Finalize the system documentation outputs for implementation readiness. Time Frame: 2 weeks	**9th Sprint** Set up the development environment and core system foundation. Time Frame: 2 weeks	**10th Sprint** Build the authentication and user access features of the system. Time Frame: 2 weeks
Plan	Listed the technical documentation outputs needed for the study, including the ERD, data dictionary, and related technical tables.	Determined the technical architecture, network structure, and technology requirements of the proposed system.	Organized the remaining documentation outputs needed before implementati-on.	The team outlined the setup needed for the web platform, Android application, database, and IoT integration.	User login flow, account structure, and role-based access for the Administrator and Engineer were identified.

Table 3.4

AGILE SPRINT PLAN cont'd

Design	Modeled the system data structure, entity relationships, and documentation format for the technical outputs.	Mapped the network model, network topology, architectural layout, block diagram, and technology stack.	Specified the structure of the software specification, hardware specification, program specification, and list of modules.	An initial project structure was arranged to define the folders, component flow, and connection of the major system parts.	The authentication process, account forms, and user access interface were drafted.
Develop	Created the ERD, data dictionary, and related technical tables and diagrams.	Documented the technical design diagrams and supporting explanations.	Compiled the documentation outputs for the system specifications and implementation-related sections.	Repositories, development tools, database connection, and starter project files for the web, Android, and IoT components were prepared.	Core account features such as login, logout, and user access management were implemented.
Test	Inspected whether the data structure and technical outputs were complete, consistent, and aligned with the proposed system.	Confirmed whether the technical design outputs were aligned with the proposed system functions and components.	Reviewed whether the documentation outputs were complete, organized, and consistent with the proposed system.	The initial setup was checked to confirm that the tools, dependencies, and connections were functioning properly.	Authentication behavior, account access, and user restrictions were verified.
Deploy	N/A for manuscript phase.	N/A for manuscript phase.	N/A for manuscript phase.	An initial local development build was prepared for internal setup checking.	A build with authentication features was released for internal testing.

Table 3.5

AGILE SPRINT PLAN cont'd

Review	Refined the technical documentation outputs for corrections and improvement.	Evaluated the accuracy and consistency of the technical architecture outputs.	Finalized the readiness of the manuscript and technical documentation before the next phase.	The setup results were examined to identify issues that needed to be resolved before feature development.	The team reviewed the account flow and noted improvements needed for usability and access control.
SDLC Phases	**11th Sprint** Develop the bridge and sensor management modules. Time Frame: 2 weeks	**12th Sprint** Develop the real-time vibration monitoring module. Time Frame: 2 weeks	**13th Sprint** Develop and integrate the AI-based anomaly classification component. Time Frame: 2 weeks	**14th Sprint** Implement threshold checking, alerts, notifications, and map-based monitoring features. Time Frame: 2 weeks	**15th Sprint** Integrate the full system and conduct pilot testing. Time Frame: 2 weeks
Plan	The needed functions for bridge profiles, bridge records, and sensor management were identified.	Requirements for live monitoring, vibration display, and real-time data viewing were mapped out.	The machine learning workflow for classifying vibration patterns and detecting possible anomalies was prepared.	Threshold logic, alert behavior, notification flow, email support, and map-based monitoring needs were determined.	The full integration of hardware, backend, web, Android, and monitoring features was prepared for pilot use.
Design	Layouts for bridge management, sensor configuration, and related data flow were prepared.	The monitoring dashboard, vibration graph, sensor status display, and live data layout were arranged.	A structure for data preparation, feature selection, model training, and anomaly output was organized.	The threshold settings flow, alerts page, notification interface, email process, and map screen were drafted.	An integrated system flow for actual monitoring and pilot deployment was organized.

Table 3.6

AGILE SPRINT PLAN cont'd

Develop	The bridge management and sensor management modules were created and connected to the system structure.	Pages for real-time vibration monitoring were built and connected to the incoming sensor readings.	The vibration dataset was prepared, required features were processed, the Random Forest model was trained, and its output was connected to the system.	Threshold-based checking, in-app alerts, notification handling, email support, and map-based viewing were added to the system.	Major modules were combined, and the sensor node, backend, web platform, Android application, and analysis features were connected into one working system.
Test	Handling of bridge records, sensor assignment, and management actions was checked.	Live display behavior, graph updates, and page responsiveness were observed and checked.	Model behavior was evaluated using available monitoring data to see whether the classification results matched expected anomaly conditions.	Alert triggering, notification delivery, email sending, and map interaction were checked for correct behavior.	Integration testing and pilot testing were carried out for the complete monitoring workflow.
Deploy	A working build containing the bridge and sensor modules was released for review.	The monitoring build was released for internal evaluation.	An AI-enabled build was made available for internal testing and model checking.	A build containing the alert and map features was released for internal testing.	A pilot-test version of the integrated ViBridge system was released.

Table 3.7

AGILE SPRINT PLAN cont'd

Review	The output of the modules was assessed and possible refinements were identified.	Findings from the monitoring output were reviewed to improve the display and overall usability.	Model performance and classification output were studied to identify possible improvements in feature handling and interpretation.	The clarity, responsiveness, and usefulness of the alerts and map outputs were reviewed.	Pilot results were examined to determine the changes needed for the final build.
SDLC Phases	**16th Sprint** Finalize the system for presentation and defense readiness. Time Frame: 2 weeks				
Plan	Final revisions, validation tasks, and defense preparation needs were identified.				
Design	The last refinements for the interface, presentation flow, and system outputs were arranged.				
Develop	Final fixes, refinements, and performance improvements were applied to the finish system				

Table 3.8

AGILE SPRINT PLAN cont'd

Test	Acceptance testing was performed to confirm that the system and outputs were ready.				
Deploy	The final capstone-ready version of ViBridge was released.				
Review	The finished outputs were checked for system readiness and defense preparation.				

Planning/Concept-Initiation Phase

The Planning or Concept-Initiation Phase marks the beginning of the "ViBridge" project. This phase establishes the project's purpose, objectives, and scope, ensuring that the proposed system aligns with bridge monitoring and maintenance need. It identifies challenges faced by DPWH engineers and bridge monitoring personnel in monitoring small bridges, such as limited resources, infrequent inspections, and risks posed by environmental hazards like seismic activity, high salinity, and typhoons.

Business Model Canvas

The Business Model Canvas provides a strategic overview of ViBridge's operations, value proposition, and growth approach. It identifies target users, key activities, resources, and revenue considerations, serving as a framework for project sustainability and scalability.

Table 4

BUSINESS MODEL CANVAS

<table>
<tr>
<td rowspan="2">Key Partners

• DPWH Region VII bridge monitoring and maintenance offices
• Hardware and sensor suppliers</td>
<td>Key Activities

• Sensor deployment and calibration
• AI algorithm development
• Real-time data analysis
• Dashboard and mobile app development
• Field testing</td>
<td rowspan="2">Value Proposition

• Real-time bridge vibration monitoring
• AI-assisted anomaly detection
• Localized alerts
• Enhanced safety for commuters
• Proactive maintenance planning</td>
<td>Customer Relationships

• Technical support for Department of Public Works and Highways (DPWH)
• Periodic training
• Continuous updates on system performance</td>
<td rowspan="2">Customer Segments

• DPWH Planning – Admin
• DPWH Planning-Engineer</td>
</tr>
<tr>
<td>Key Resources

• ESP32 microcontroller
• ADXL355 accelerometer (vibration monitoring)
• Edge computing devices
• AI algorithms (Random Forest)
• Local Bridge Datasets
• Software & Hardware Integration</td>
<td>Channels

• Web-based dashboard
• Android mobile application
• Direct coordination with DPWH bridge monitoring and maintenance offices</td>
</tr>
<tr>
<td colspan="2">Cost Structure

• Hardware procurement
• Software development
• Installation and maintenance
• Training and support
• Pilot testing logistics</td>
<td colspan="3">Revenue Streams

• Government Procurement
• Annual Maintenance Contract
• Installation and Hardware Sales
• Training and Onboarding Services</td>
</tr>
</table>

Table 4 presents the Business Model Canvas of the proposed ViBridge system. The model is centered on the DPWH Planning - Admin and DPWH Planning - Engineer as the primary

customer segments of the system. These users are directly involved in bridge monitoring, planning, maintenance coordination, and decision support. The DPWH Planning - Admin is responsible for managing bridge records, sensor nodes, monitoring configurations, alerts, and system records, while the DPWH Planning - Engineer focuses on monitoring bridge conditions, reviewing vibration data, receiving alerts, and supporting inspection-related activities.

The key partners of ViBridge include the DPWH Region VII bridge monitoring and maintenance offices and hardware and sensor suppliers, which may support the coordination, acquisition, and deployment of the system's monitoring components. The key activities include sensor deployment and calibration, AI algorithm development, real-time data analysis, dashboard and mobile application development, and field testing. These activities support the system's value proposition, which focuses on real-time bridge vibration monitoring, AI-assisted anomaly detection, localized alerts, improved bridge safety awareness, and proactive maintenance planning.

Customer relationships will be maintained through technical support for DPWH, periodic training, and continuous updates on system performance. The system will reach its users through a web-based dashboard, an Android mobile application, and direct coordination with DPWH bridge monitoring and maintenance offices. Its key resources include the ESP32 microcontroller, ADXL355 accelerometer for vibration monitoring, edge computing devices, AI algorithms using Random Forest, local bridge datasets, and software and hardware integration.

The cost structure includes hardware procurement, software development, installation and maintenance, training and support, and pilot testing logistics. Revenue streams may come from government procurement, annual maintenance contracts, installation and hardware sales, and training and onboarding services. Overall, the Business Model Canvas presents ViBridge as a technology-driven bridge vibration monitoring solution that may support DPWH Planning personnel in monitoring small single-span bridges, identifying possible anomalies, and improving maintenance planning through organized vibration data and alerts

Program Workflow

Program workflow shows the presentation of the ViBridge system application in a sequential graphical form which serves as a guide for the system design. The workflow illustrates how the tasks in the application are accomplished from start to end using standardized symbols and processes. It also helps the developers understand how the administrator interacts with the system and how the monitoring process operates within the application.

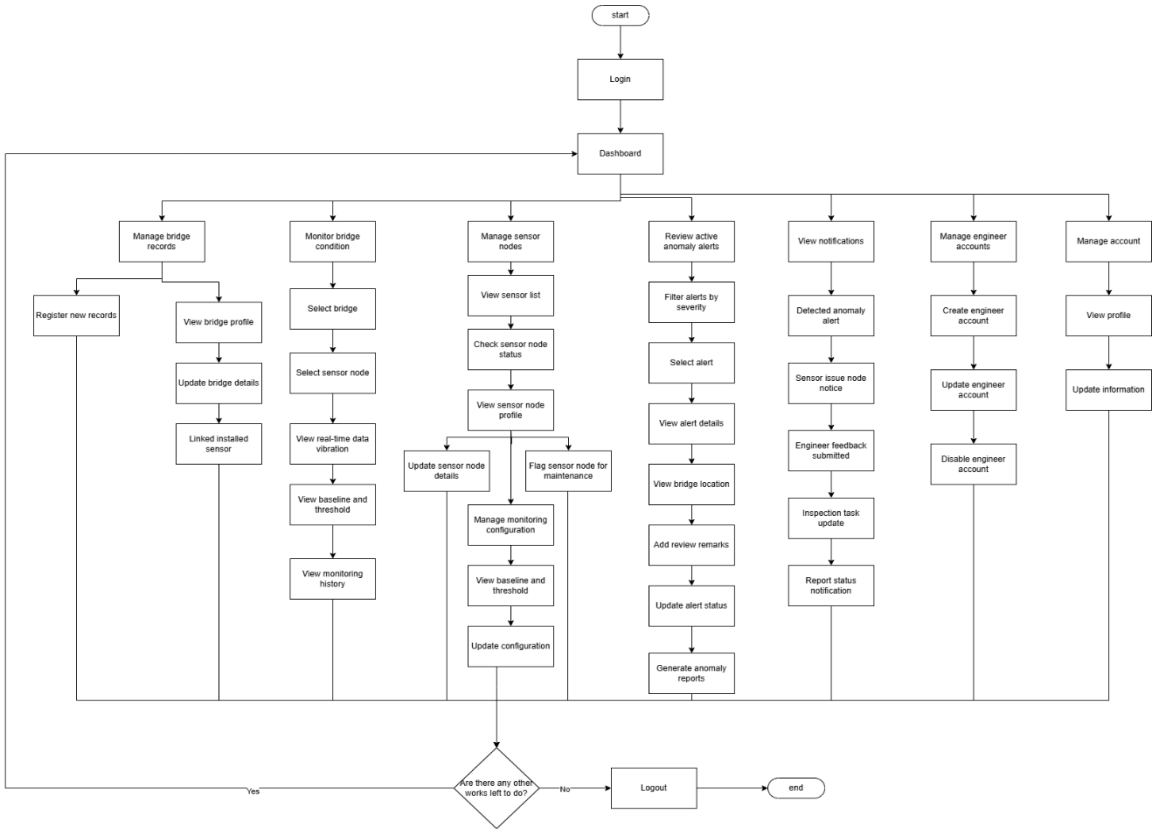


Figure 3: **Program Workflow (DPWH Administrator)**

Figure 3 The Admin workflow starts when the administrator logs into the ViBridge web system and accesses the dashboard. From the dashboard, the administrator can manage bridge records, monitor bridge condition, manage sensor nodes, review active anomaly alerts, view notifications, manage engineer accounts, and manage the admin account. After completing a task, the administrator decides whether to perform another task or log out of the system.

In Manage Bridge Records, the administrator can register new records, view bridge profiles, update bridge details, and link installed sensors. In Monitor Bridge Condition, the administrator selects a bridge and sensor node, then views real-time vibration data, baseline and threshold values, and monitoring history. In Manage Sensor Nodes, the administrator can view the sensor list, check node status, open the sensor node profile, update node details, flag a node for maintenance, and manage monitoring configuration.

In Review Active Anomaly Alerts, the administrator filters alerts by severity, selects an alert, views alert details and bridge location, adds review remarks, updates alert status, and generates anomaly reports. In View Notifications, the administrator receives notices such as detected anomaly alerts, sensor issue notices, engineer feedback submissions, inspection task

updates, and report status notifications. In Manage Engineer Accounts and Manage Account, the administrator can create, update, or disable engineer accounts, as well as view and update the admin profile.

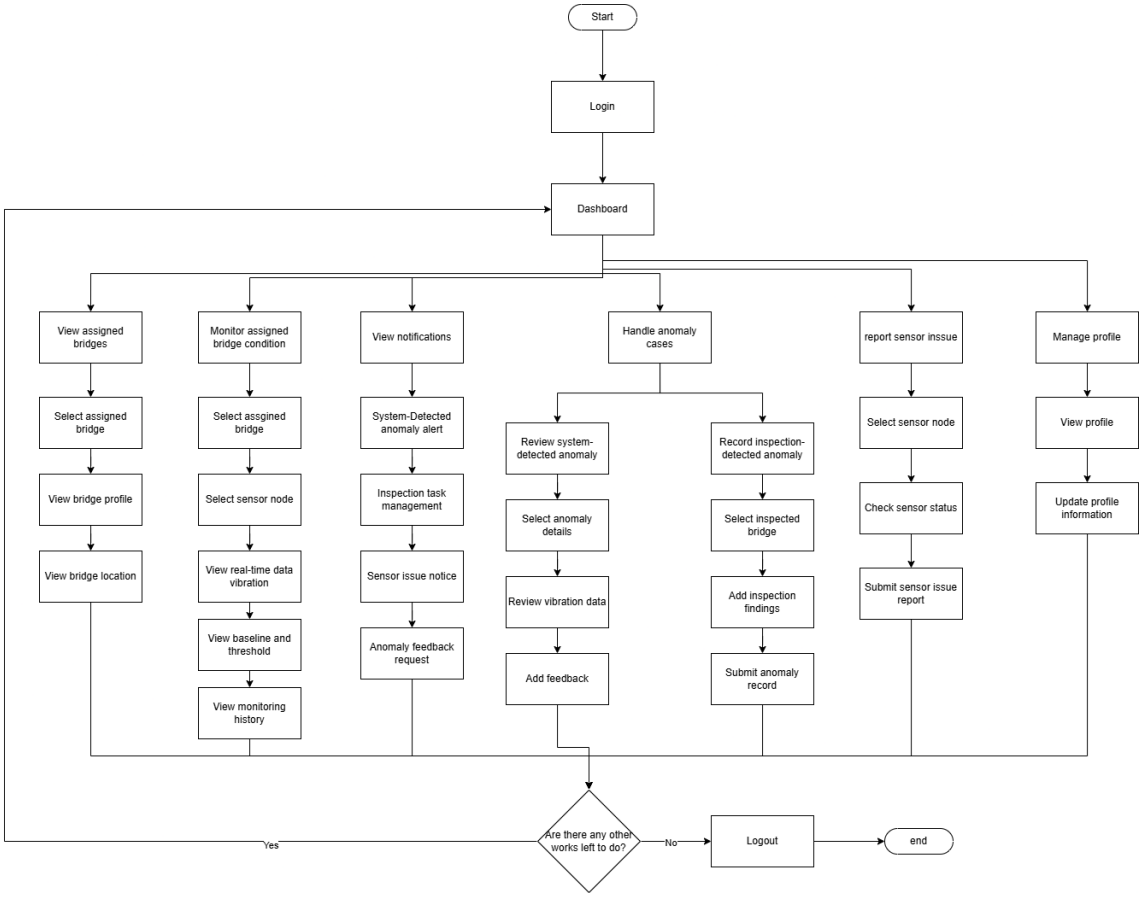


Figure 4: **Program Workflow (DPWH Engineer)**

Figure 4 illustrates the program workflow of the engineer in the ViBridge system. The workflow shows the sequence of operations performed by the engineer within the mobile monitoring application.

The process begins when the engineer logs in to the system using authorized credentials. After successful authentication, the engineer is redirected to the dashboard, which provides access to assigned bridges, monitoring functions, notifications, anomaly handling, sensor issue reporting, and profile management.

From the dashboard, the engineer can view assigned bridges by selecting a bridge, viewing its profile, and checking its location. The engineer can also monitor the assigned bridge condition by selecting a bridge, viewing real-time vibration data, and reviewing the monitoring history. The

engineer can view notifications such as system-detected anomaly alerts, inspection task updates, sensor issue notices, and anomaly feedback requests.

The engineer can also handle anomaly cases by reviewing system-detected anomalies, selecting anomaly details, reviewing vibration data, and adding feedback. For inspection-detected anomalies, the engineer can select the inspected bridge, add inspection findings, and submit an anomaly record. The engineer may also report sensor issues by selecting a sensor node, checking its status, and submitting a sensor issue report. Lastly, the engineer can manage the profile by viewing and updating profile information. After completing the required tasks, the engineer may choose to perform another task or log out of the system. The workflow ends once the engineer logs out.

Validation Board (Stages 1 and 2)

Table 5.1

VALIDATION BOARD

Experiments	1	2
User	DPWH Administrator	DPWH Engineer
Problem	Difficulty overseeing multiple bridges and making fast decisions due to lack of centralized monitoring.	Difficulty in monitoring bridge conditions due to reliance on manual inspections and limited tools.
Solution	Web-based dashboard with real-time monitoring, anomaly alerts, and bridge overview.	IoT-based vibration monitoring system with mobile access for real-time data.

Table 5.2

VALIDATION BOARD cont'd

Riskiest Assumptions	Administrators will use a centralized system for monitoring and decision-making.	Engineers will adopt sensor-based monitoring instead of relying only on manual inspection.
Method & Success Criteria	Survey at least 20+ respondents; majority must see value in monitoring systems and alerts.	Survey at least 20+ engineers; at least 70% must show interest in automated monitoring.
Result & Destination	Majority of respondents prefer real-time monitoring, alerts, and remote access → Proceed.	25 out of 26 respondents are interested in automated monitoring → Strong validation, proceed.
Learnings	Users prioritize real-time monitoring, alerts, and remote access, not complex manual processes.	Many engineers are only somewhat or not very confident in identifying structural issues → need system support.

The table presents the validation overview of the ViBridge system. The identified problem highlights the challenges faced by bridge management authorities in continuously monitoring structural conditions and detecting early signs of potential damage, as traditional inspection methods are often manual, time-consuming, and infrequent. The proposed solution is the implementation of an IoT-based bridge monitoring system that provides real-time vibration data, anomaly detection, and automated alert notifications to support timely decision-making. The riskiest assumptions include the reliability and accuracy of sensor data, as well as the willingness of administrators and engineers to trust and adopt automated monitoring systems. The validation method and success criteria were based on user feedback, including interviews and evaluation of the system's proposed features. The results indicate a strong acceptance of the system's

functionality and its potential to improve monitoring efficiency. The findings suggest that the ViBridge system can enhance infrastructure monitoring by enabling early detection of structural anomalies, reducing manual inspection efforts, and supporting more proactive maintenance strategies.

Gantt Chart

This section presents the researchers' Gantt chart for the ViBridge project, illustrating the systematic timeline for integrating IoT hardware, cloud infrastructure, and mobile/web applications. The chart visually maps the project's lifecycle, from the initial gathering of structural health data in Cebu to the final implementation of the anomaly detection algorithm and notification system.

Table 6.1

GANTT CHART

Task ID	Task Name	Task Lead	Start Date	End Date	JAN 2026	FEB 2026				MAR 2026				APR 2026				MAY 2026				AUG 2026				SEPT 2026				OCT 2026				NOV 2026			
					4	1	2	3	4	1	2	3	4	1	2	3	4	1	2	3	4	1	2	3	4	1	2	3	4	1	2	3	4	1	2	3	4
1	Title Formulation	All	February 16	March 1																																	
2	Problem Identification	All	February 16	March 1																																	
3	Background of the Study Development	Lepatan	February 16	March 1																																	
4	Objectives Formulation	All	February 16	March 1																																	
5	Scope and Limitation Development	All	February 16	March 1																																	
6	Review of Related Literature and Studies	Lepatan	February 16	March 1																																	
7	Related Studies	Lepatan	February 16	March 1																																	
8	Comparative Analysis	Rosales	February 23	March 1																																	
9	Conceptual Framework	Lepatan	February 23																																		
10	Methodology	Beltran	February 23																																		

Table 6.2

GANTT CHART cont'd

Task ID	Task Name	Task Lead	Start Date	End Date	JAN 2026	FEB 2026				MAR 2026				APR 2026				MAY 2026				AUG 2026				SEPT 2026				OCT 2026				NOV 2026			
					4	1	2	3	4	1	2	3	4	1	2	3	4	1	2	3	4	1	2	3	4	1	2	3	4	1	2	3	4	1	2	3	4
11	System Planning	Rosales	February 23	March 8																																	
12	Business Model Canvas	Carombana	February 23	March 8																																	
13	User Workflow	Carombana	March 13	March 16																																	
14	Use Case Diagram	Beltran	March 13	March 16																																	
15	Functional Decomposition Diagram	Lepatan	March 13	March 16																																	
16	Database Design	Carombana	March 19	March 22																																	
17	System Architecture	Carombana	March 21	March 24																																	
18	Network and IoT Design	All	March 25	March 29																																	
19	Software Specification	All	April 9	April 13																																	
20	Hardware Specification	All	April 9	April 13																																	
21	Program Specification	All	April 9	April 13																																	
22	Interface Design	Beltran, Rosales	Aug 10	Aug 16																																	
23	Development Setup	All	Aug 19	Aug 23																																	
24	User Authentication	Lepatan	Aug 19	Aug 23																																	
25	Bridge Management	All	Aug 21	Aug 26																																	
26	Sensor Management	All	Aug 26	Aug 26																																	

Table 6.3

GANTT CHART cont'd

Task ID	Task Name	Task Lead	Start Date	End Date	JAN 2026	FEB 2026				MAR 2026				APR 2026				MAY 2026				AUG 2026				SEPT 2026				OCT 2026				NOV 2026			
					4	1	2	3	4	1	2	3	4	1	2	3	4	1	2	3	4	1	2	3	4	1	2	3	4	1	2	3	4	1	2	3	4
27	Real-Time Monitoring	Carombana	Sept 2	Sept 12																																	
28	Anomaly Detection	Carombana	Sept 2	Sept 12																																	
29	Alerts and Notifications	Beltran, Rosales	Sept 2	Sept 12																																	
30	Map Integration	Carombana	Sept 14	Sept 21																																	
31	System Integration	All	Sept 14	Sept 21																																	
32	System Testing	All	Sept 30	Oct 7																																	
33	Pilot Testing	All	Sept 30	Oct 7																																	
34	System Evaluation	Lepatan	Nov 9	Nov 17																																	
35	Final Revision	All	Nov 23	Nov 30																																	
36	Defense Preparation	All	Nov 28	Nov 30																																	

LEGEND	COMPLETE
	ONGOING
	INCOMPLETE

This section displays tasks and their associated timeframes in a horizontal bar format, enabling researchers to organize, track, and coordinate activities efficiently. It provides a comprehensive overview of the project's progress, improves communication, and ensures that project requirements are completed on schedule. Using the Agile Methodology, the Gantt chart is helpful in presenting the planned activities, sprint schedules, and iterative development tasks of the project. It allows the researchers to organize, monitor, and track the progress of each sprint while providing flexibility for revisions, testing, and improvements throughout the development process.

Functional Decomposition Diagram

The Functional Decomposition Diagram (FDD) illustrates the breakdown of the proposed ViBridge system into smaller functional components. It provides a hierarchical representation of

the major processes involved in the system, starting from the main function down to its sub-functions. The purpose of the FDD is to clearly define how the system operates by dividing the overall process into manageable and understandable modules.

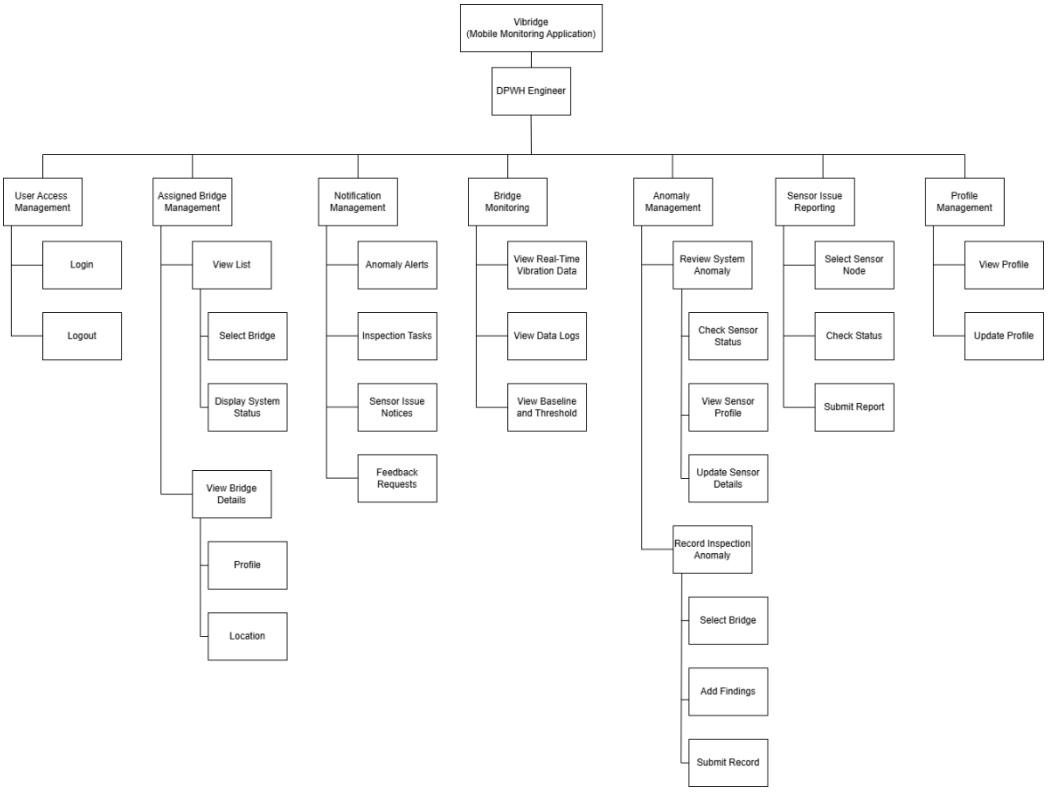


Figure 5: **Functional Decomposition Diagram (ViBridge Mobile Application)**

Figure 5 shows the Functional Decomposition Diagram of the ViBridge Mobile Monitoring Application. The mobile application is designed for the DPWH Engineer, who is responsible for monitoring assigned bridges, reviewing notifications, handling anomaly cases, reporting sensor issues, and managing profile information. In this diagram, the system is divided into seven (7) major functional components, namely User Access Management, Assigned Bridge Management, Notification Management, Bridge Monitoring, Anomaly Management, Sensor Issue Reporting, and Profile Management. Each component is further subdivided into specific sub-functions that represent the actions available to the engineer within the mobile application.

The first component, User Access Management, consists of the Login and Logout sub-functions. The Login sub-function allows engineers to securely access the system using authorized credentials, while the Logout sub-function ensures proper termination of the session. The second component, Assigned Bridge Management, includes View List, Select Bridge, Display System

Status, View Bridge Details, Profile, and Location. These sub-functions allow engineers to view assigned bridges, select a specific bridge, and access its basic information and location.

The third component, Notification Management, includes Anomaly Alerts, Inspection Tasks, Sensor Issue Notices, and Feedback Requests. These sub-functions allow engineers to receive and review important system updates related to bridge monitoring and inspection activities. The fourth component, Bridge Monitoring, includes View Real-Time Vibration Data, View Data Logs, and View Baseline and Threshold. These functions allow engineers to observe live vibration readings, review historical monitoring records, and compare bridge readings with established safety references.

The fifth component, Anomaly Management, includes Review System Anomaly, Check Sensor Status, View Sensor Profile, Update Sensor Details, Record Inspection Anomaly, Select Bridge, Add Findings, and Submit Record. These sub-functions support the engineer in reviewing system-detected anomalies and recording inspection-based findings. The sixth component, Sensor Issue Reporting, includes Select Sensor Node, Check Status, and Submit Report. These functions allow engineers to report problems related to sensor nodes when irregularities or device issues are observed. The seventh component, Profile Management, includes View Profile and Update Profile. These sub-functions allow engineers to access and update their personal profile information within the mobile application.

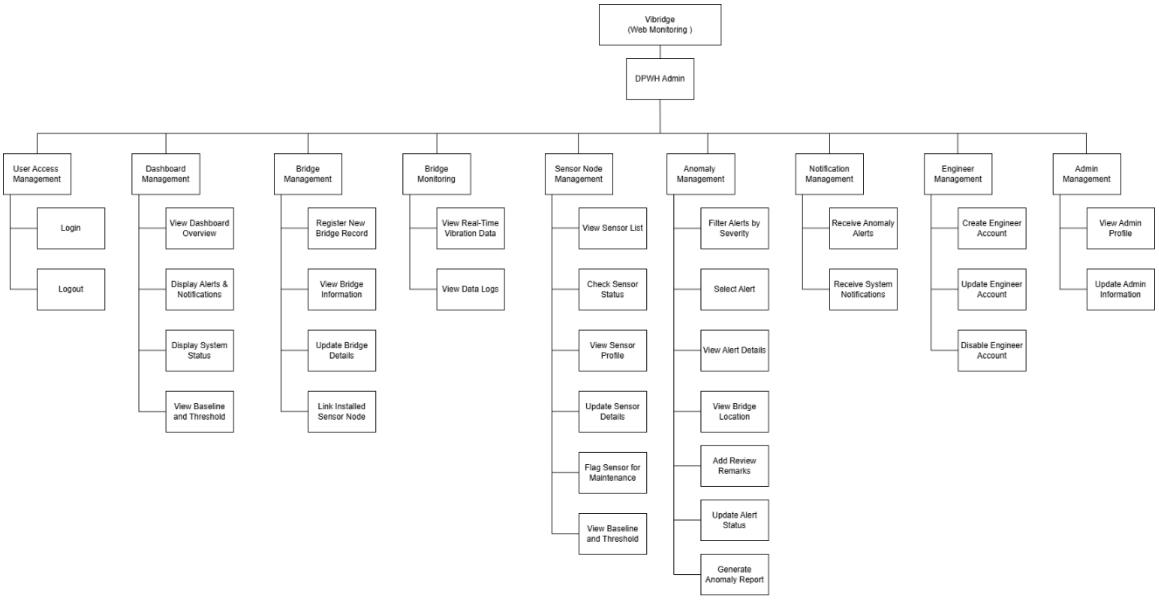


Figure 6: **Functional Decomposition Diagram (ViBridge Web Application)**

Figure 6 shows the Functional Decomposition Diagram of the ViBridge Web Monitoring Application. The web application is designed for the DPWH Administrator, who is responsible for managing the monitoring system, overseeing bridge condition data, and handling system-wide operations. Unlike the mobile application which focuses on field monitoring, the web application provides comprehensive administrative functions such as managing bridges, sensor nodes, users, alerts, and notifications. In this diagram, the system is divided into nine (9) major functional components, namely User Access Management, Dashboard Management, Bridge Management, Bridge Monitoring, Sensor Node Management, Anomaly Management, Notification Management, Engineer Management, and Admin Management. Each component is further subdivided into specific sub-functions that represent the actions available to the administrator within the system.

The first component, User Access Management, consists of the Login and Logout sub-functions. The Login sub-function allows administrators to securely access the system using authorized credentials, while the Logout sub-function ensures proper termination of the session.

The second component, Dashboard Management, includes View Dashboard Overview, Display Alerts and Notifications, Display System Status, and View Baseline and Threshold. These sub-functions provide administrators with a centralized interface to monitor overall system activity, including current alerts, system health, and configured monitoring thresholds.

The third component, Bridge Management, includes Register New Bridge Record, View Bridge Information, Update Bridge Details, and Link Installed Sensor Node. These sub-functions allow administrators to maintain accurate records of bridges and ensure proper association with installed sensor devices. The fourth component, Bridge Monitoring, includes View Real-Time Vibration Data and View Data Logs. These enable administrators to observe live sensor readings and review historical monitoring data for analysis and decision-making.

The fifth component, Sensor Node Management, consists of View Sensor List, Check Sensor Status, View Sensor Profile, Update Sensor Details, Flag Sensor for Maintenance, and View Baseline and Threshold. These functions allow administrators to manage sensor devices, monitor their operational status, and configure their monitoring parameters. The sixth component, Anomaly Management, includes Filter Alerts by Severity, Select Alert, View Alert Details, View Bridge Location, Add Review Remarks, Update Alert Status, and Generate Anomaly Report. These sub-functions support the analysis and handling of detected anomalies, enabling administrators to assess issues and document findings.

The seventh component, Notification Management, includes Receive Anomaly Alerts and Receive System Notifications. These ensure that administrators are promptly informed of important system events and abnormal conditions. The eighth component, Engineer Management, includes

Create Engineer Account, Update Engineer Account, and Disable Engineer Account. These functions allow administrators to manage engineer access and responsibilities within the system. The ninth component, Admin Management, includes View Admin Profile and Update Admin Information. These sub-functions allow administrators to manage their own account details within the system.

Architectural Layout

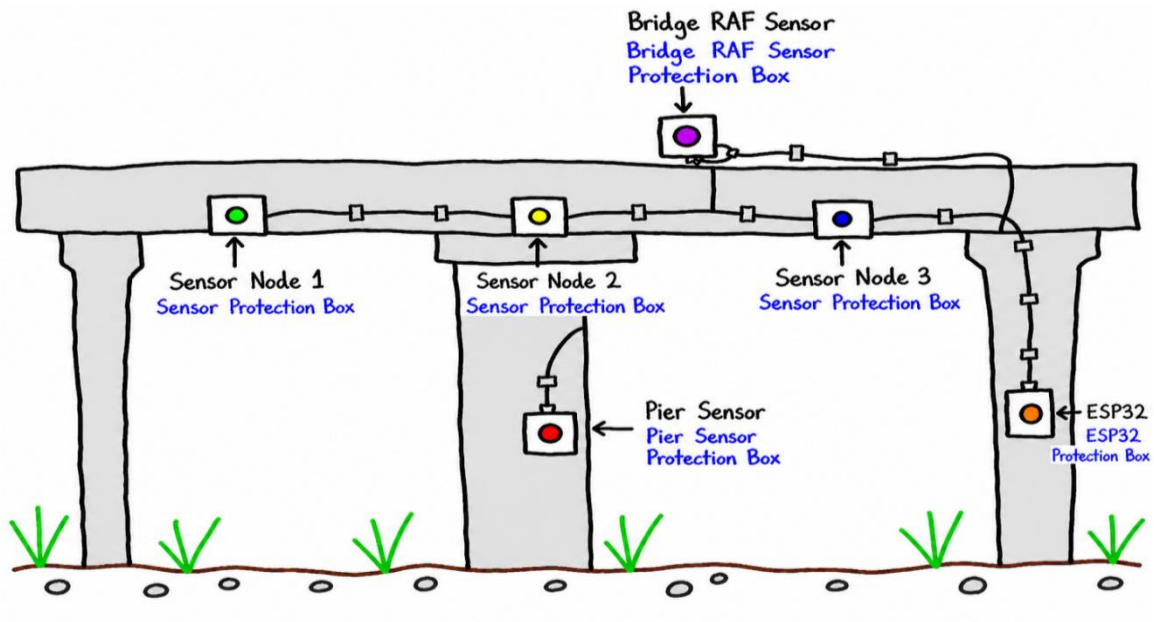


Figure 7: **Side View**

Figure 7 shows the side view of the proposed ViBridge architectural layout. This perspective presents how the monitoring components are arranged along the side portion of the bridge structure, making the placement of the sensor nodes, bridge RAF sensor, pier sensor, and ESP32 unit easier to understand. The figure also highlights that each major component is installed with its corresponding protection box, which helps protect the sensors and controller from rain, dust, debris, and other possible environmental exposure. Through this view, the arrangement of the monitoring devices and their connection to the bridge structure are presented more clearly for the proposed system setup.

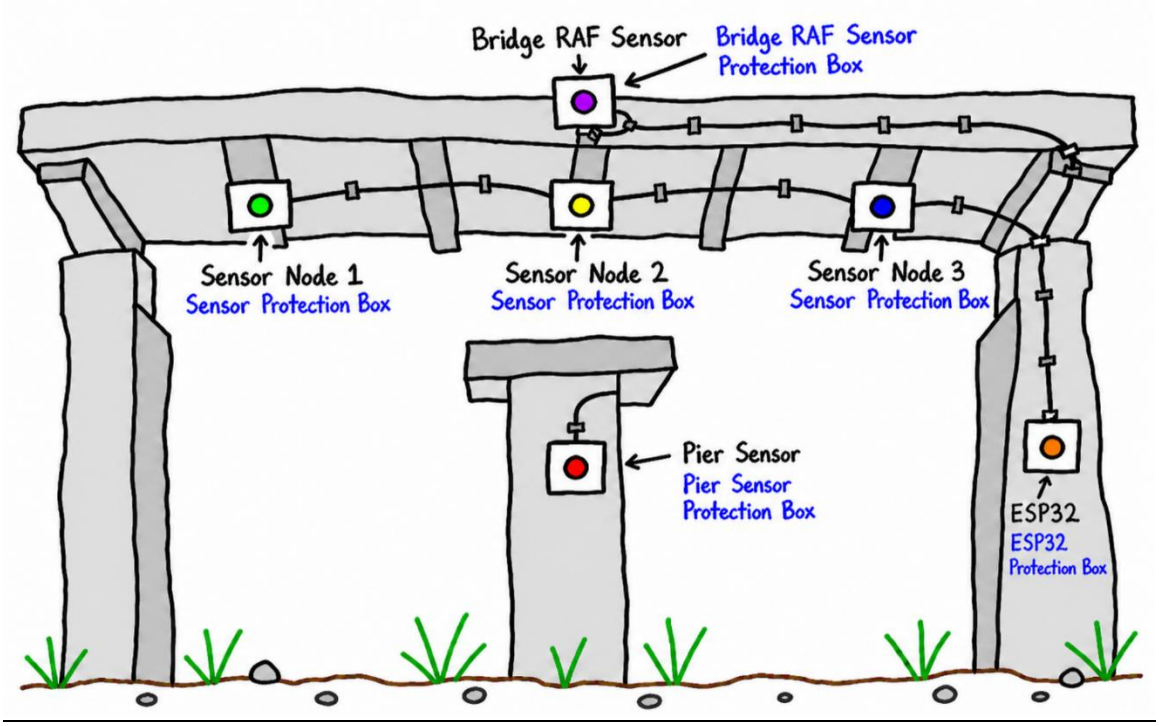


Figure 8: **UnderSide/Bottom View**

Figure 8 shows the underside or bottom view of the proposed ViBridge architectural layout. This perspective presents the arrangement of the monitoring components installed beneath the bridge, including the sensor nodes placed along the bridge span, the bridge RAF sensor, the pier sensor, and the ESP32 unit. The figure makes it easier to see how the devices are positioned from below and how they are connected within the monitoring setup. It also shows that the major components are enclosed in their respective protection boxes to help protect them from environmental exposure such as rain, dust, debris, and possible flooding. Through this view, the underside placement of the monitoring devices is shown more clearly, helping explain how the proposed system will be installed on the bridge structure.

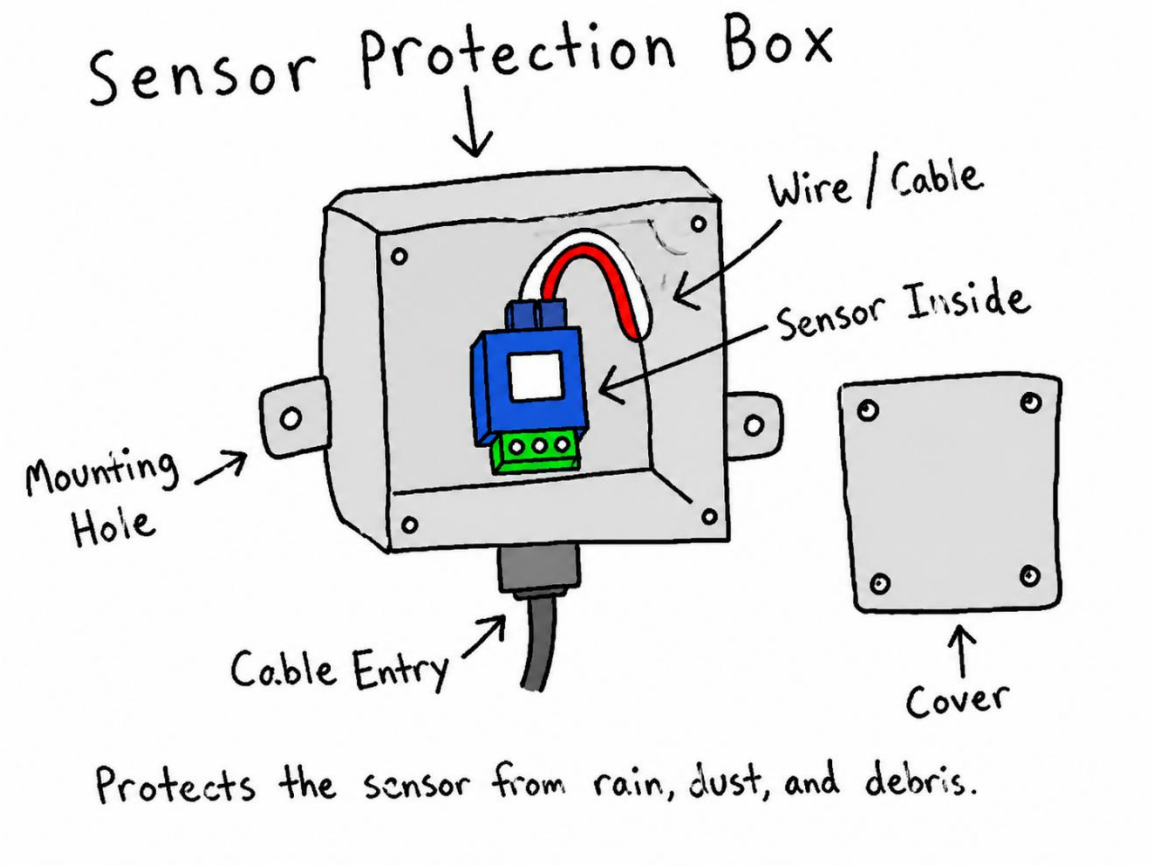


Figure 9: **Protection Box**

Figure 9 presents the box that houses and protects the installed monitoring components, helping shield them from external conditions such as rain, dust, debris, and possible flood exposure. This protective casing is important because it helps maintain the safety and durability of the sensors and other electronic parts while they are deployed on the bridge. The figure also helps the viewer understand that the protection box is part of the physical installation setup, serving as a simple enclosure to secure the monitoring device and support reliable data collection in the actual bridge environment.

Block Diagram

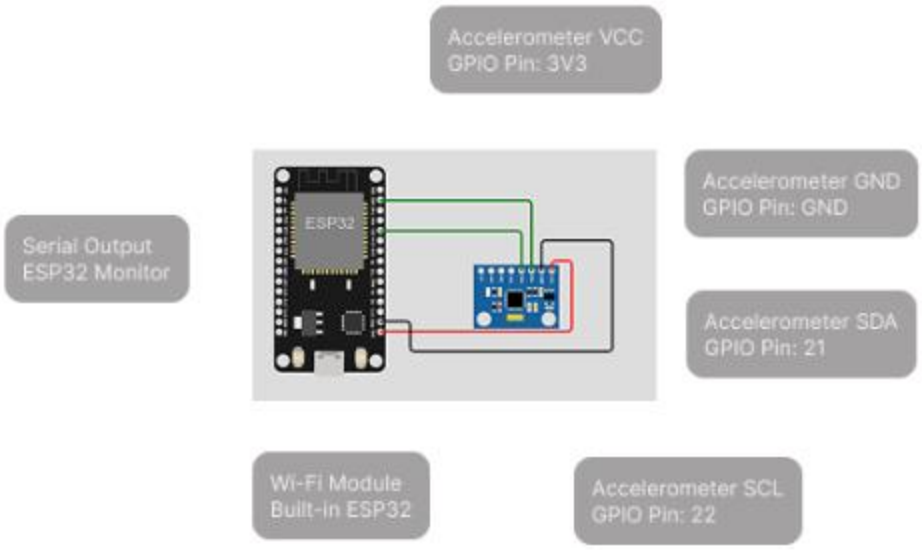


Figure 10: **Block Diagram**

Figure 10 illustrates the block diagram of the "ViBridge" system. The system uses an ADXL355 triaxial MEMS accelerometer connected to the ESP32 sensor node to collect bridge vibration data. The ADXL355 is powered through the 3.3V and GND pins of the ESP32. For communication, the sensor uses the SPI connection, where SCLK is connected to GPIO Pin 18, MISO is connected to GPIO Pin 19, MOSI is connected to GPIO Pin 23, and CS (Chip Select) is connected to GPIO Pin 5. These connections allow the ESP32 to read vibration data from the accelerometer in real time. The ESP32 then uses its built-in Wi-Fi module to transmit the collected data to the cloud database or server for storage and processing. From there, the vibration data can be accessed through the web monitoring platform and Android mobile application, allowing authorized users to view bridge condition information and receive alerts or notifications when unusual vibration behavior is detected.

Analysis-Design Phase

This phase includes developing the use case diagram, user interface diagram, storyboard, database design, entity-relationship diagram, and the system's data dictionary.

Use Case Diagram

This section presents the use case diagram, which offers a visual representation of the system’s functional requirements and user interactions. For "ViBridge", the diagram outlines

various scenarios where users engage with the platform to monitor real-time vibration data, manage structural health assessments, and execute localized emergency alerts for bridge safety.

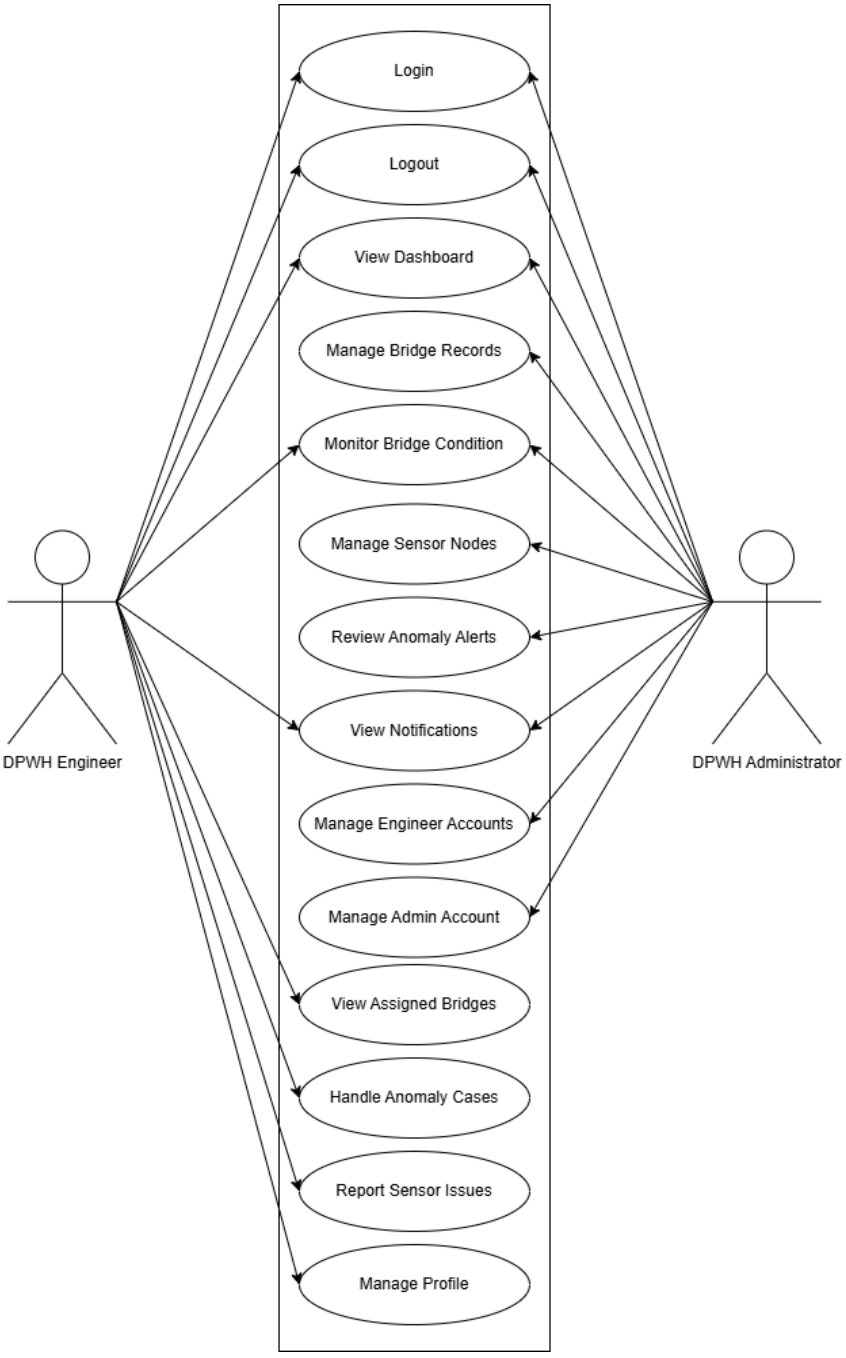


Figure 11: **Use Case Diagram of ViBridge**

Figure 11 presents the use case diagram of the ViBridge system, which includes two primary actors: the DPWH Administrator and the DPWH Engineer. Both actors share common

functionalities such as logging in, logging out, viewing the dashboard, monitoring bridge conditions, reviewing anomaly alerts, and viewing notifications. The DPWH Administrator has broader system control, including managing bridge records, managing sensor nodes, managing engineer accounts, and managing the administrator account. In contrast, the DPWH Engineer focuses on operational and field-related tasks, such as viewing assigned bridges, handling anomaly cases, reporting sensor issues, and managing their profile. This division of responsibilities ensures that administrative control and technical monitoring are properly distributed within the system.

User Interface Design

This section presents a visual overview of the proposed web and mobile application, demonstrating how users interact with the system and outlining its key features and functions.

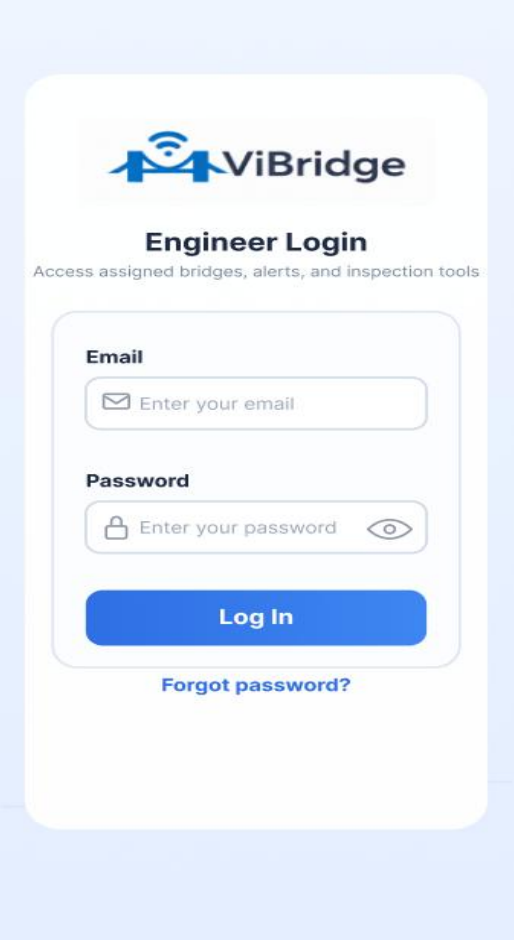


Figure 12: **Engineer Login Page**

Figure 12 presents the login screen, which follows the splash screen. It prompts users to enter their email and password to proceed. Once the correct credentials are submitted and the “Log

In" button is clicked, the system verifies the account. If valid, the user is directed to the dashboard. Otherwise, the user remains on the login page and may recover their account through the "Forgot password?" option.

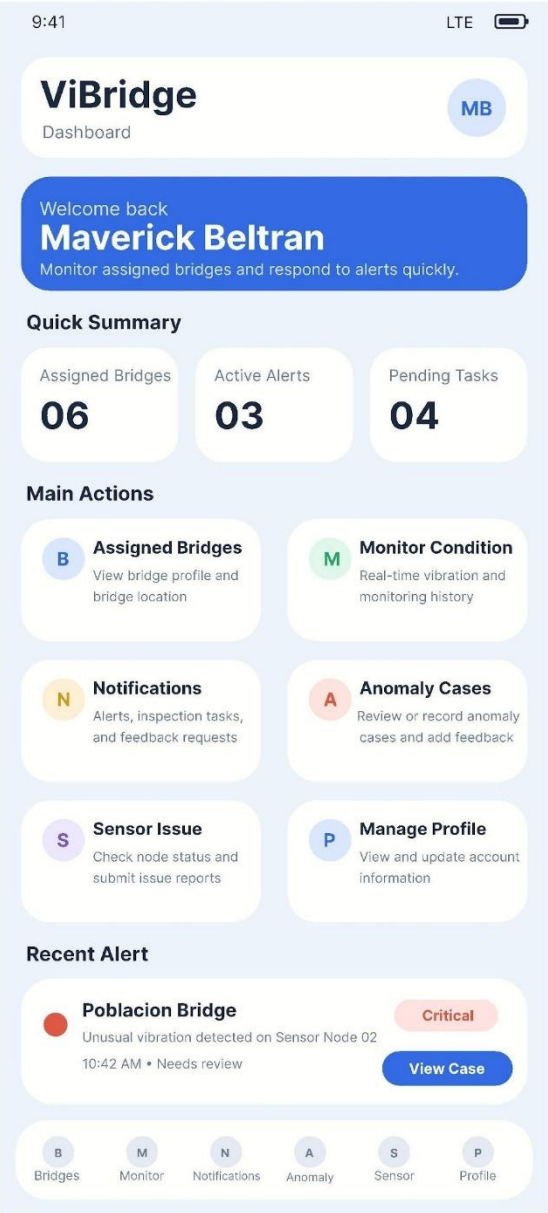


Figure 13: **Dashboard Screen Page**

Figure 13 presents the Engineer Dashboard, which serves as the main landing screen after a successful login. It provides the engineer with an overview of assigned bridges, active alerts, and pending tasks for quick monitoring and decision-making. The screen also contains the main action cards, which allow the engineer to directly access key functions such as viewing assigned bridges,

monitoring bridge condition, checking notifications, handling anomaly cases, reporting sensor issues, and managing profile information. In addition, the recent alert section highlights the latest critical bridge-related concern that may require immediate attention. Through the bottom navigation bar, the engineer can move easily between major application sections. This screen helps the engineer quickly identify priorities and access important monitoring activities within the system.

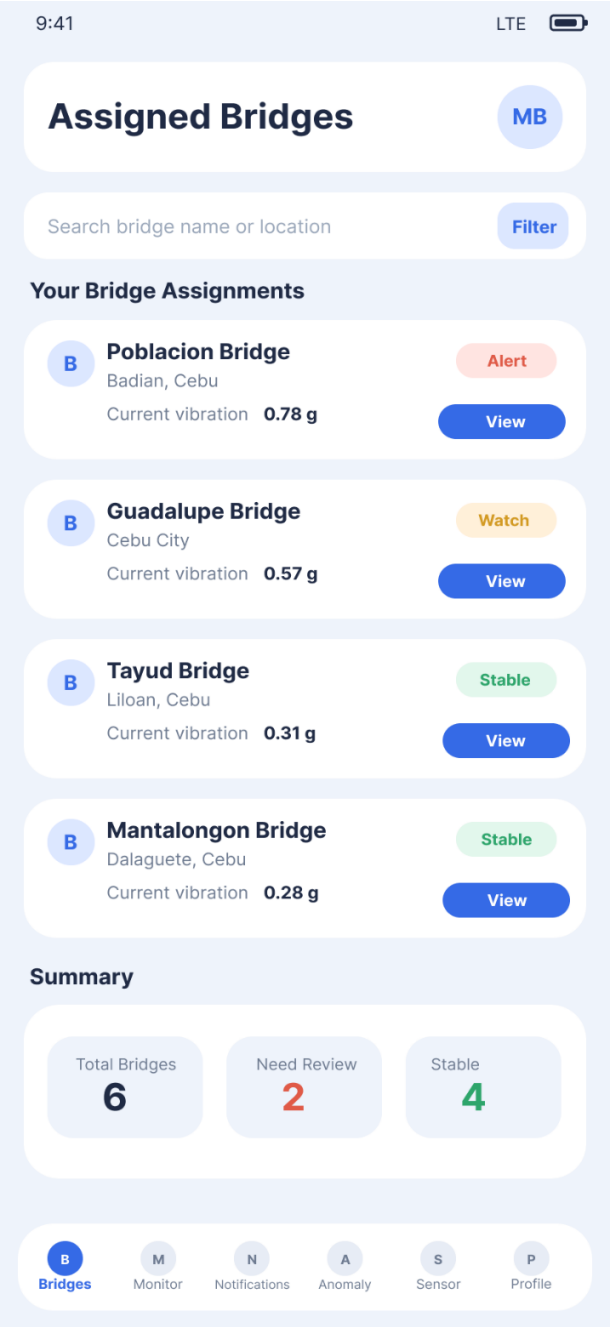


Figure 14: **Assigned Bridges Screen Page**

Figure 14 presents the Assigned Bridges screen, where the engineer can view all bridges assigned to them for monitoring. It displays each bridge name, location, current vibration reading, and status so the engineer can quickly identify which bridge needs attention. The search bar and filter option help narrow the list, while the “View” button opens the selected bridge for further monitoring actions. The summary section at the bottom gives a quick count of total bridges, bridges needing review, and stable bridges.

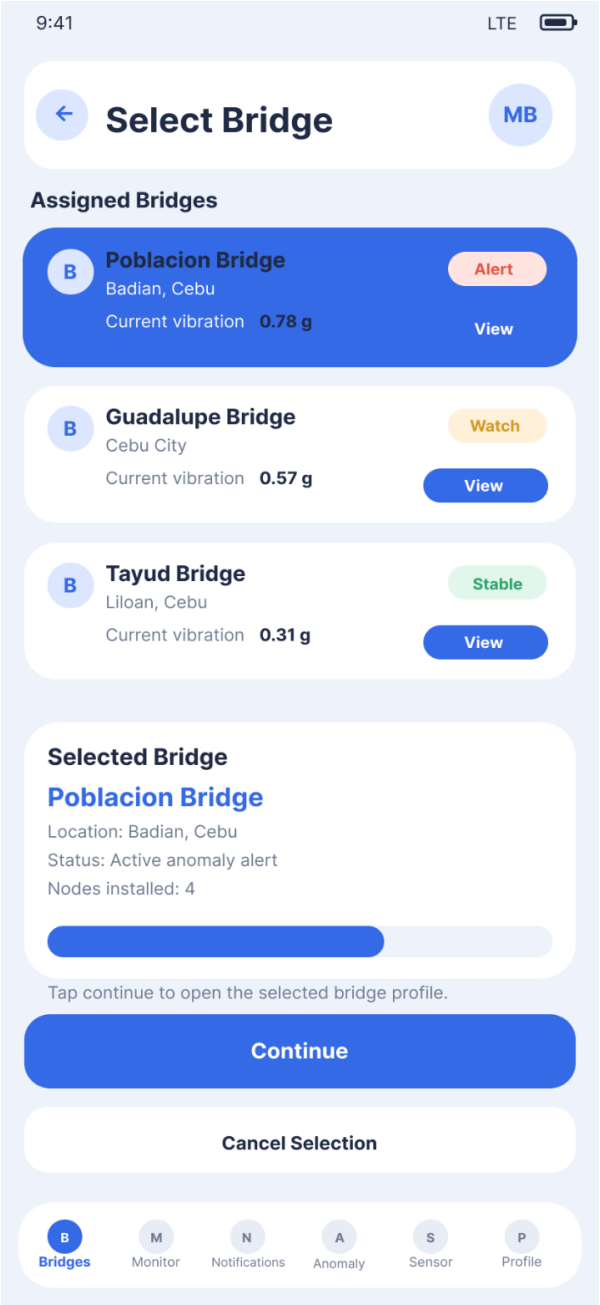


Figure 15: **Select Bridges Page**

Figure 15 presents the Select Bridge screen, where the engineer chooses one assigned bridge for monitoring. It highlights the selected bridge and shows key details such as location, status, and number of installed nodes. The “Continue” button opens the chosen bridge for the next monitoring step, while “Cancel Selection” clears the current choice.

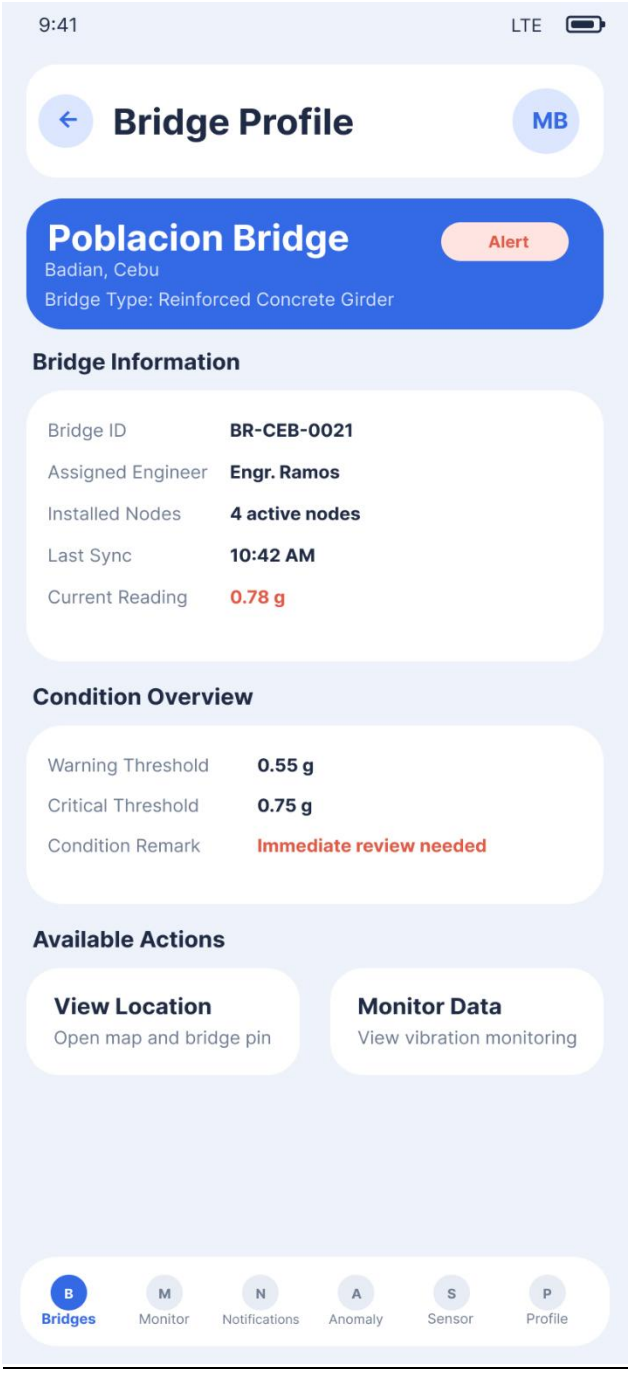


Figure 16: **Bridge Profile Page**

Figure 16 presents the Bridge Profile screen, where the engineer can view the complete details of the selected bridge. It shows the bridge name, location, type, assigned engineer, installed sensor nodes, last sync, and current reading. It also displays the warning threshold, critical threshold, and condition remark to help the engineer assess the bridge status. The available actions at the bottom allow the engineer to open the bridge location or proceed to vibration monitoring.

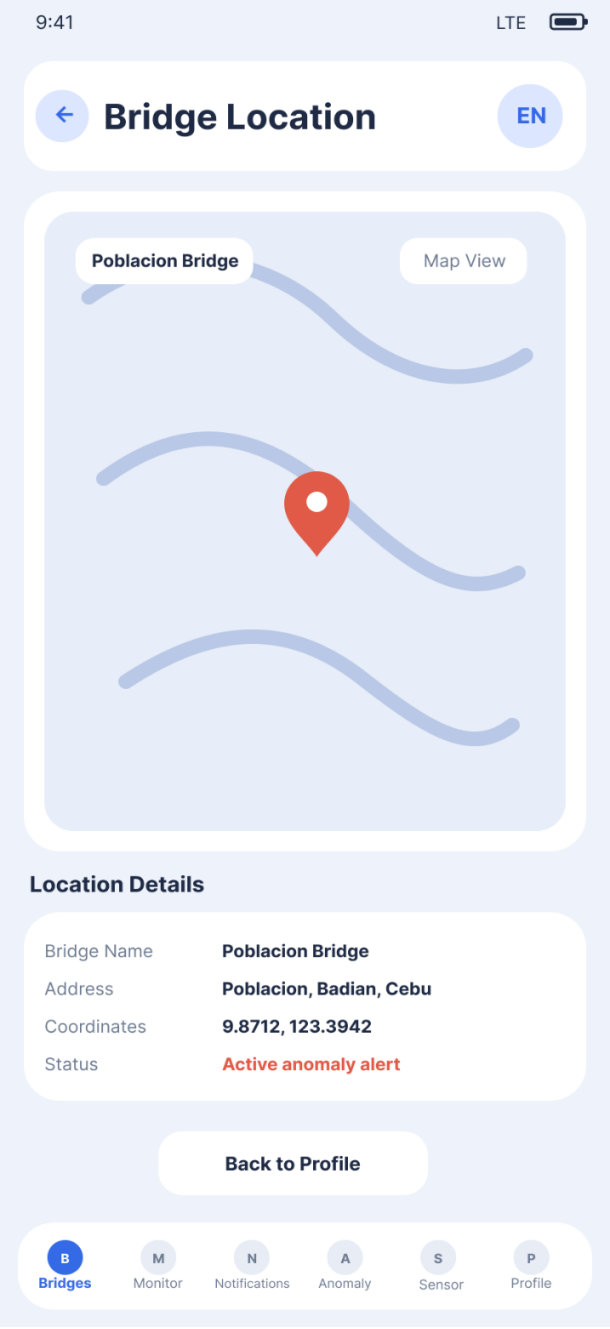


Figure 17: **View Bridge Location Page**

Figure 17 presents the Bridge Location screen, where the engineer can view the mapped location of the selected bridge. It shows the bridge pin on the map and displays the bridge name, address, coordinates, and current status. The Back to Profile button allows the engineer to return to the Bridge Profile screen.

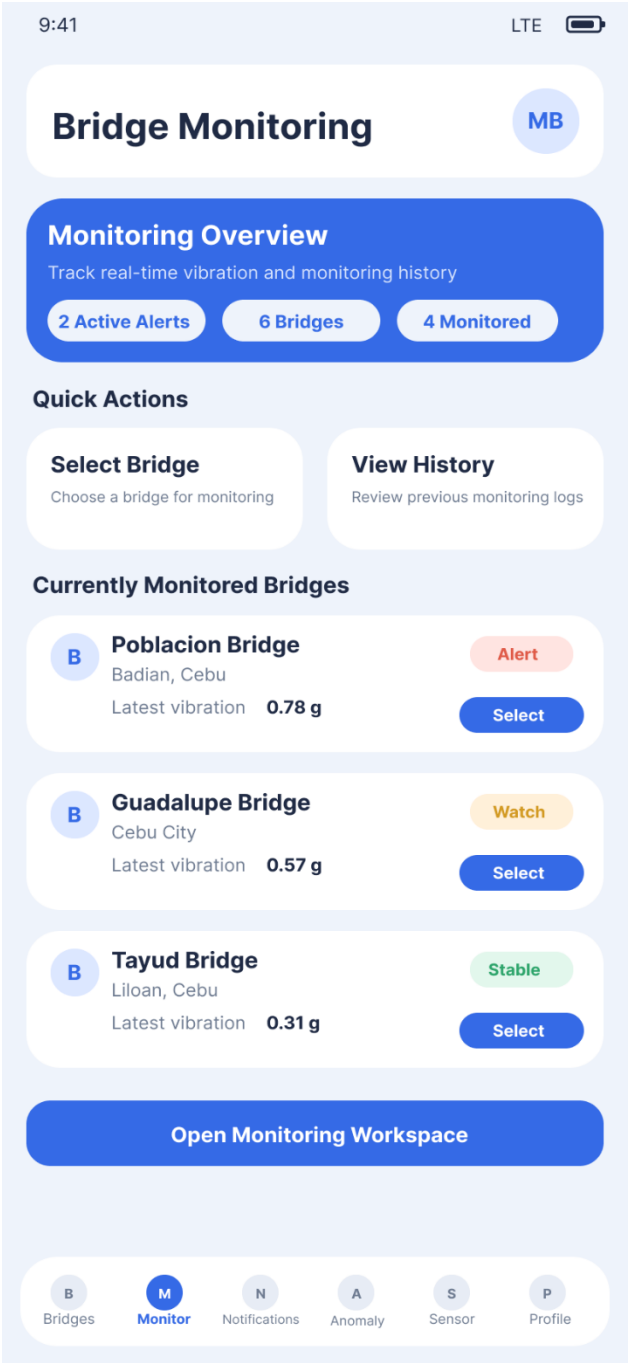


Figure 18: **Bridge Monitoring Page**

Figure 18 presents the Monitor Assigned Bridge Condition screen, where the engineer can access the monitoring workflow for assigned bridges. It shows a quick overview of active alerts, total bridges, and currently monitored bridges. The screen also provides quick actions for selecting a bridge and viewing monitoring history. Below, the list of currently monitored bridges displays the latest vibration reading and status of each bridge. The Select button allows the engineer to choose a bridge for monitoring, while Open Monitoring Workspace proceeds to the next monitoring steps.

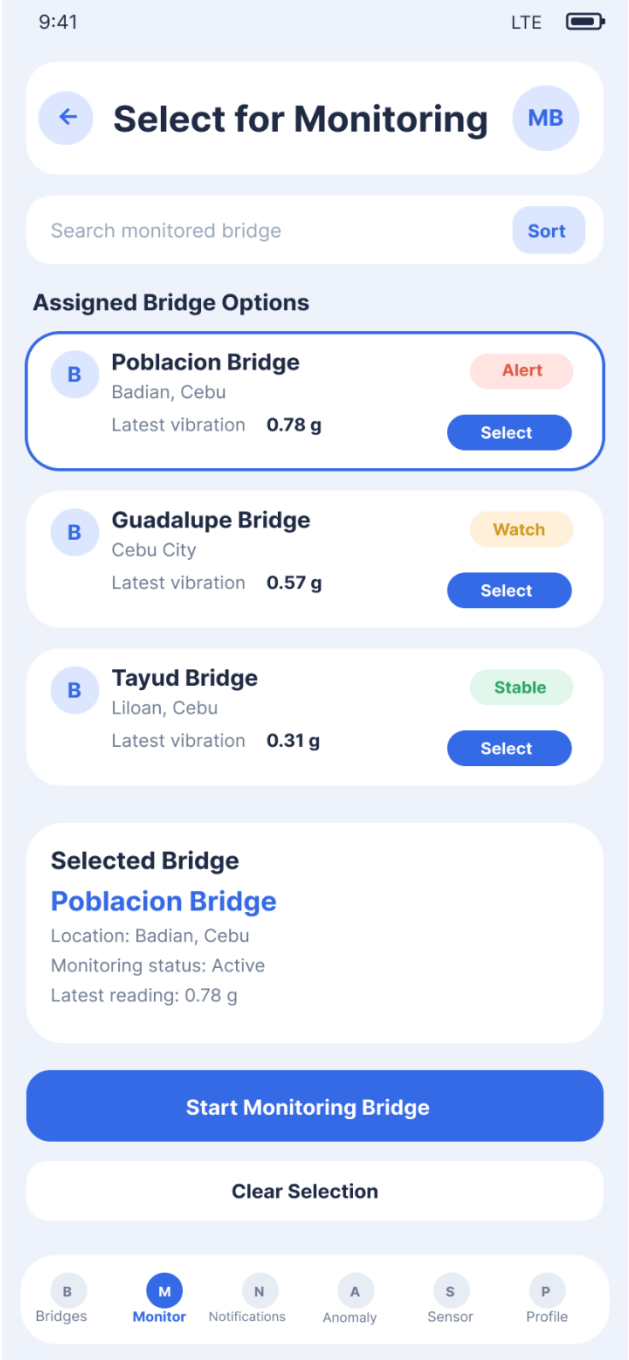


Figure 19: **Select Monitoring Page**

Figure 19 presents the Select for Monitoring screen, where the engineer chooses one assigned bridge to monitor. It displays the available bridge options with their latest vibration reading and current status. The selected bridge is shown in a separate summary section for

confirmation. The Start Monitoring Bridge button proceeds to the next monitoring step, while Clear Selection removes the current choice.

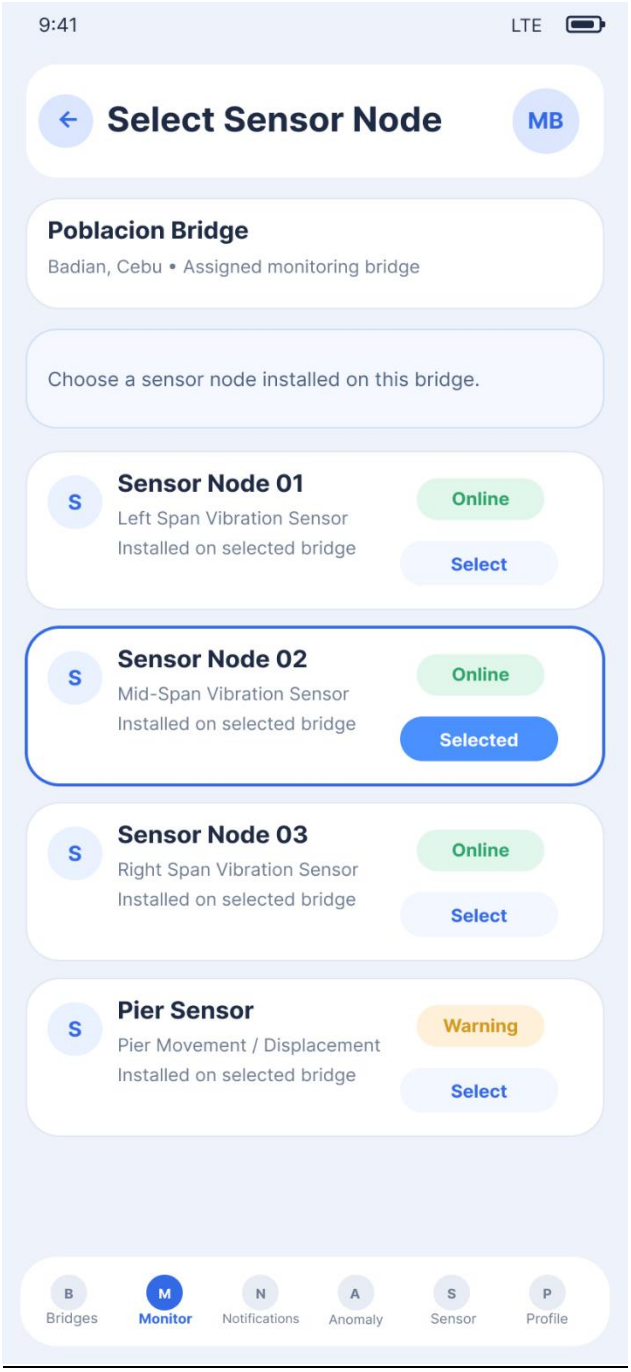


Figure 20: **Select Sensor Page**

Figure 20 presents the Select Sensor Node screen, where the engineer chooses the installed sensor node connected to the selected bridge. It lists the available sensor nodes together with their

type and current status. The selected node is highlighted to confirm the engineer's choice. This screen ensures that the correct sensor source is chosen before viewing real-time vibration data.

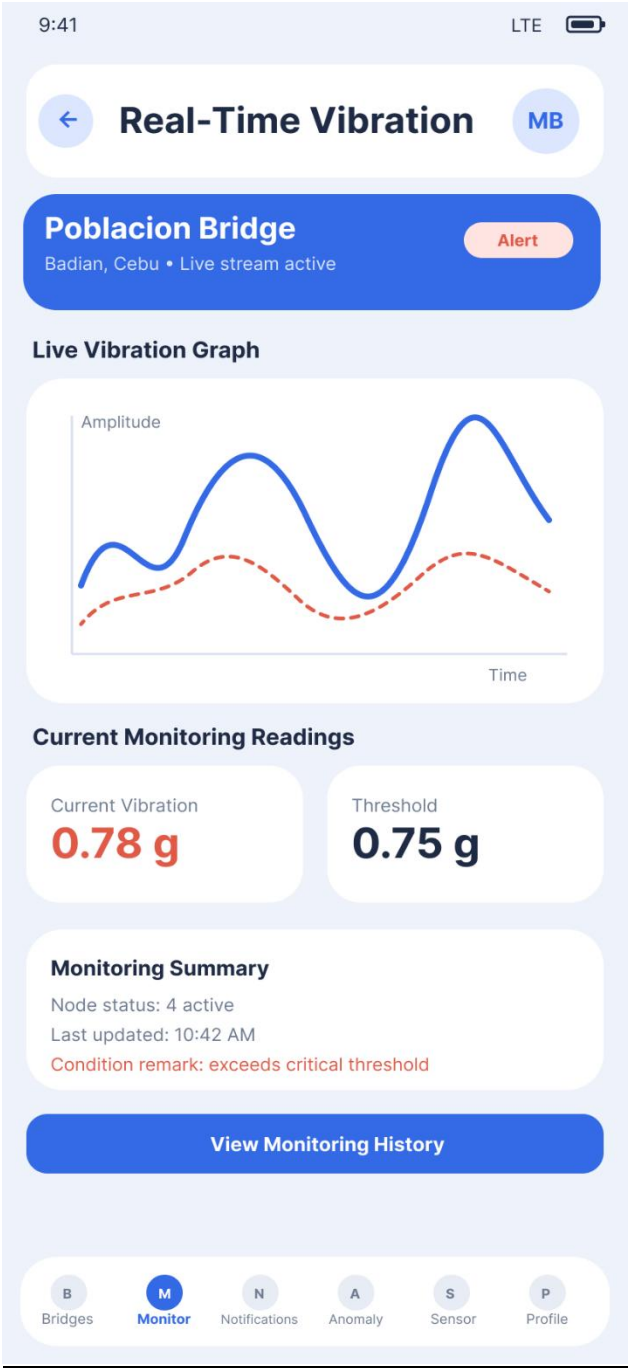


Figure 21: **View Real Time Vibration Page**

Figure 21 presents the View Real-Time Data Vibration screen, where the engineer can monitor the live vibration reading of the selected sensor node. It displays the current bridge,

selected sensor node, node status, and the real-time vibration trend graph. The screen also shows the current reading, peak reading, and overall condition to help the engineer assess the bridge status. From this screen, the engineer may proceed to view the baseline and threshold values or open the monitoring history.

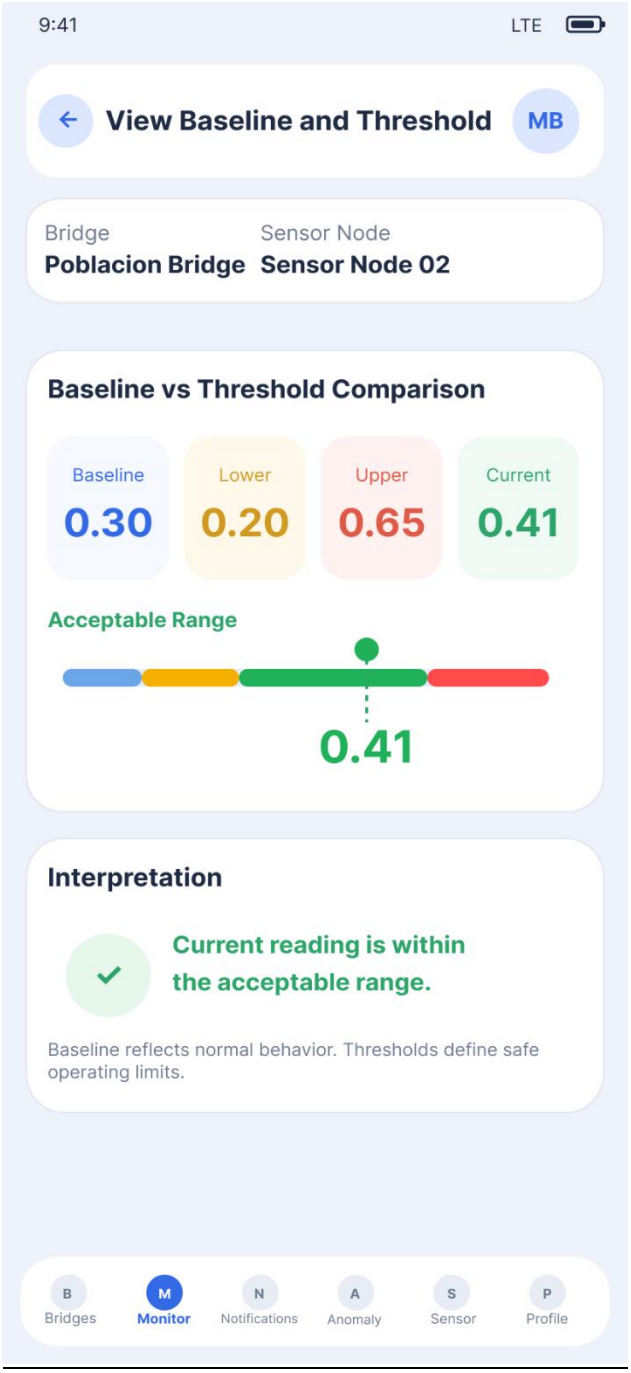


Figure 22: **View Baseline and Threshold**

Figure 22 presents the View Baseline and Threshold screen, where the engineer can compare the current sensor reading with the saved baseline, lower threshold, and upper threshold values. It also provides a visual range indicator and a short interpretation of the result. This screen helps the engineer determine whether the current reading is still within the acceptable monitoring range.

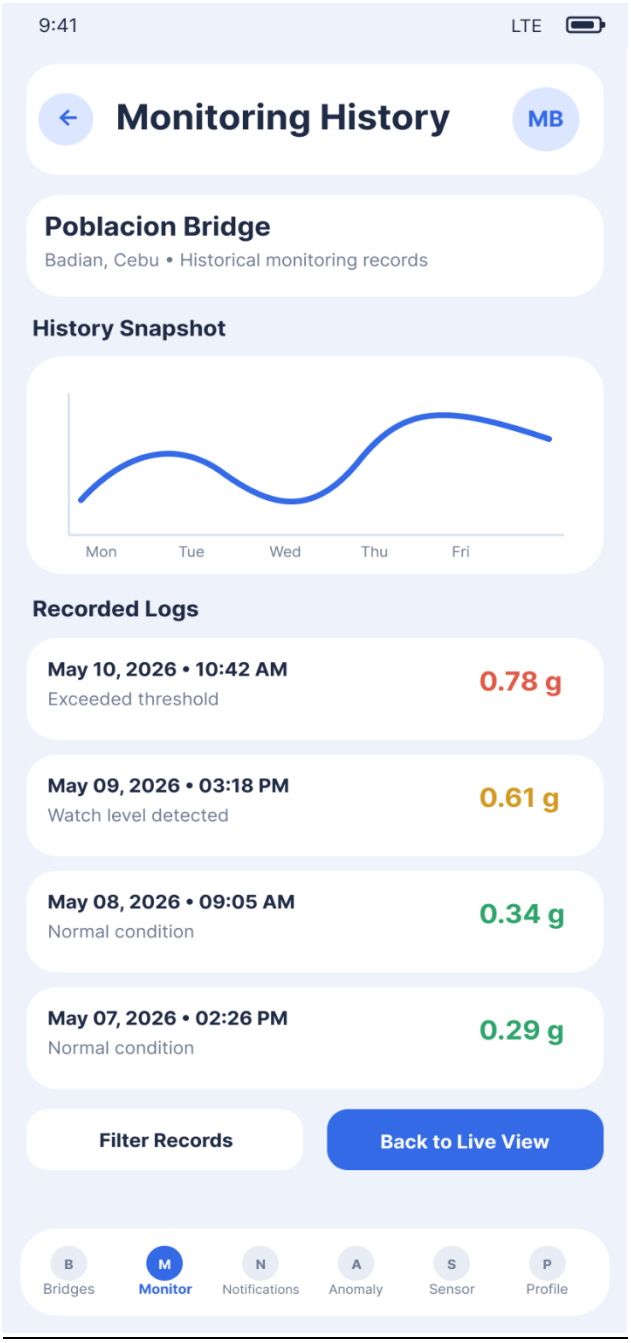


Figure 23: **View Monitoring History Page**

Figure 23 presents the Monitoring History screen, where the engineer can review the past monitoring records of the selected bridge. It shows a history snapshot graph and recorded logs with corresponding dates, times, readings, and condition remarks. The Filter Records button allows the engineer to narrow the displayed records, while Back to Live View returns to the real-time monitoring screen.

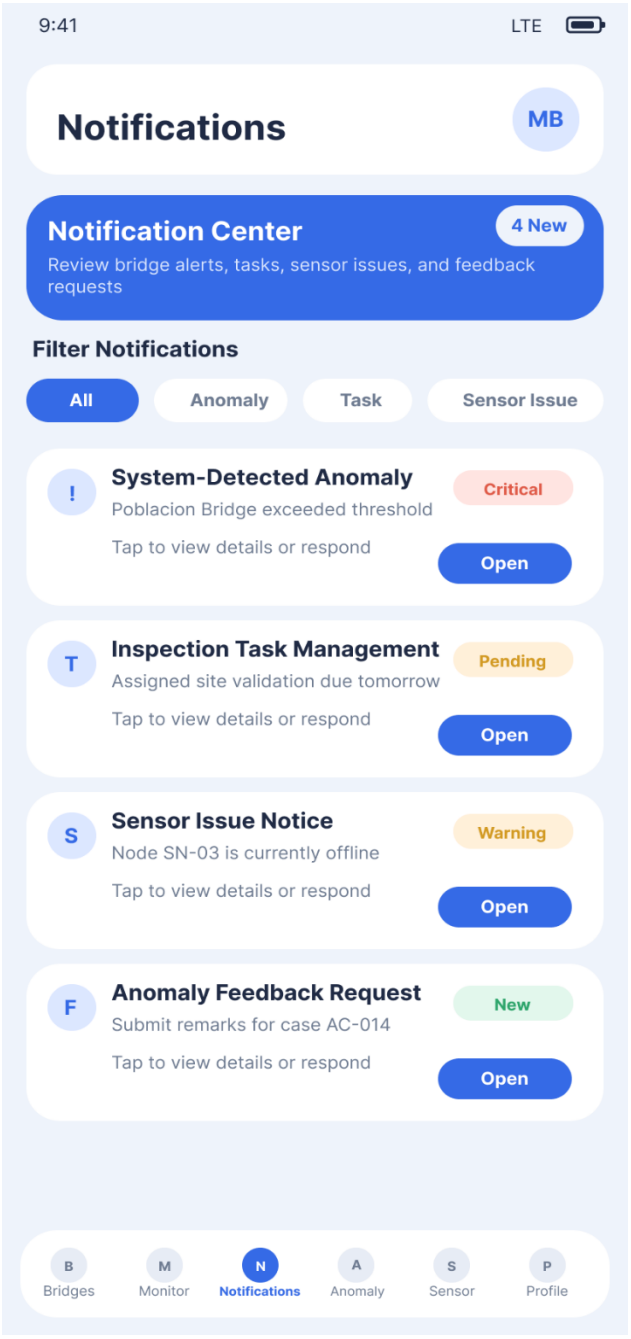


Figure 24**: Notification Page**

Figure 24 presents the Notifications screen, where the engineer can view system-detected anomalies, inspection tasks, sensor issue notices, and anomaly feedback requests. It also provides filter options so the engineer can sort notifications by category. The Open button allows the engineer to access the selected notification and respond to the required action.

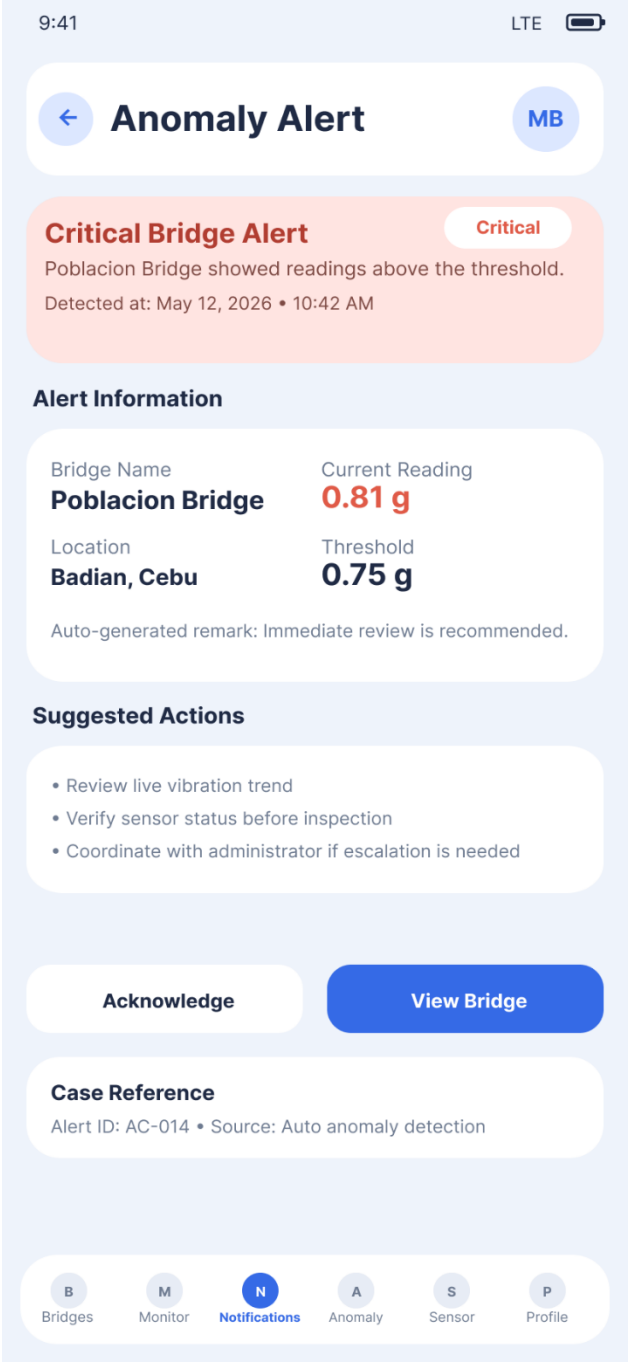


Figure 25: **System Detected Anomaly Alert Page**

Figure 25 presents the System-Detected Anomaly Alert screen, where the engineer can view the details of a critical anomaly detected by the system. It shows the affected bridge, current reading, threshold value, alert time, and suggested actions. The Acknowledge button confirms the alert, while View Bridge opens the related bridge record for further review.

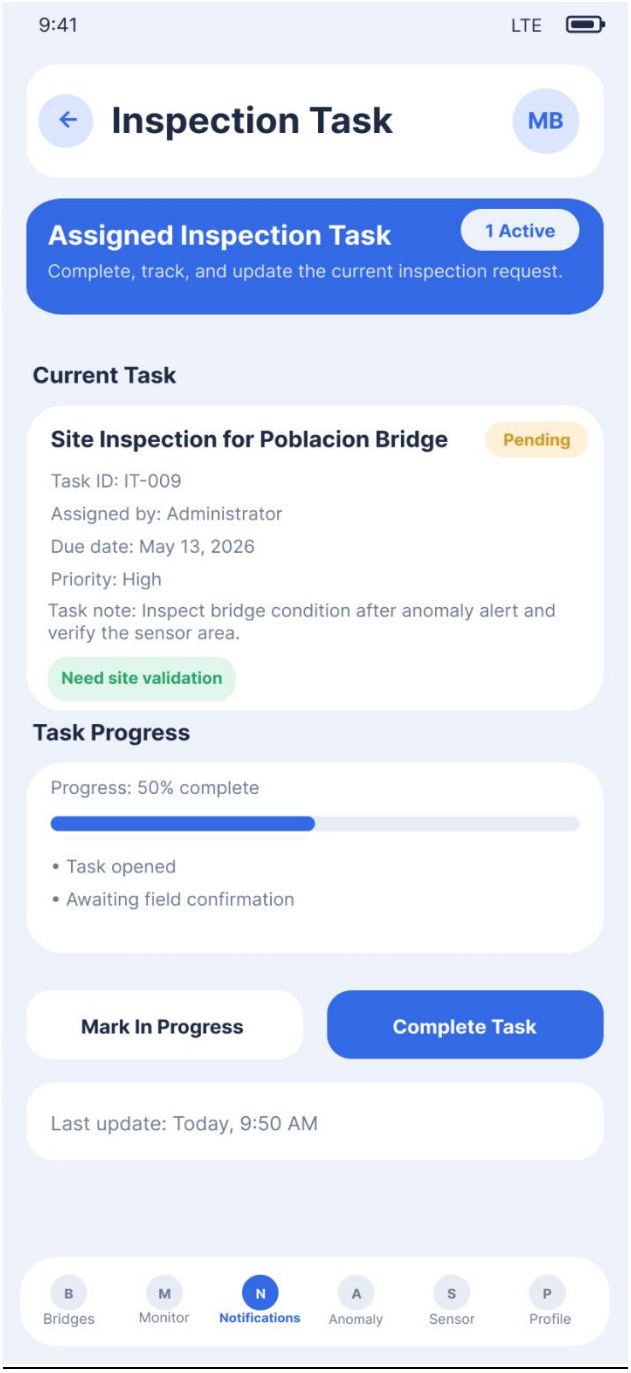


Figure 26: **Inspection Task Page**

Figure 26 presents the Inspection Task Management screen, where the engineer can view and manage the currently assigned inspection task. It shows the task details, priority, due date, note, and progress status. The Mark in Progress button updates the task status, while Complete Task records the task as finished.

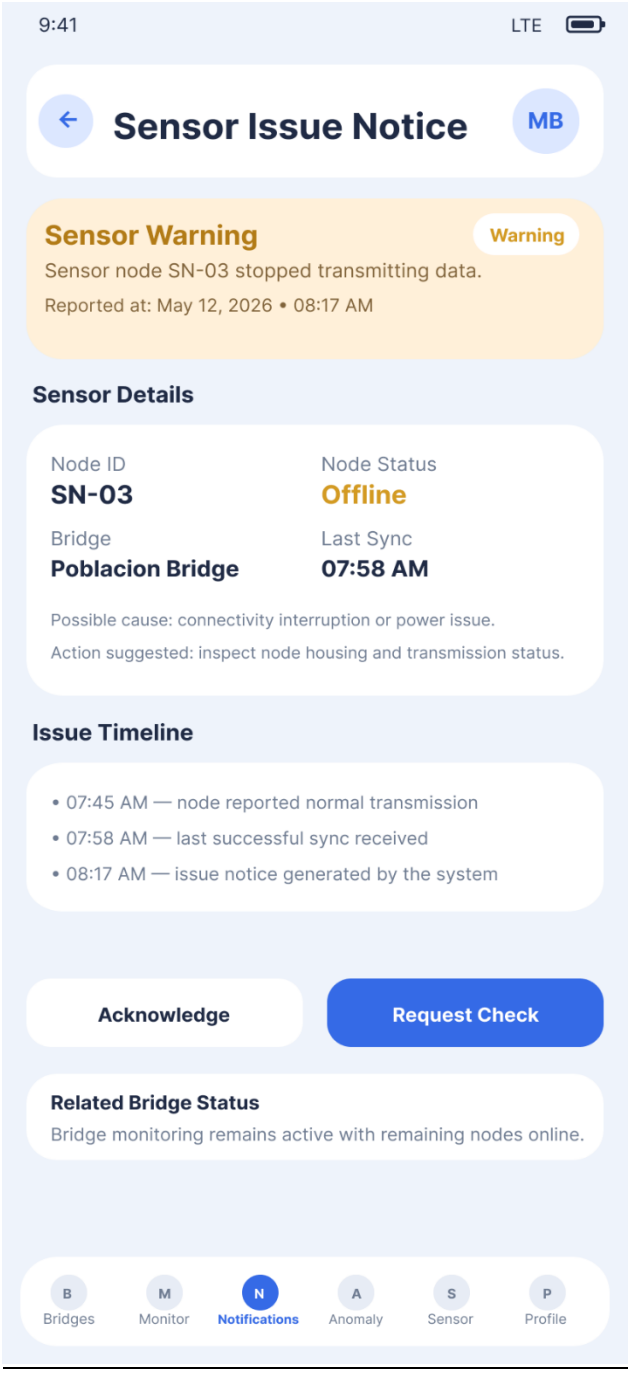


Figure 27: **Sensor Issue Notice Page**

Figure 27 presents the Sensor Issue Notice screen, where the engineer can view the details of a reported sensor problem. It shows the affected node, bridge, node status, last sync, possible cause, and issue timeline. The Acknowledge button confirms the notice, while Request Check allows the engineer to initiate further inspection of the sensor.

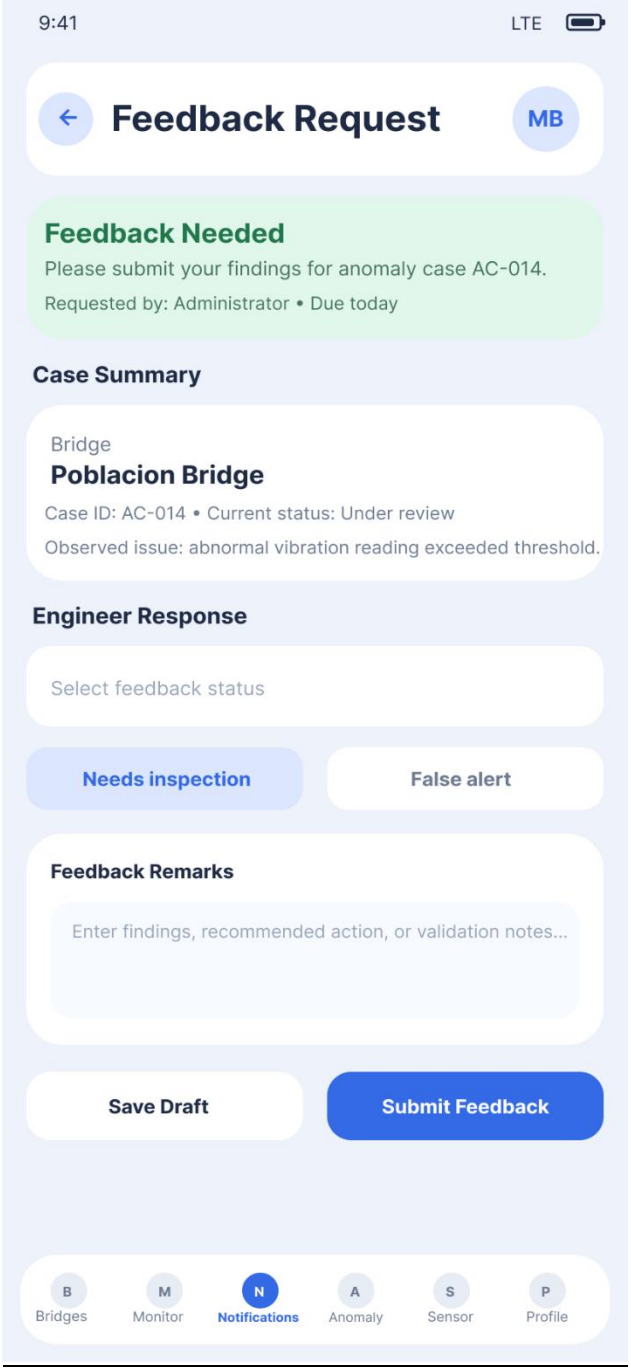


Figure 28: **Anomaly Feedback Request Page**

Figure 28 presents the Anomaly Feedback Request screen, where the engineer can provide findings and remarks for a specific anomaly case. It shows the case summary, lets the engineer choose a feedback status, and includes a field for entering validation notes or recommended action. The Save Draft button stores the response temporarily, while Submit Feedback sends the completed feedback for review.

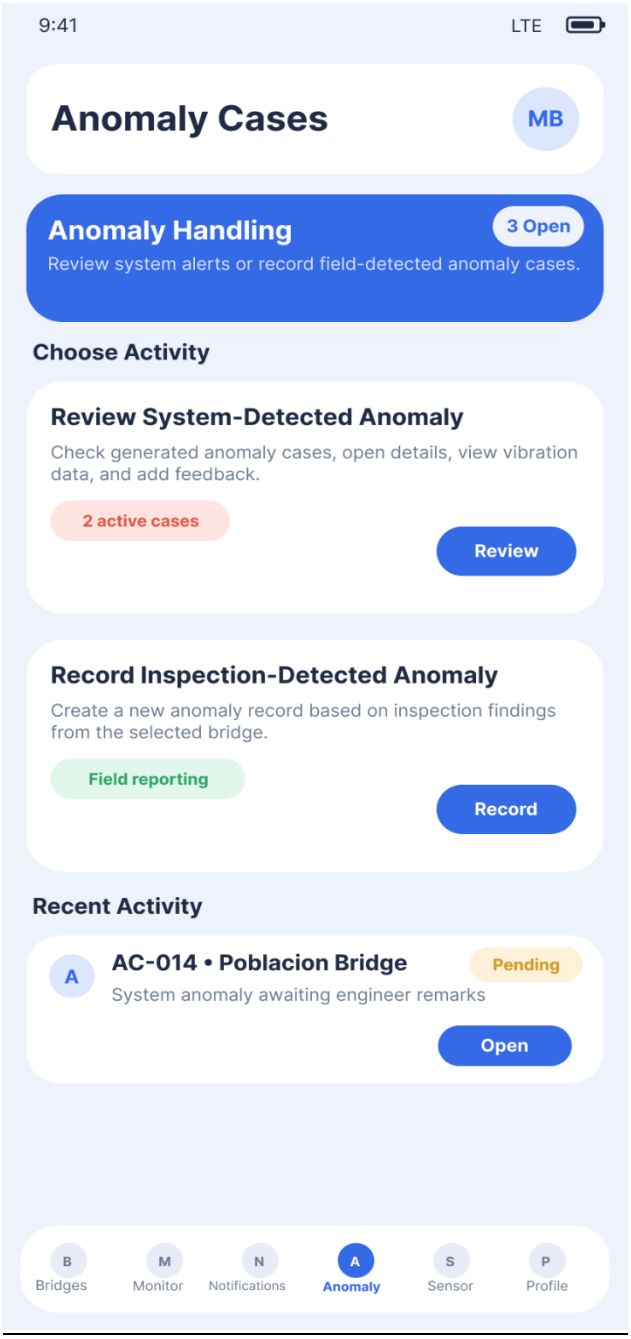


Figure 29: **Anomaly Cases Page**

Figure 29 presents the Handle Anomaly Cases screen, where the engineer can choose whether to review system-detected anomalies or record inspection-detected anomalies. It also shows recent anomaly-related activity for quick access. The Review button opens generated anomaly cases, while Record allows the engineer to create a new anomaly record based on field findings.

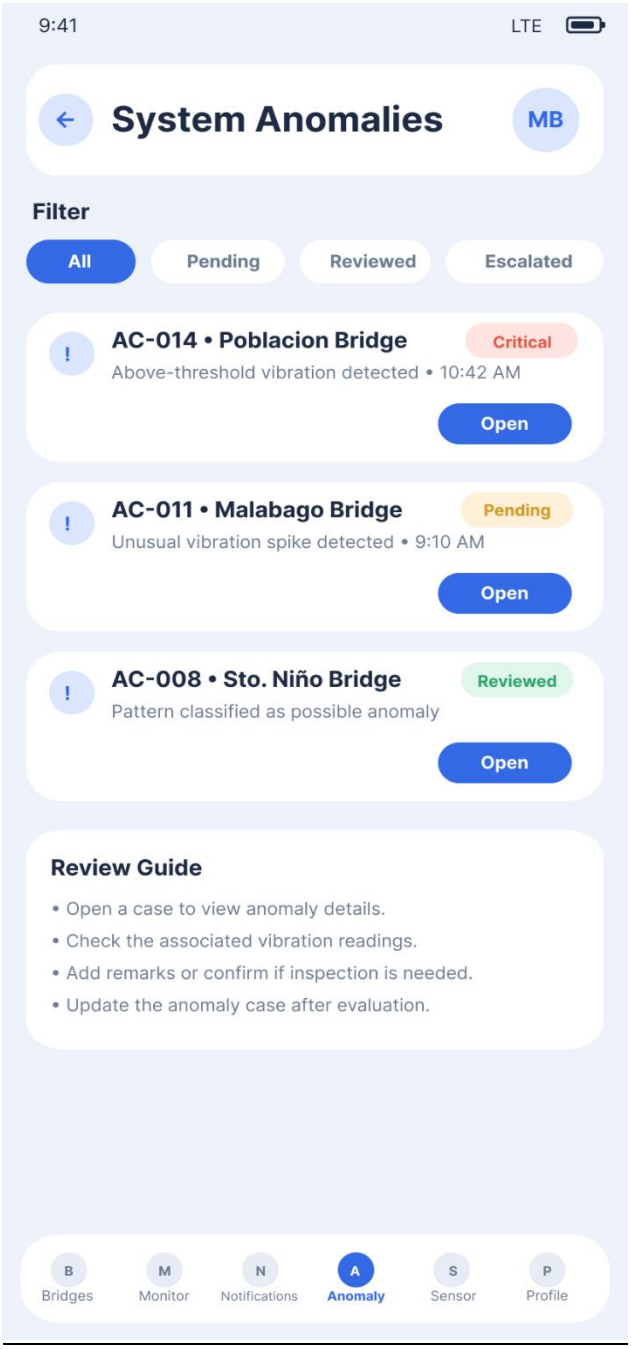


Figure 30: **System Anomalies Page**

Figure 30 presents the Review System-Detected Anomaly screen, where the engineer can view system-detected anomaly cases and filter them by status. It lists each anomaly case with its bridge name, detected issue, and current case status. The Open button allows the engineer to access the selected anomaly case for detailed review and evaluation.

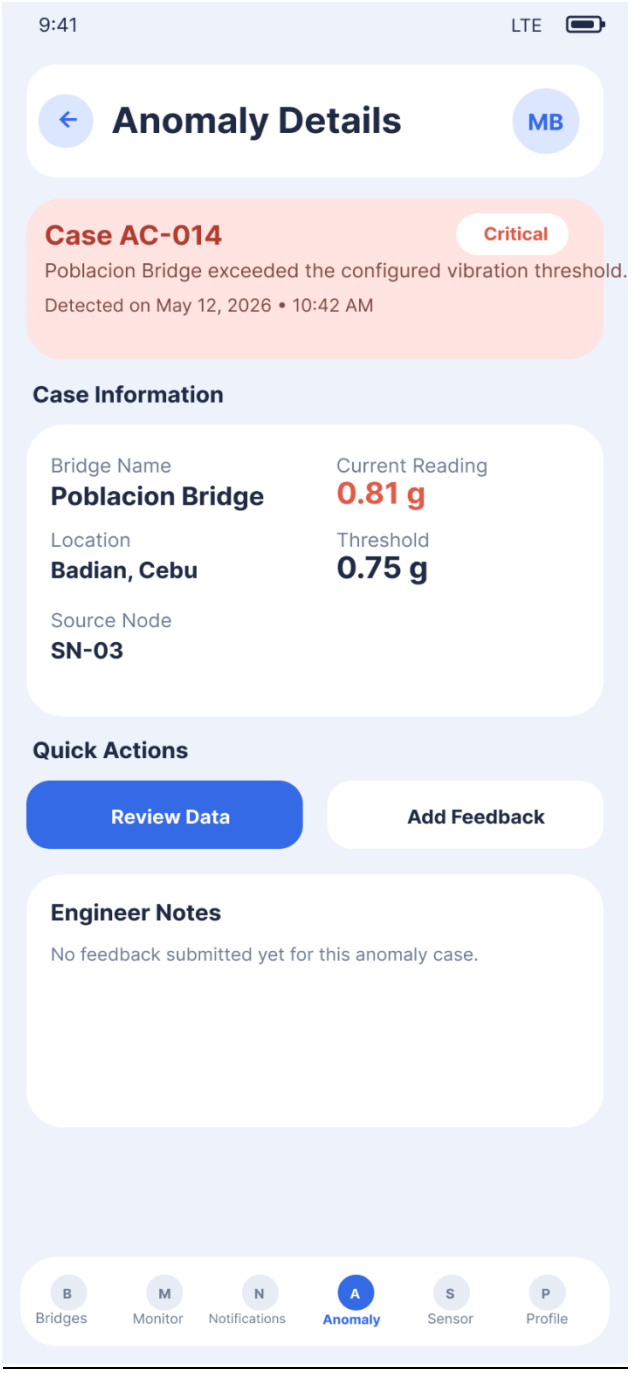


Figure 31: **Anomaly Details Page**

Figure 31 presents the Select Anomaly Details screen, where the engineer can view the detailed information of the selected anomaly case. It shows the case ID, affected bridge, location, current reading, threshold value, and source node. The Review Data button opens the vibration data for further analysis, while Add Feedback allows the engineer to submit remarks for the anomaly case.



Figure 32: **Review Vibration Data Page**

Figure 32 presents the Review Vibration Data screen, where the engineer can examine the vibration readings linked to the selected anomaly case. It displays the live vibration trend, the threshold line, the peak reading, and the assessment result. This screen helps the engineer evaluate whether the anomaly requires further review before submitting feedback.

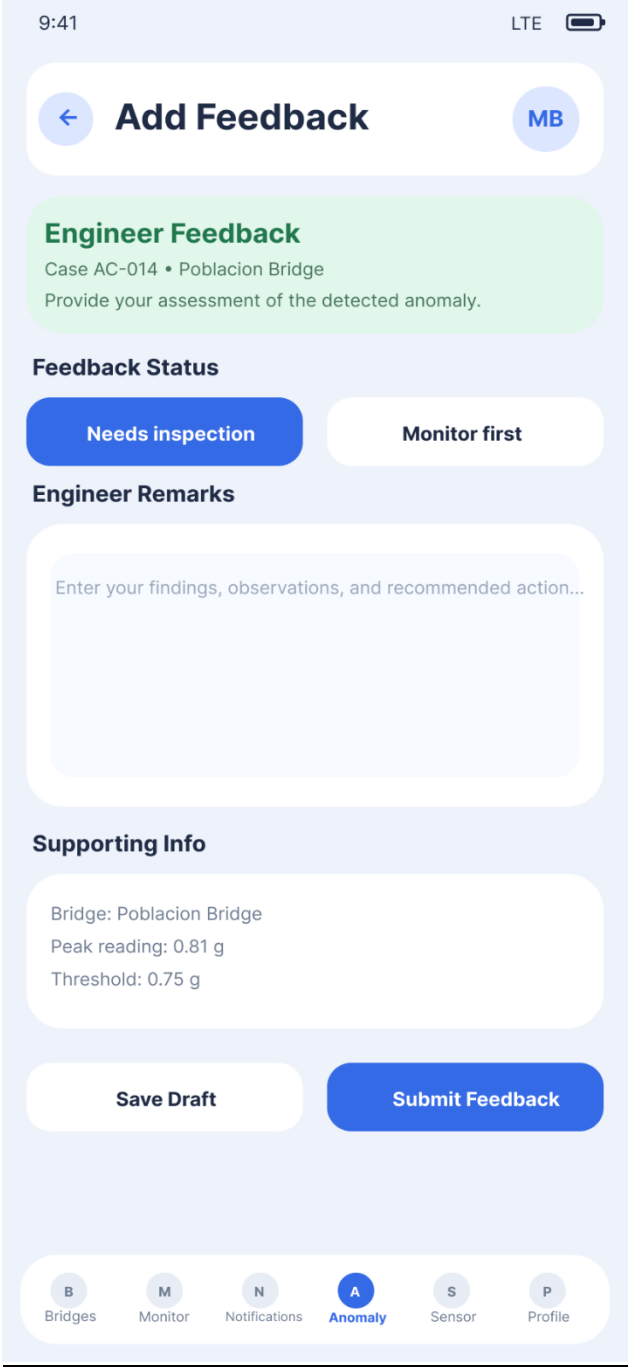


Figure 33: **Add Feedback Anomaly Page**

Figure 33 presents the Add Feedback screen, where the engineer can provide remarks for the selected anomaly case. It allows the engineer to choose a feedback status, enter findings and recommended action, and review supporting information such as the bridge, peak reading, and threshold. The Save Draft button stores the response temporarily, while Submit Feedback sends the completed feedback for review.

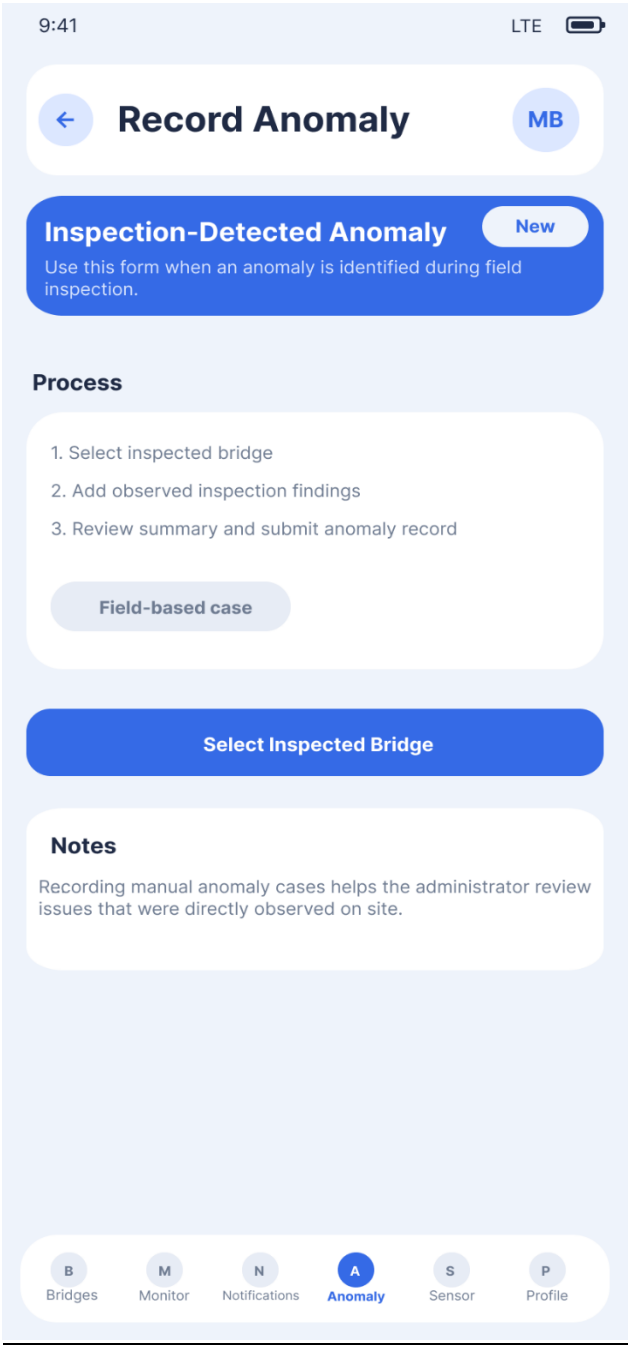


Figure 34: **Record Anomaly Detected Page**

Figure 34 presents the Record Inspection-Detected Anomaly screen, where the engineer begins creating a new anomaly record based on field inspection. It outlines the reporting steps and provides a button to select the inspected bridge. This screen serves as the starting point for documenting anomalies directly observed on site.

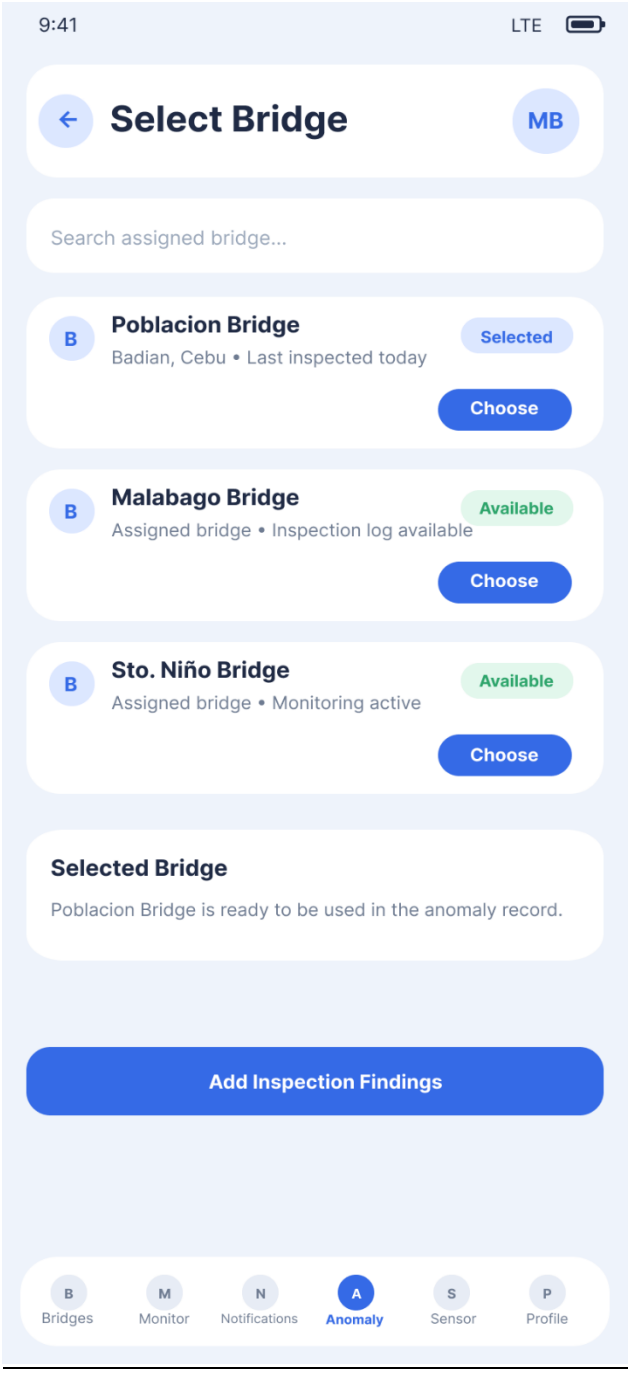


Figure 35: **Select Inspected Bridge Page**

Figure 35 presents the Select Inspected Bridge screen, where the engineer chooses the bridge to be used in creating an anomaly record. It lists the available assigned bridges and highlights the selected one for confirmation. The Add Inspection Findings button proceeds to the next step of recording the observed anomaly.

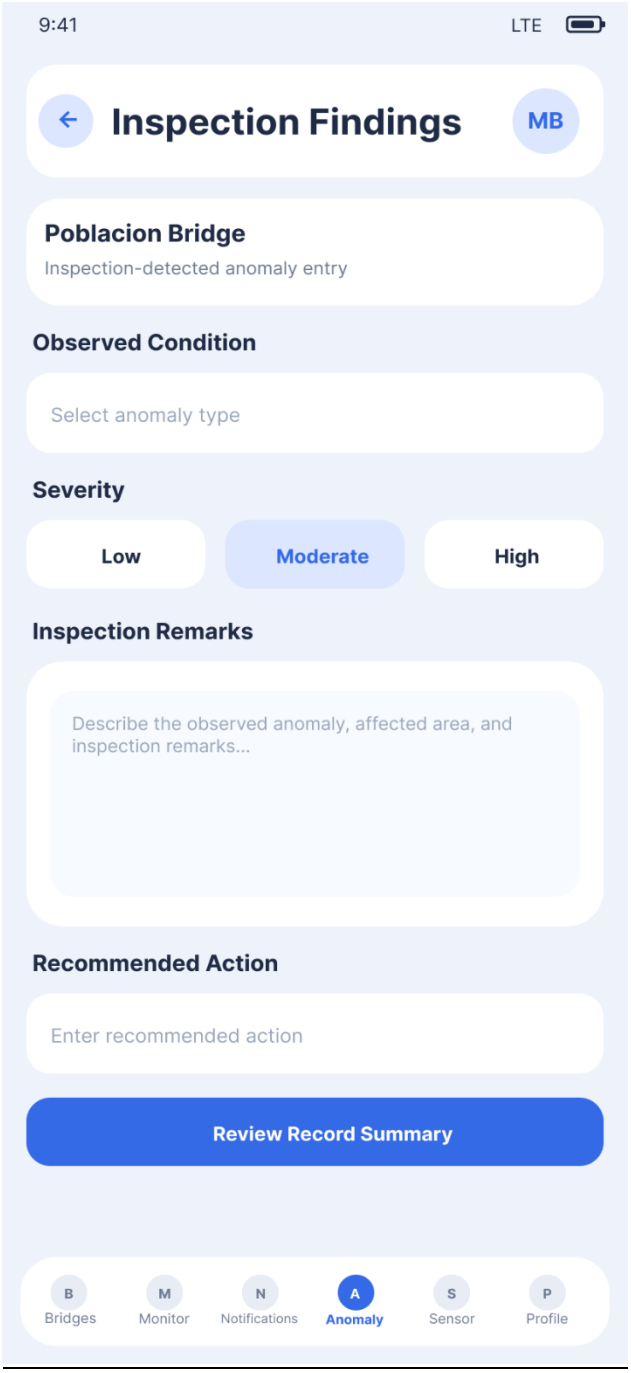


Figure 36: **Inspection Findings Page**

Figure 36 presents the Add Inspection Findings screen, where the engineer records the observed anomaly during inspection. It allows the engineer to select the anomaly type, choose the severity level, enter inspection remarks, and provide a recommended action. The Review Record Summary button proceeds to the final review of the anomaly record.

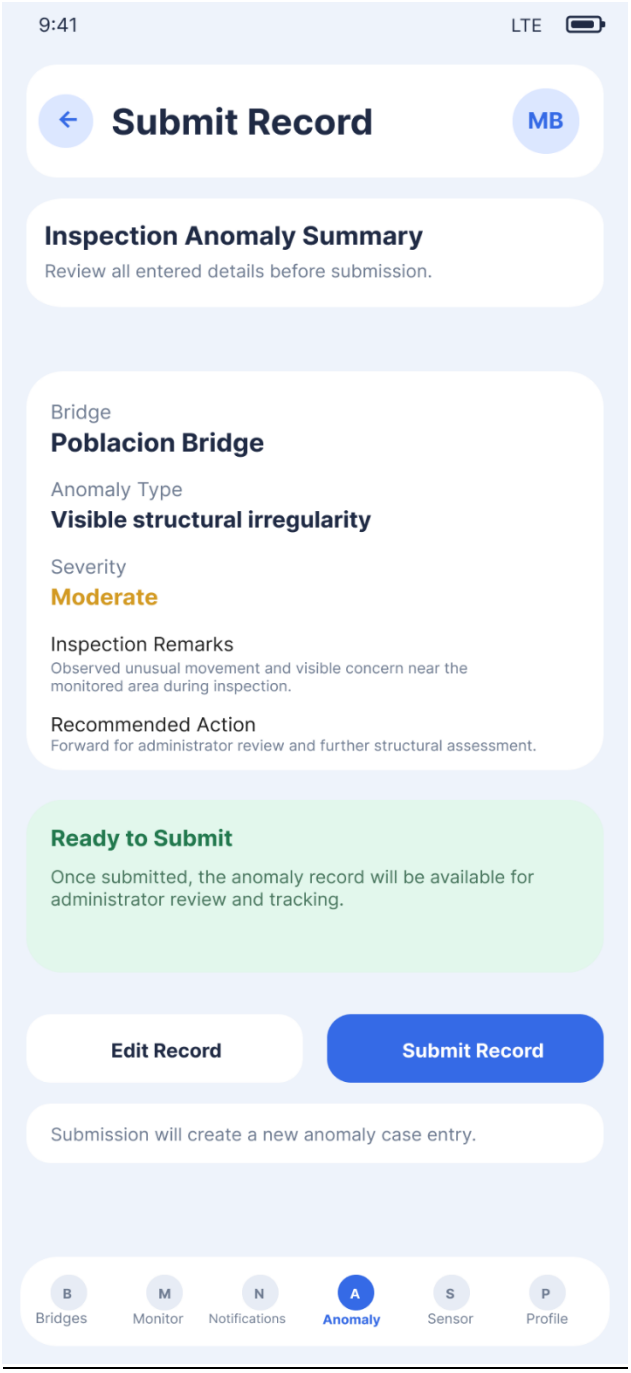


Figure 37**: Submit Anomaly Record Page**

Figure 37 presents the Submit Anomaly Record screen, where the engineer reviews all entered anomaly details before final submission. It summarizes the bridge, anomaly type, severity, remarks, and recommended action. The Edit Record button allows changes to the entry, while Submit Record saves the anomaly case for administrator review and tracking.

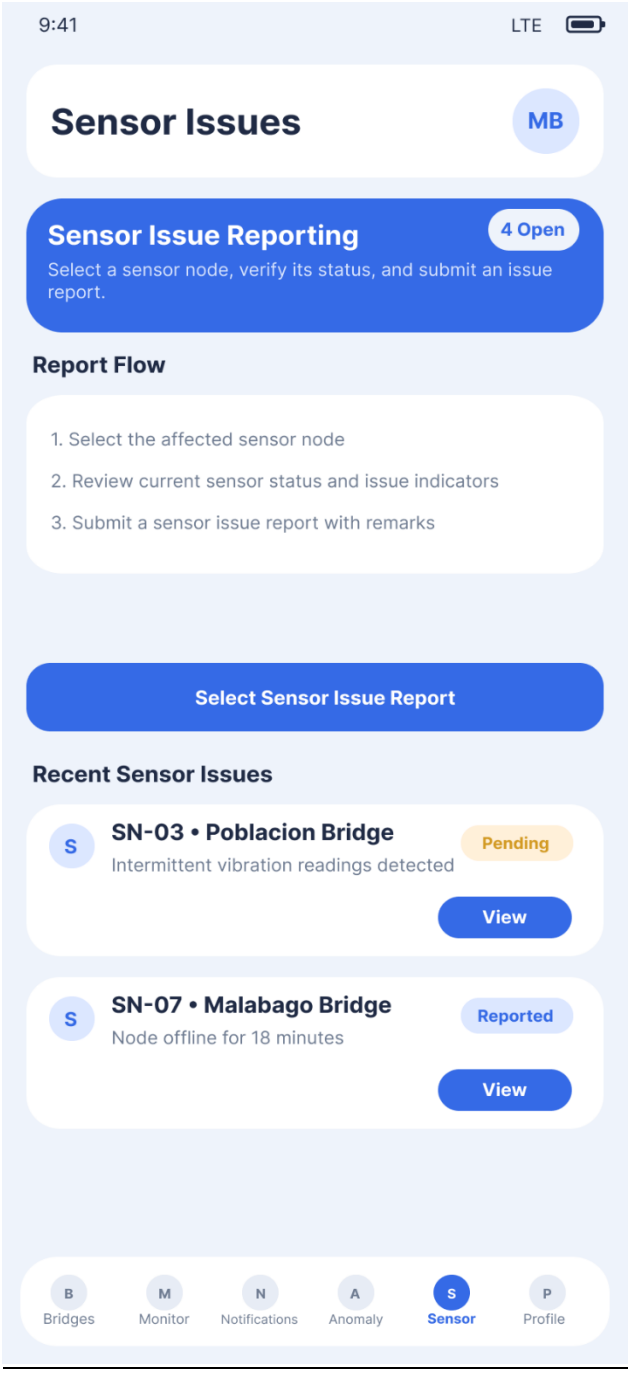


Figure 38: **Report Sensor Issues Page**

Figure 38 presents the Report Sensor Issue screen, where the engineer starts the process of reporting a sensor-related problem. It outlines the reporting steps and shows recent sensor issue records for reference. The Select Sensor Issue Report button allows the engineer to proceed with creating a new sensor issue report.

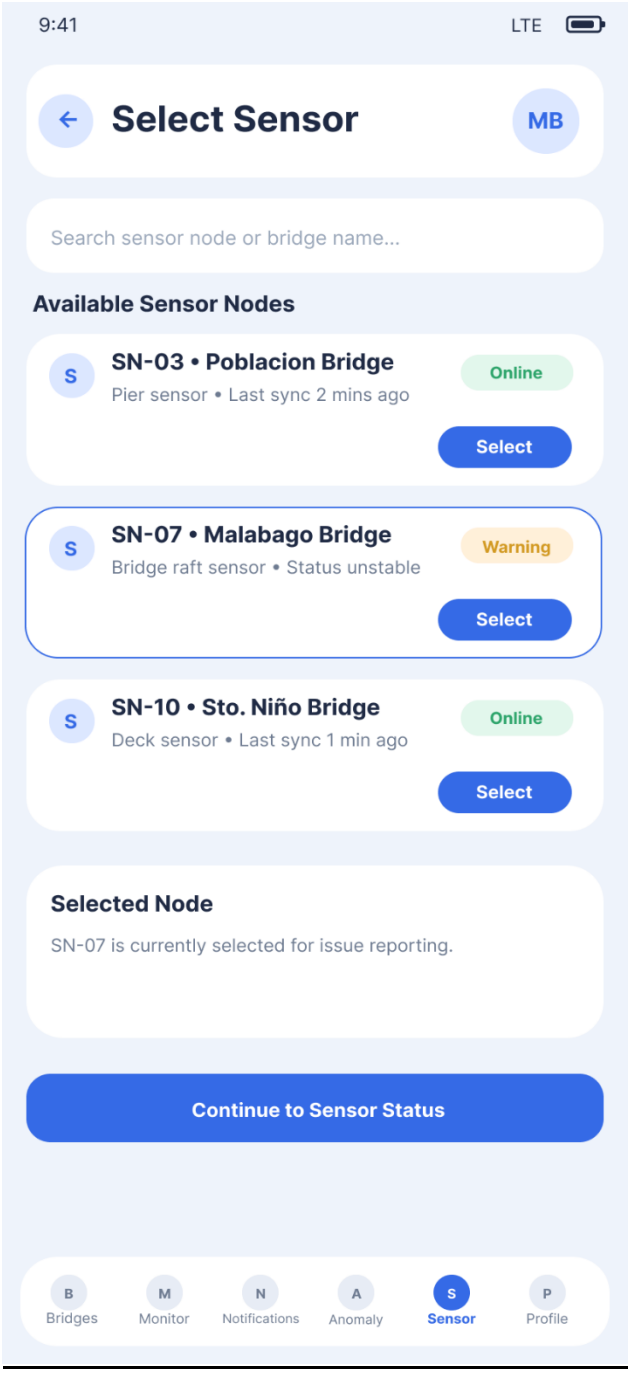


Figure 39: **Select Sensor Node Page**

Figure 39 presents the Select Sensor screen, where the engineer chooses the sensor node to be used for issue reporting. It displays the available sensor nodes together with their bridge, sensor type, sync status, and current condition. The selected node is shown in a separate summary section, while Continue to Sensor Status proceeds to the next step of checking the selected node's status.

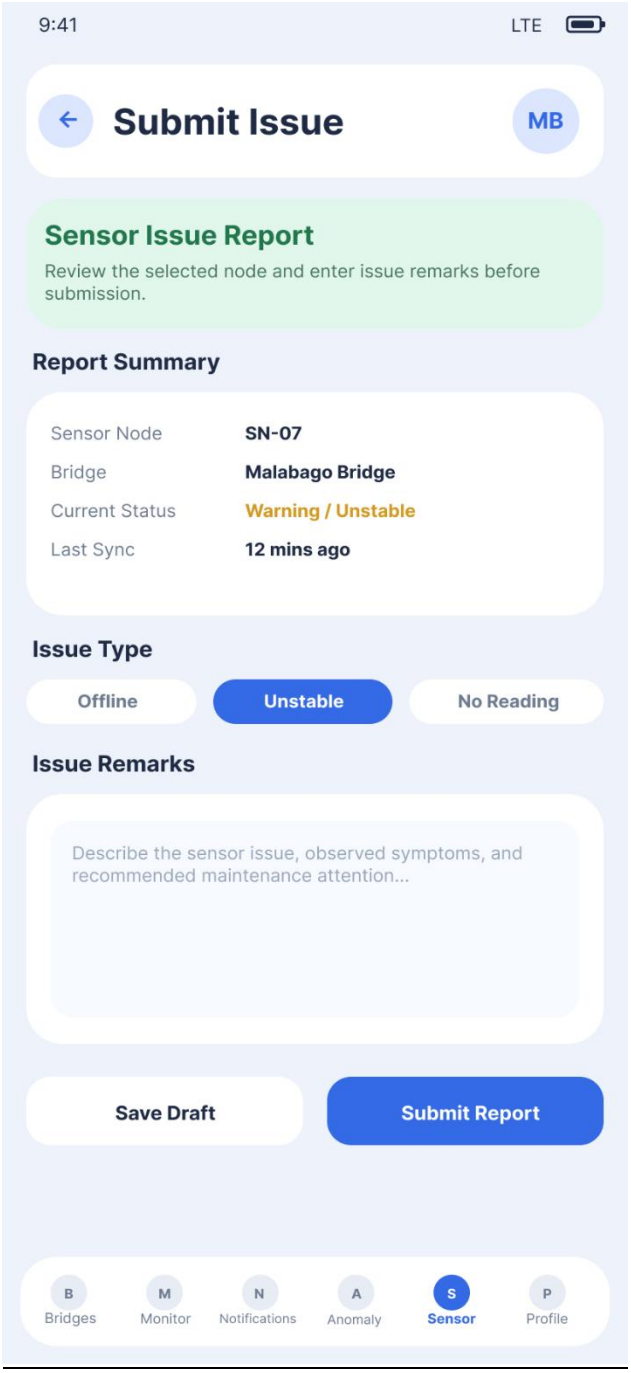


Figure 40: **Submit Sensor Issue Report Page**

Figure 40 presents the Submit Sensor Issue Report screen, where the engineer reviews the selected sensor node details and enters the issue report before submission. It shows the sensor node, bridge, current status, last sync, issue type, and remarks field for describing the problem. The Save Draft button stores the report temporarily, while Submit Report sends the completed sensor issue report to the system.

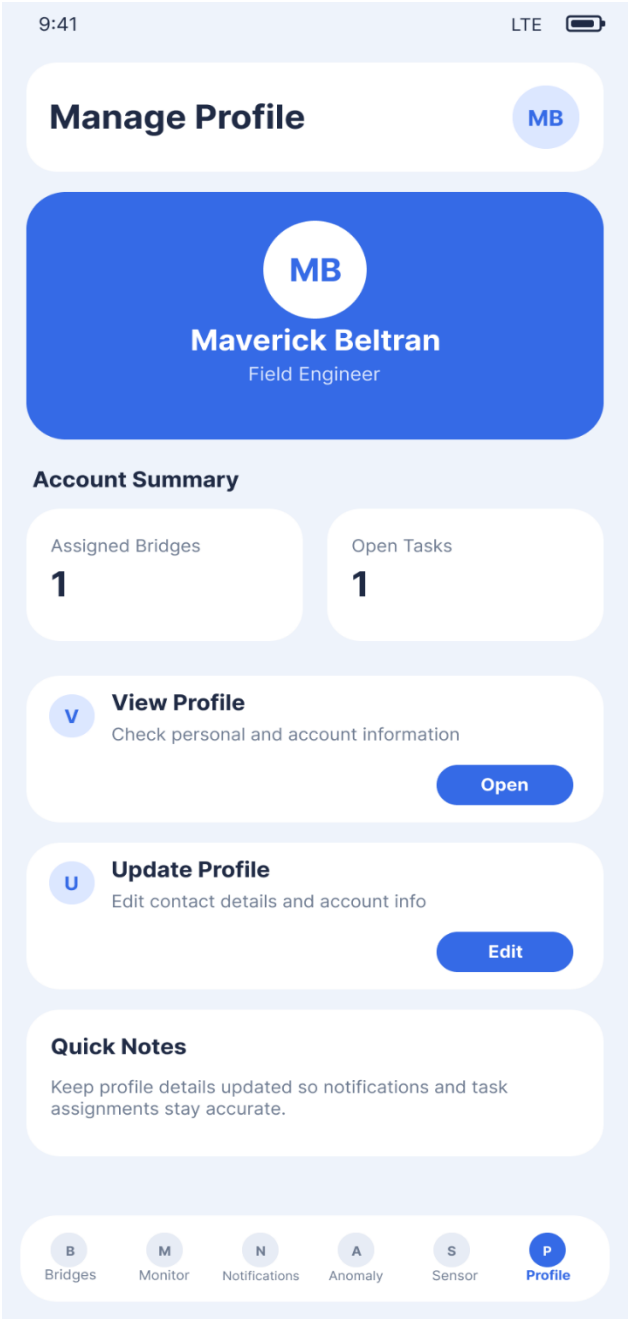


Figure 41: **Manage Profile Page**

Figure 41 presents the Manage Profile screen, where the engineer can access profile-related account functions. It shows the engineer's basic profile summary, account overview, and quick options for viewing or updating profile information. This screen serves as the main profile hub for managing personal and account details.

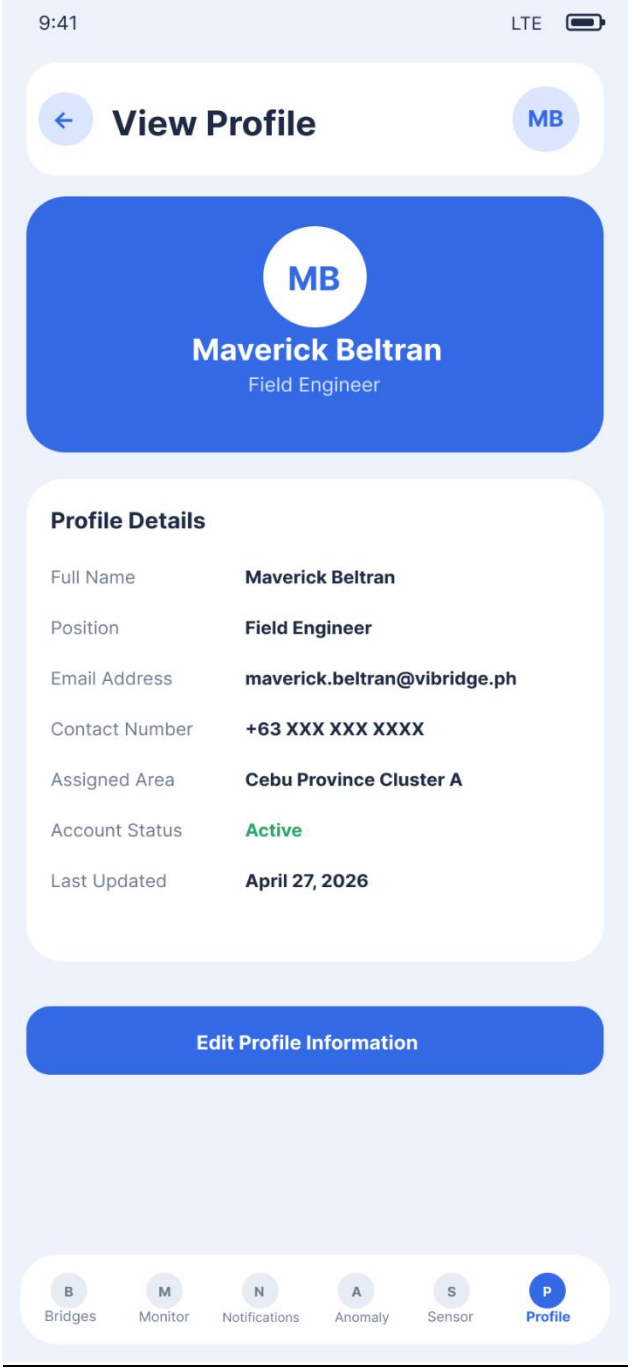


Figure 42: **View Profile Page**

Figure 42 presents the View Profile screen, where the engineer can see their personal and account information. It displays details such as full name, position, email address, contact number, assigned area, account status, and last updated date. The Edit Profile Information button allows the engineer to proceed to the profile update screen.

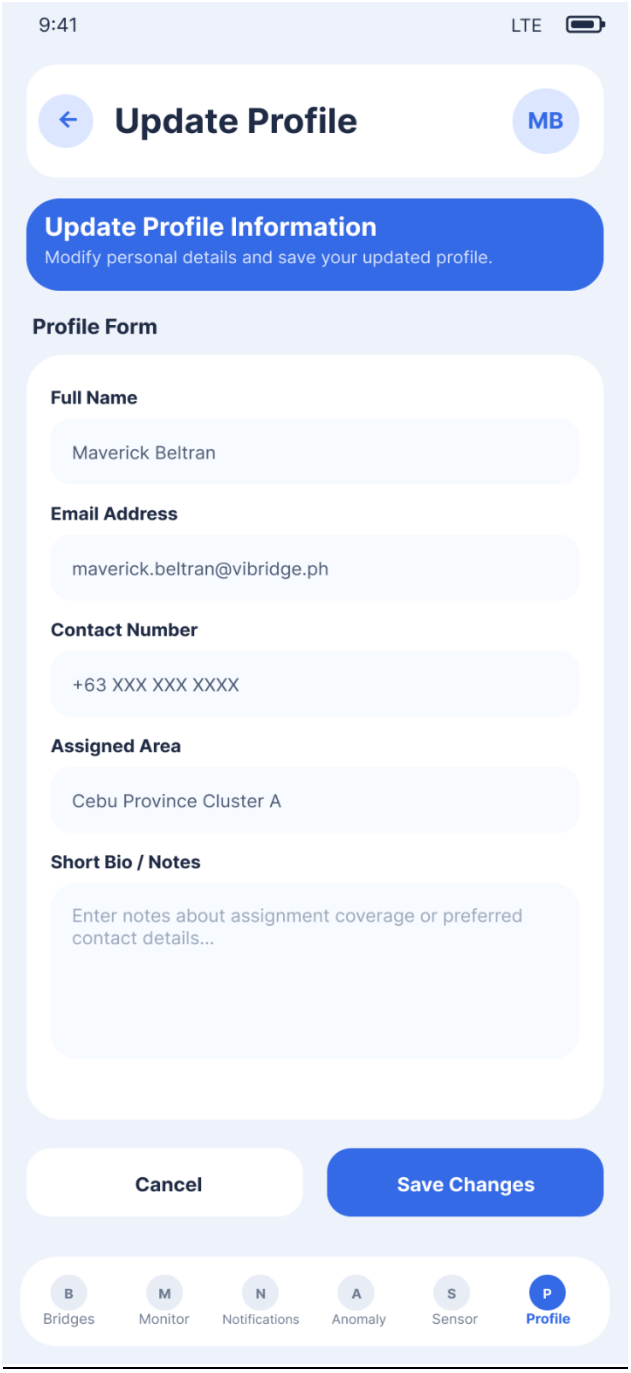


Figure 43: **Update Profile Page**

Figure 43 presents the Update Profile Information screen, where the engineer can edit personal and account details. It provides input fields for full name, email address, contact number, assigned area, and short notes. The Cancel button discards changes, while Save Changes updates the engineer's profile information.

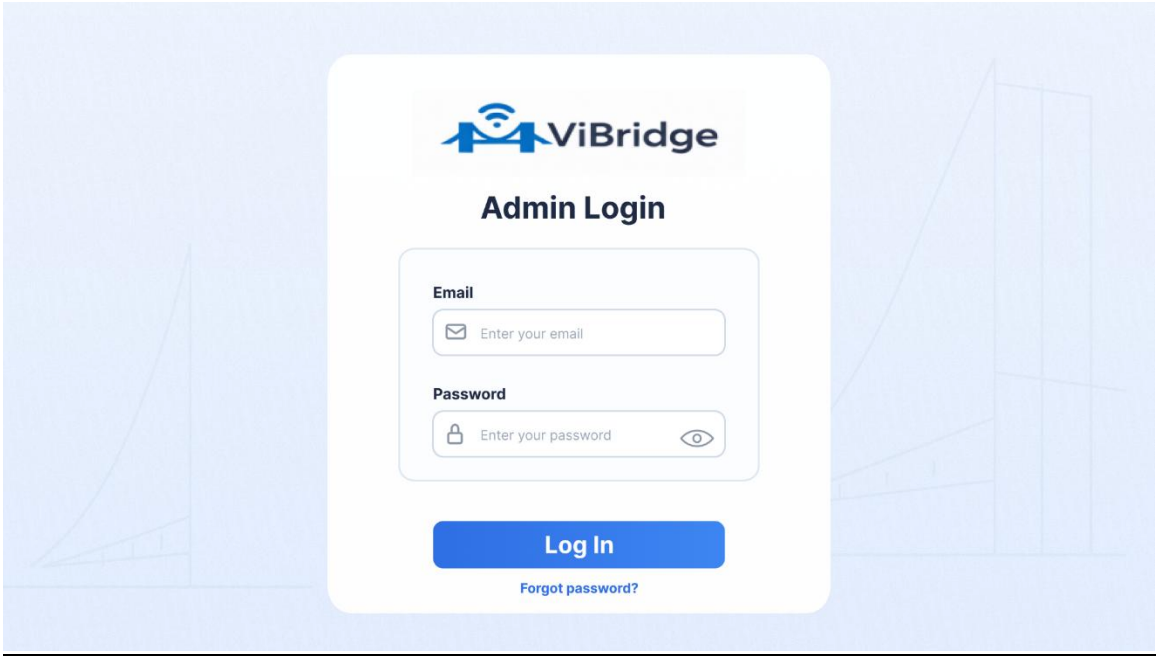


Figure 44: **Web Administration Login Page**

Figure 44 presents the Admin Login screen, where the administrator enters an email and password to access the system. It serves as the entry point to the admin platform and verifies the provided credentials before allowing access to the dashboard. The Log In button submits the account details, while Forgot password? helps the administrator recover the account if needed.

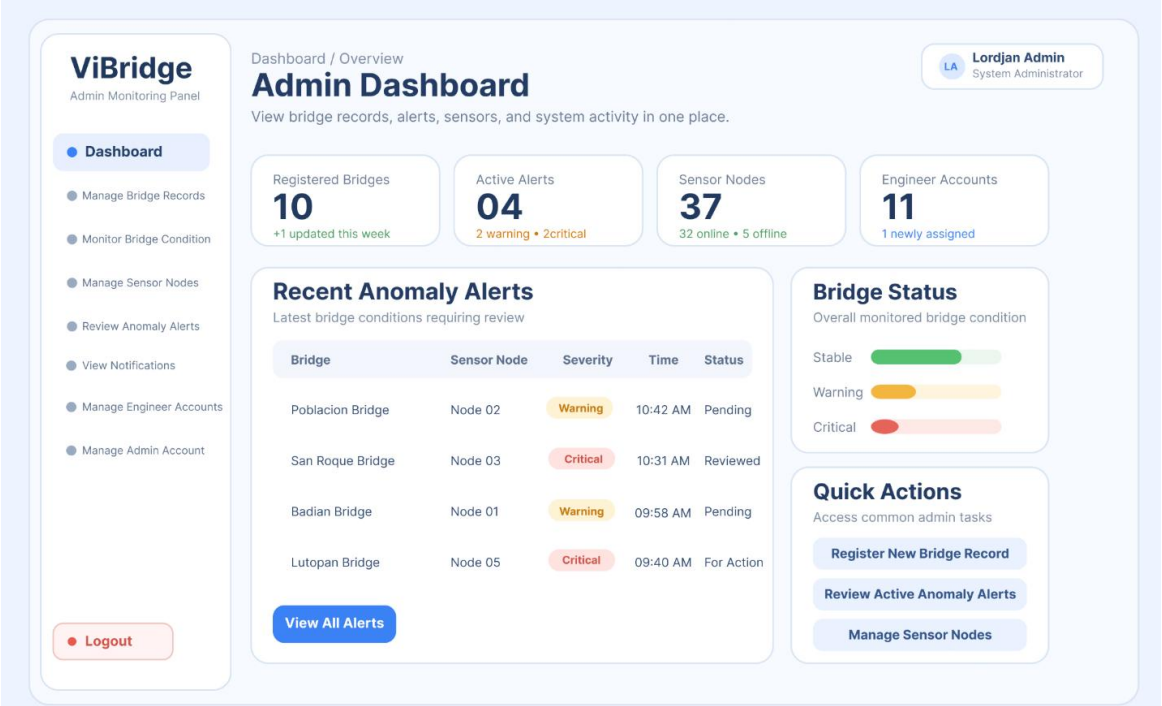


Figure 45: **Web Administration Dashboard Page**

Figure 45 presents the Admin Dashboard, which serves as the main screen after a successful login. It gives the administrator a quick overview of registered bridges, active alerts, sensor nodes, and engineer accounts. It also displays recent anomaly alerts, bridge status, and quick action buttons so the administrator can immediately access major system functions.

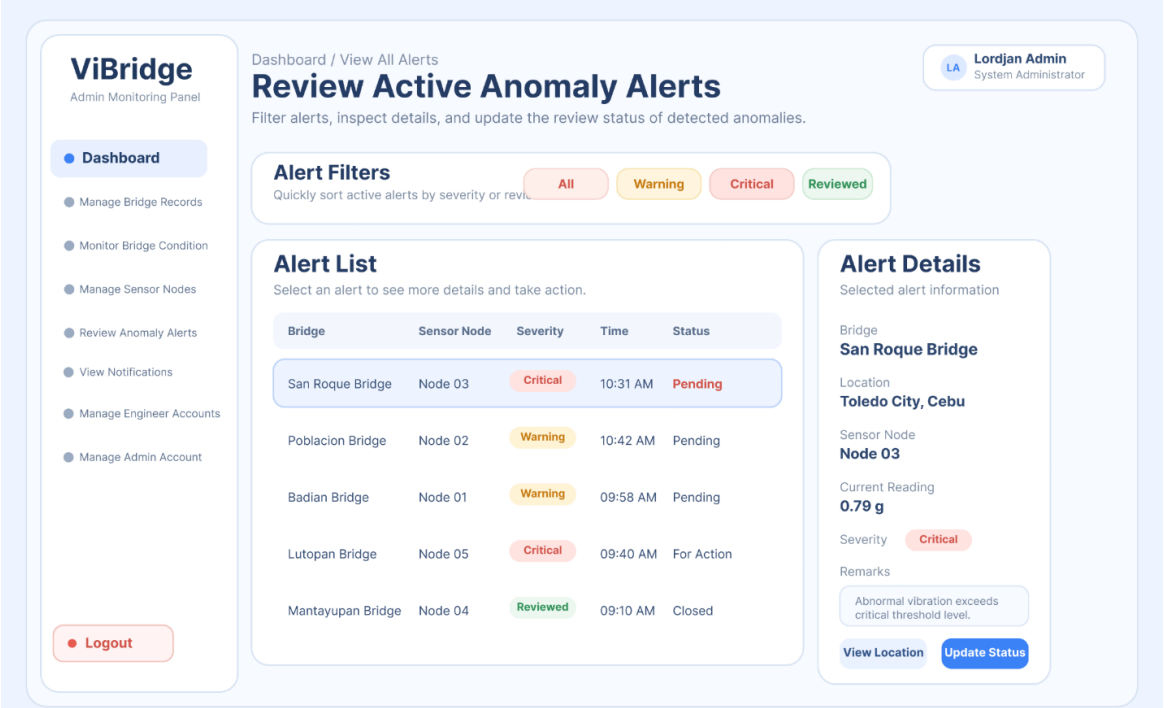


Figure 46: **Web Administration Review Anomaly Alerts Page**

Figure 46 presents the Review Active Anomaly Alerts screen, where the administrator can filter, review, and manage detected anomaly alerts. It shows the alert list with bridge, sensor node, severity, time, and status, while the details panel displays the selected alert's information for closer review. The available actions allow the administrator to view the bridge location and update the alert status.

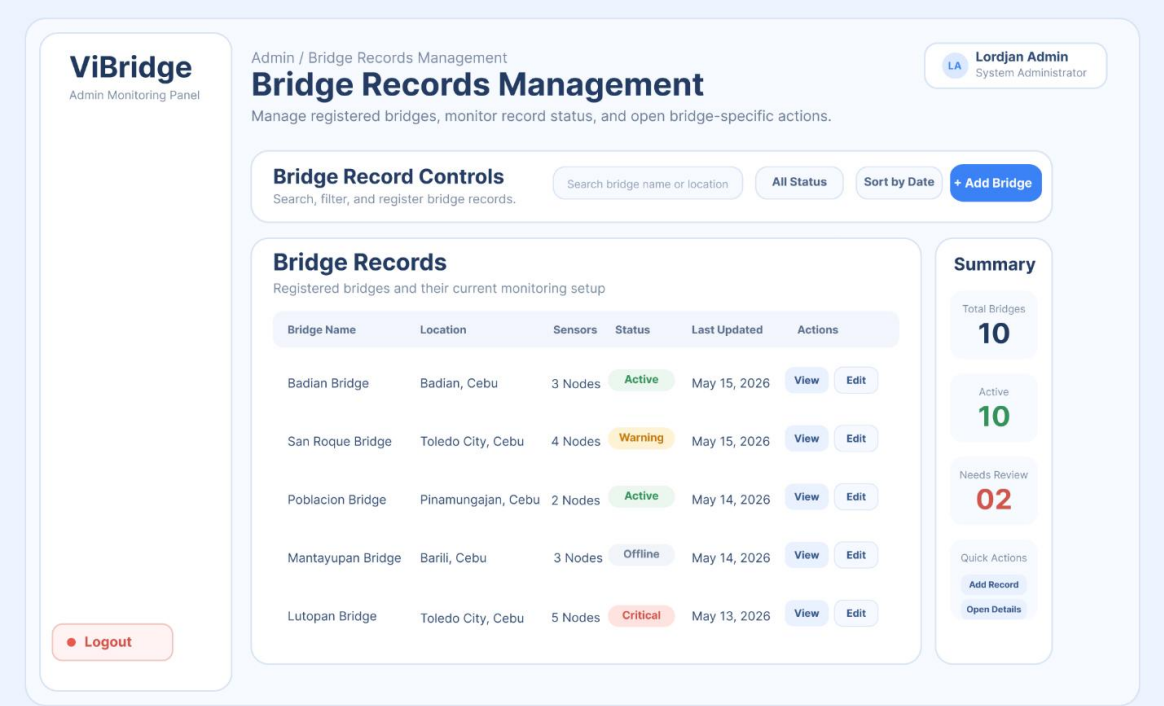


Figure 47: **Web Administration Review Bridge Records Management Page**

Figure 47 presents the Bridge Records Management screen, where the administrator can view and manage all registered bridge records. It provides controls for searching, filtering, sorting, and adding bridge entries, while the records table shows each bridge's location, sensor count, status, and last updated date. The View and Edit actions allow the administrator to open and manage a specific bridge record.

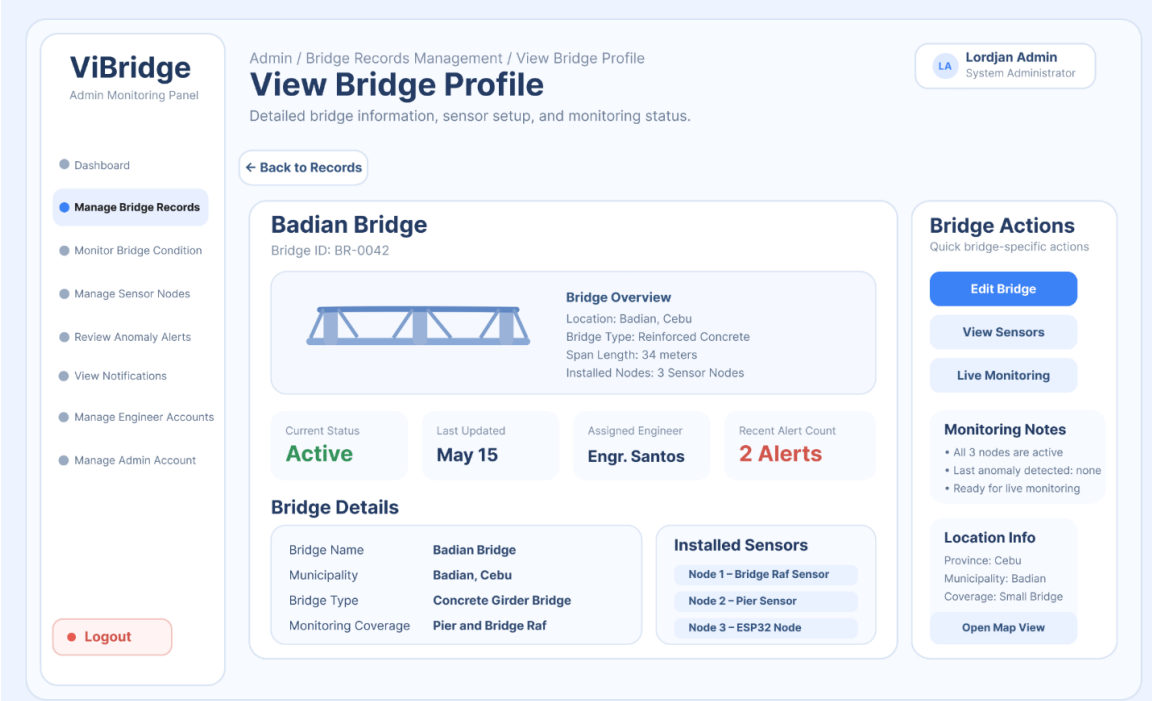


Figure 48: **Web Administration Bridge Profile Page**

Figure 48 presents the View Bridge Profile screen, where the administrator can review the complete information of a selected bridge record. It displays the bridge overview, current status, last updated details, assigned engineer, alert count, bridge details, and installed sensors. The action panel on the right also allows the administrator to edit the bridge, view sensors, access live monitoring, and open the bridge location.

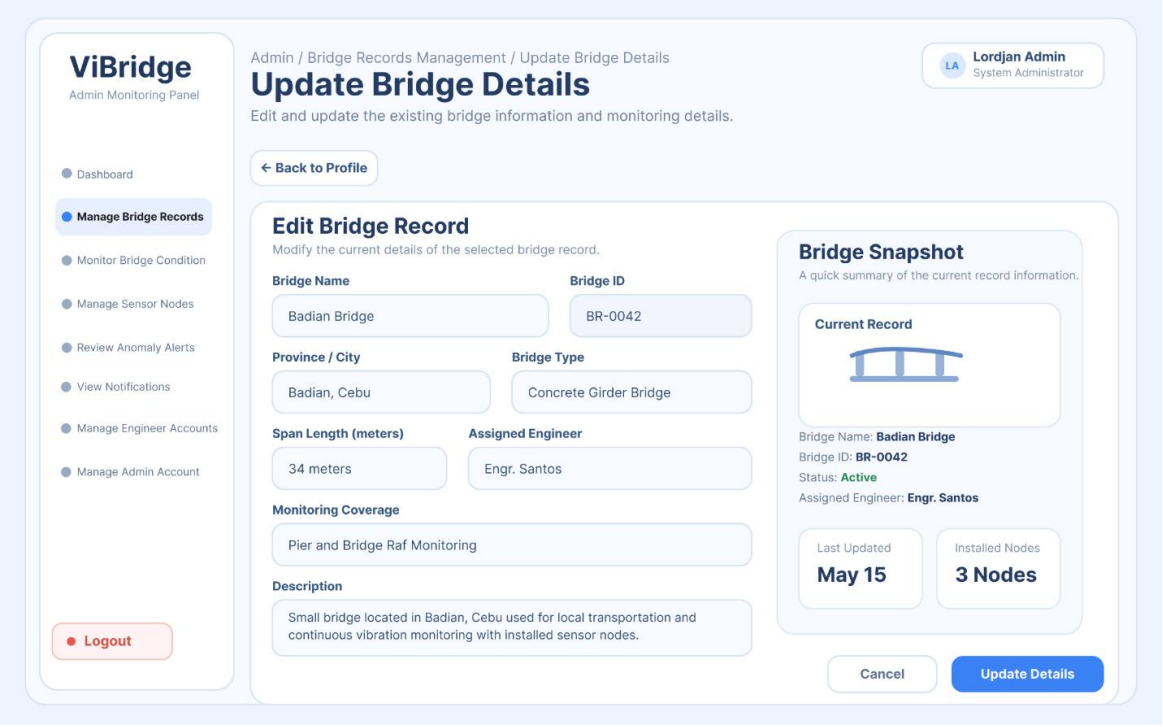


Figure 49: **Web Administration Update Bridge Details Page**

Figure 49 presents the Update Bridge Details screen, where the administrator can edit the existing information of a selected bridge record. It provides input fields for bridge name, bridge ID, location, type, span length, assigned engineer, monitoring coverage, and description. The Cancel button discards changes, while Update Details saves the updated bridge information.

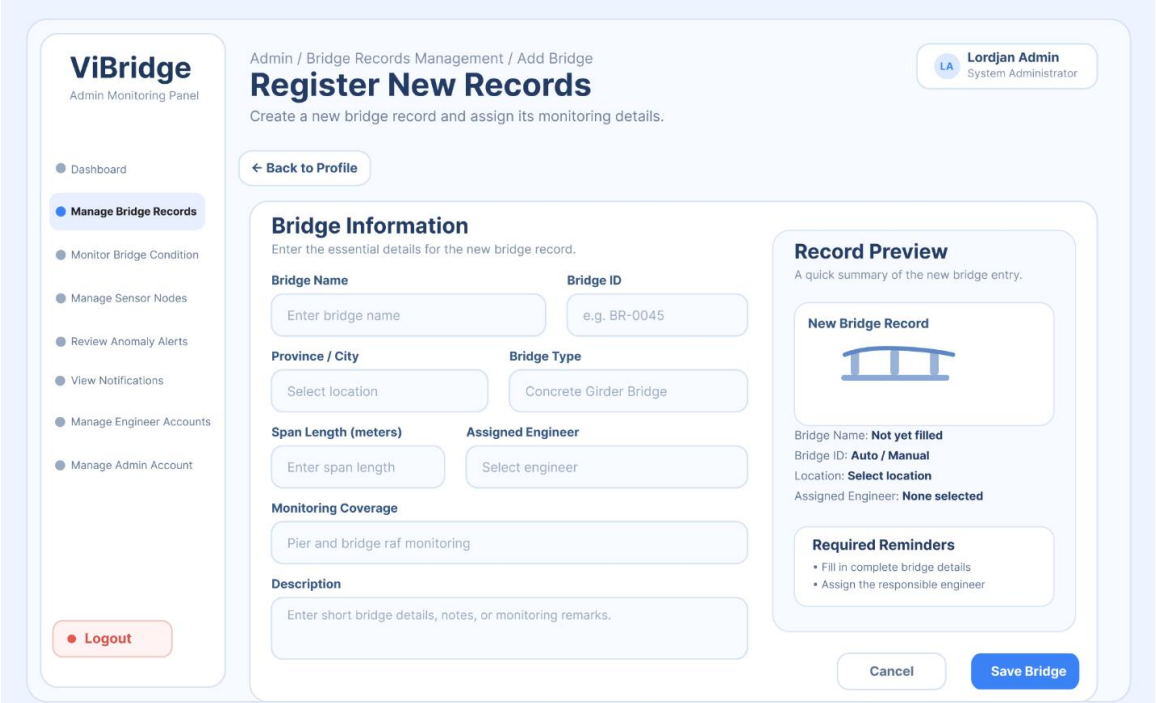


Figure 50: **Web Administration Register New Records Page**

Figure 50 presents the Register New Bridge Records screen, where the administrator can create a new bridge entry in the system. It provides input fields for bridge name, bridge ID, location, type, span length, assigned engineer, monitoring coverage, and description. The preview panel on the right summarizes the entered information, while Save Bridge records the new bridge entry and Cancel stop the process.

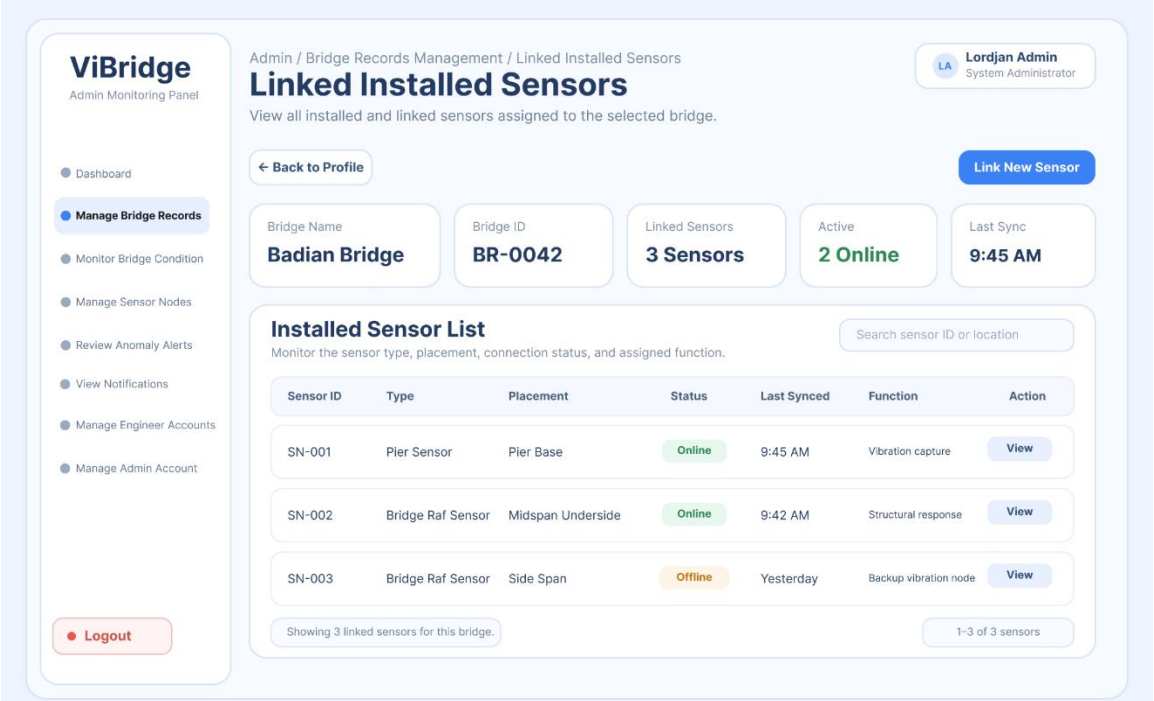


Figure 51: **Web Administration Installed Sensors Page**

Figure 51 presents the Linked Installed Sensors screen, where the administrator can view all sensors linked to the selected bridge. It shows each sensor's ID, type, placement, status, last synced time, and function for easier monitoring and management. The Link New Sensor button allows the administrator to add another sensor, while the View action opens the details of a specific sensor.

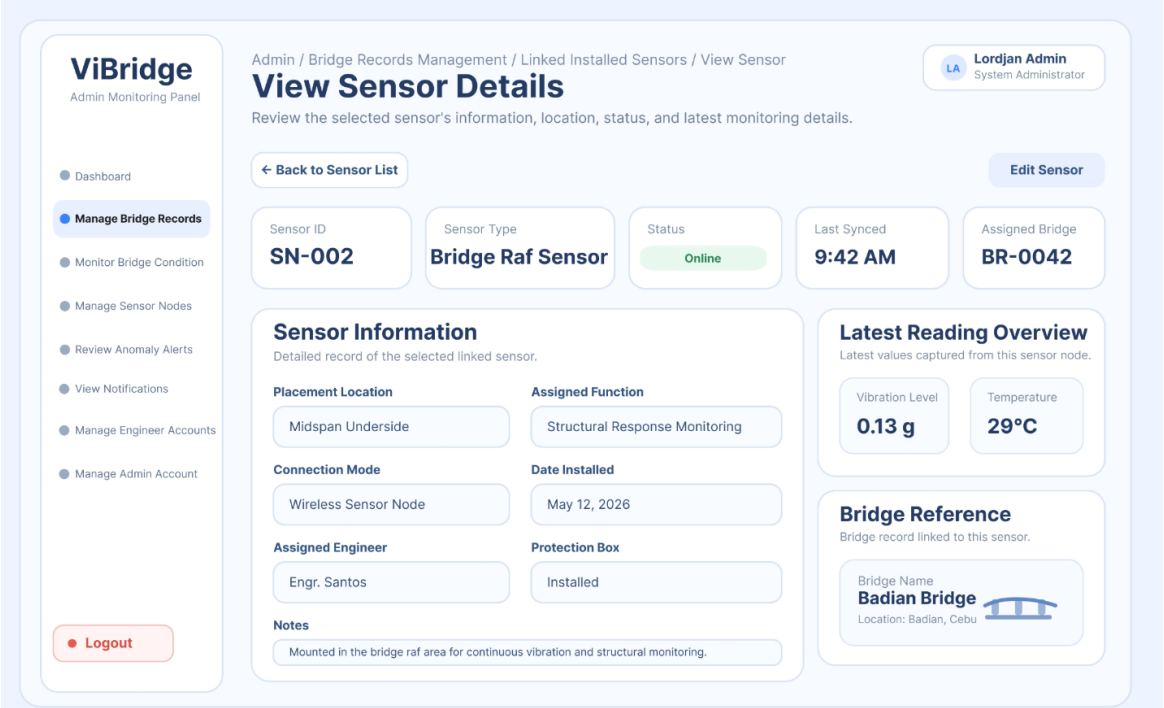


Figure 52: **Web Administration Sensors Details Page**

Figure 52 presents the View Sensor Details screen, where the administrator can review the complete information of a selected linked sensor. It displays the sensor ID, type, status, last synced time, assigned bridge, placement location, function, connection mode, installation date, assigned engineer, and other monitoring details. The Edit Sensor button allows the administrator to update the selected sensor information.

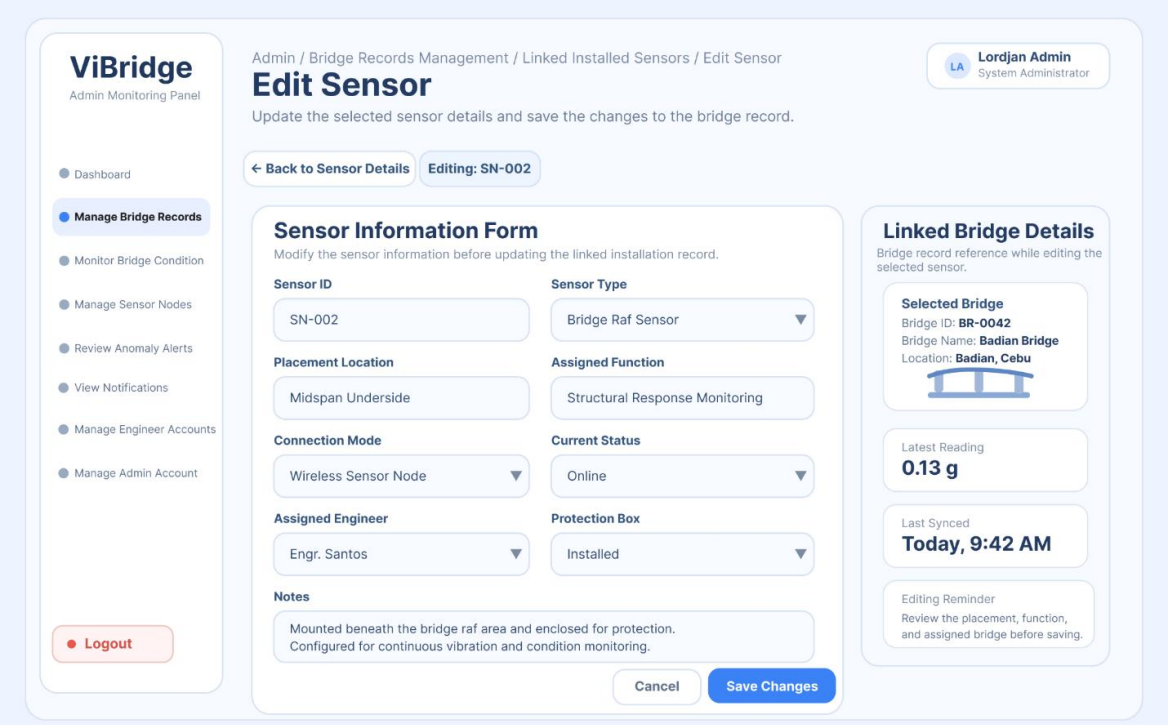


Figure 53: **Web Administration Edit Sensors Page**

Figure 53 presents the Edit Sensor screen, where the administrator can update the details of a selected linked sensor. It provides editable fields for sensor type, placement location, assigned function, connection mode, status, assigned engineer, protection box, and notes. The Cancel button discards changes, while Save Changes updates the sensor information in the system.

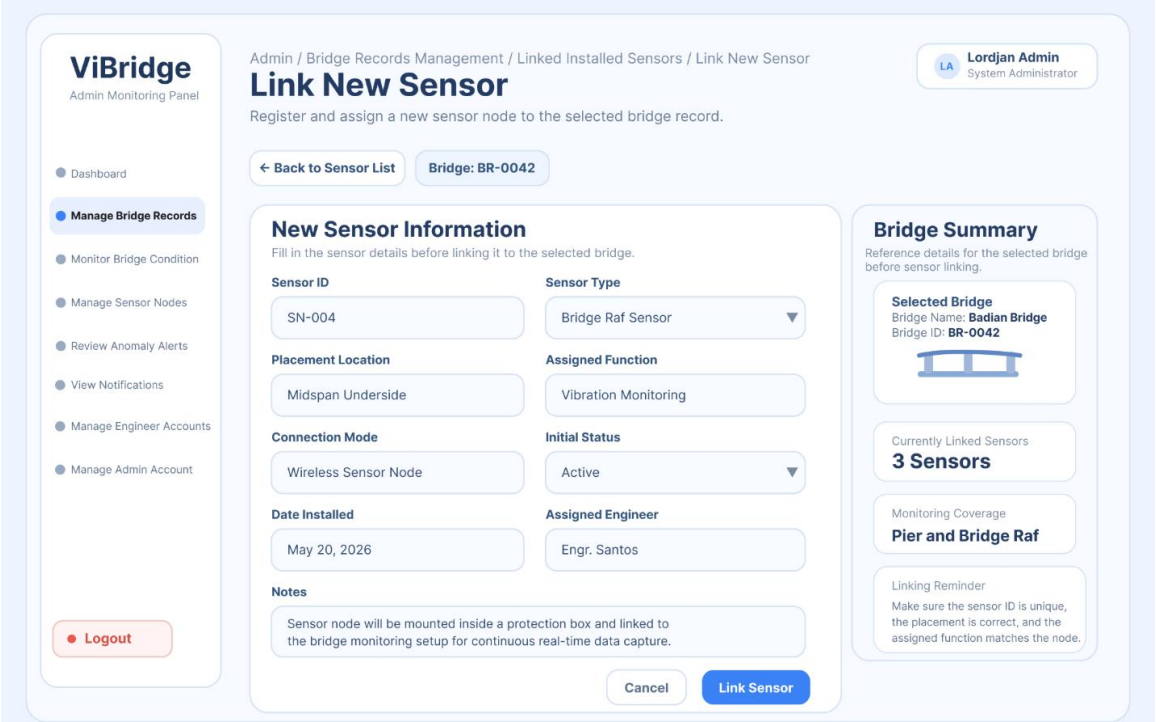


Figure 54: **Web Administration Link New Sensor Page**

Figure 54 presents the Link New Sensor screen, where the administrator can register and assign a new sensor node to the selected bridge. It provides fields for sensor ID, type, placement location, assigned function, connection mode, initial status, installation date, assigned engineer, and notes. The Cancel button stops the process, while Link Sensor saves and links the new sensor to the selected bridge.

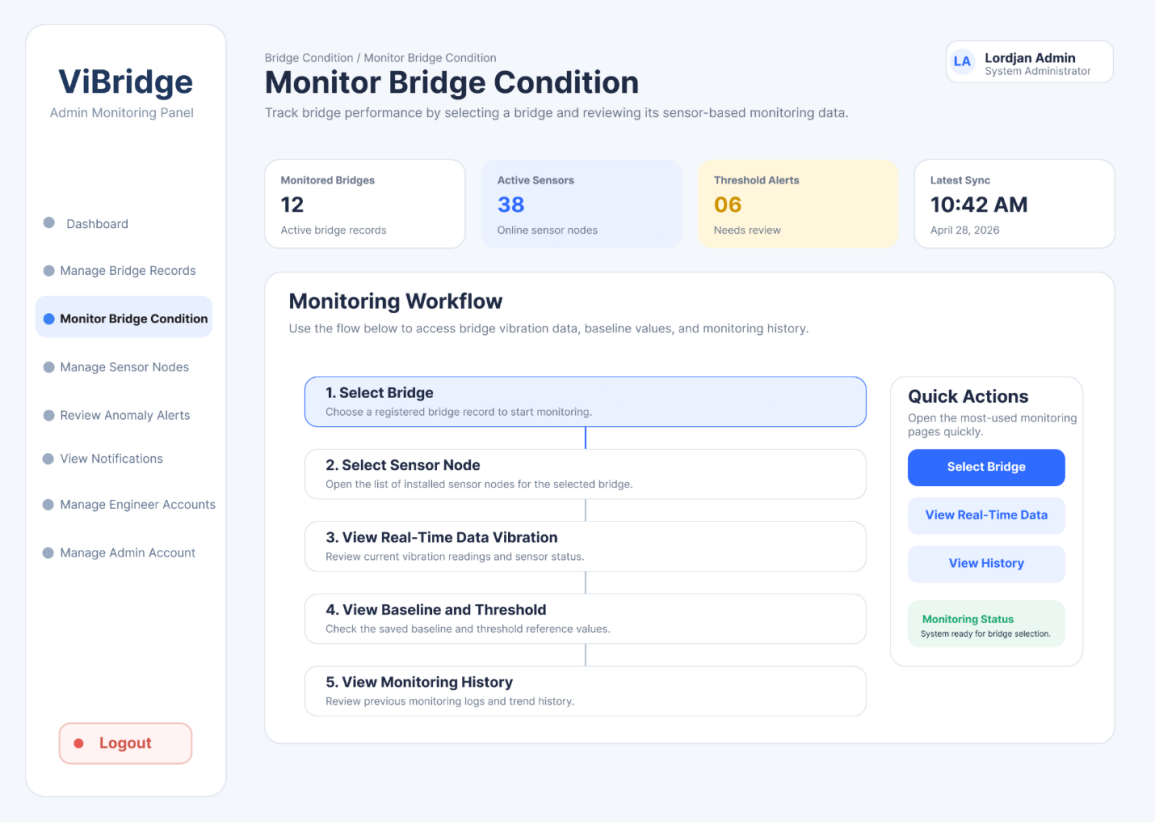


Figure 55: **Web Administration Monitor Bridge Condition Page**

Figure 55 presents the Monitor Bridge Condition screen, where the administrator follows the monitoring workflow for bridge condition assessment. It outlines the steps from selecting a bridge and sensor node up to viewing real-time vibration data, baseline and threshold values, and monitoring history. The quick action buttons on the right also allow the administrator to open key monitoring pages directly.

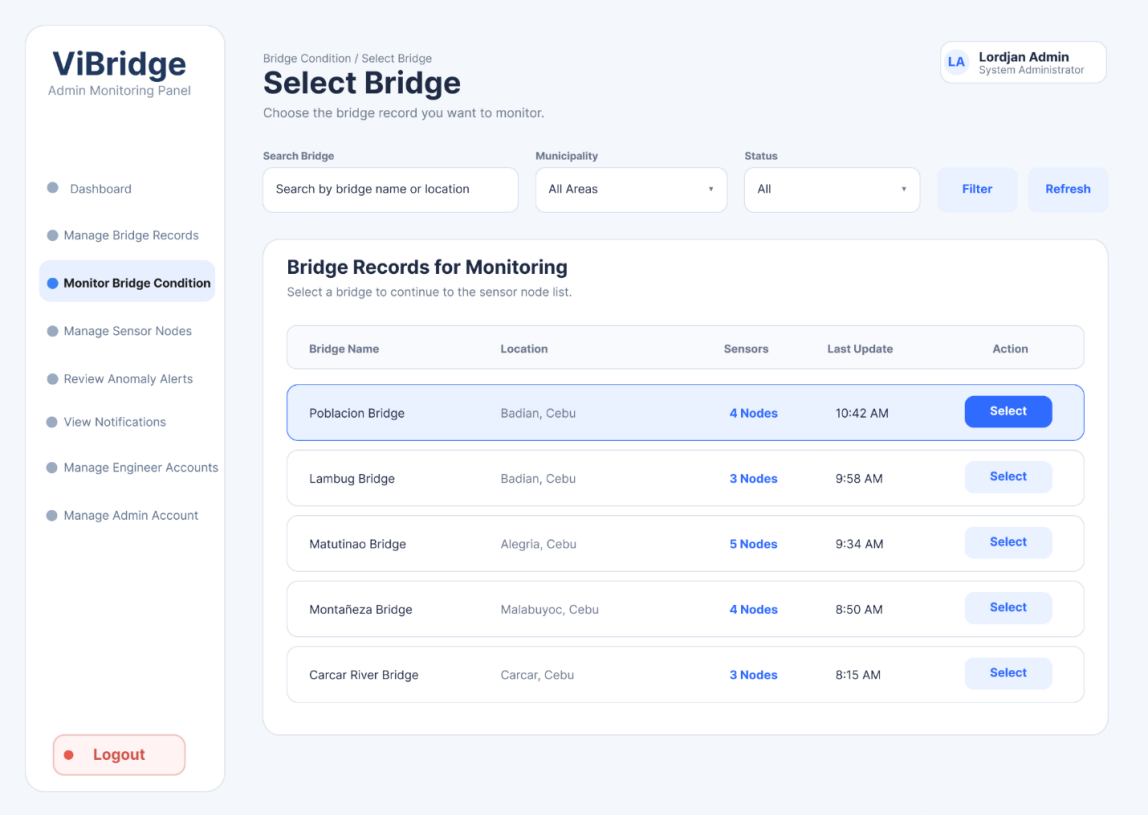


Figure 56: **Web Administration Select Bridge Page**

Figure 56 presents the Select Bridge screen, where the administrator chooses the bridge record to be monitored. It provides search, municipality, and status filters to help narrow the bridge list. The Select button allows the administrator to open the chosen bridge and continue to the next monitoring step.

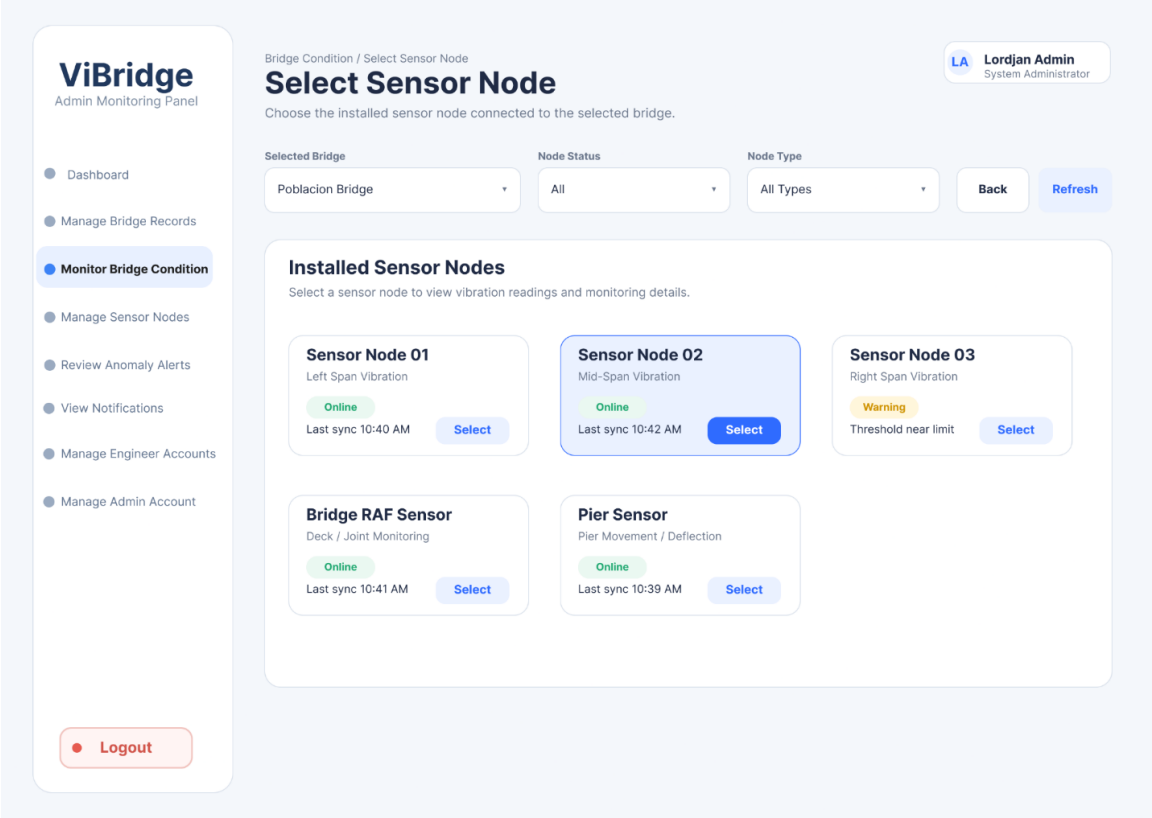


Figure 57: **Web Administration Select Sensor Node Page**

Figure 57 presents the Select Sensor Node screen, where the administrator chooses the installed sensor node connected to the selected bridge. It shows the available sensor nodes together with their type, current status, and last sync time. The Select button allows the administrator to proceed with the chosen sensor node for bridge condition monitoring.

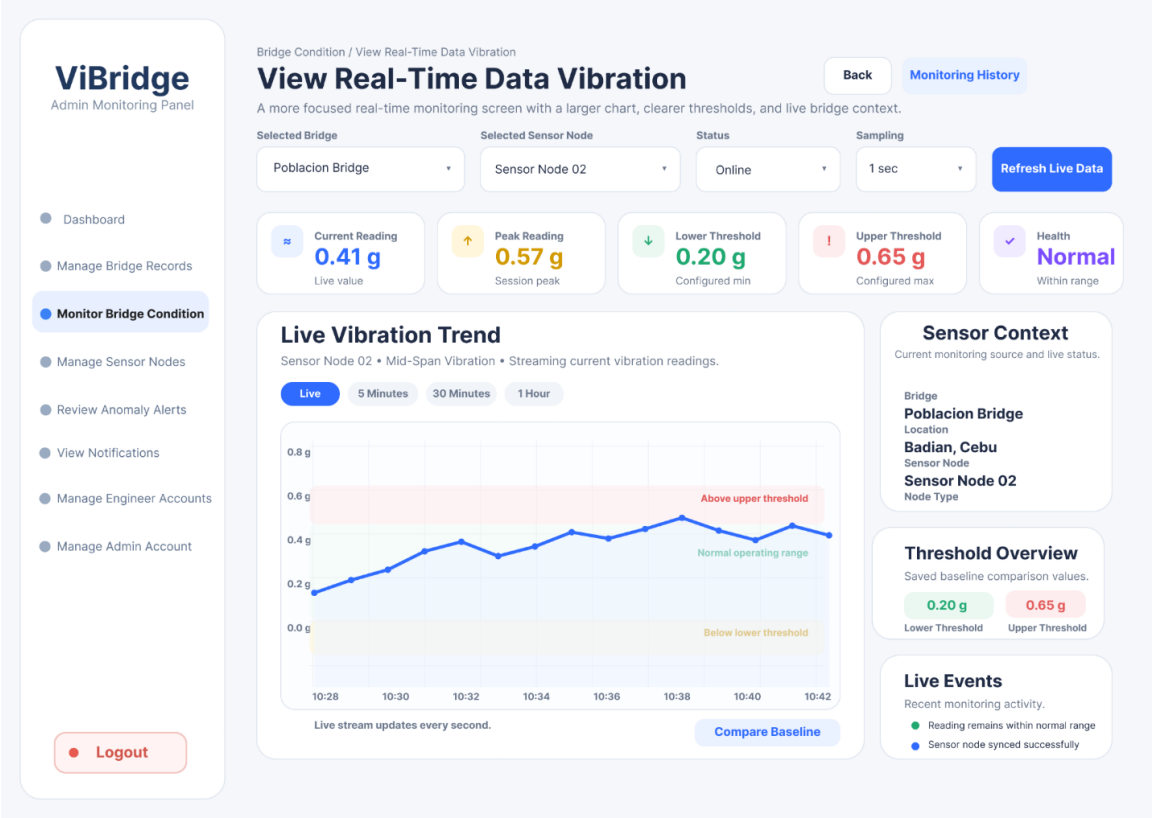


Figure 58: **Web Administration Real Time Data Vibration Page**

Figure 58 presents the View Real-Time Data Vibration screen, where the administrator can monitor the live vibration reading of the selected sensor node. It displays the current reading, peak reading, threshold values, health status, and a live vibration trend graph. The side panels also show the sensor context, threshold overview, and recent monitoring events to support real-time assessment.

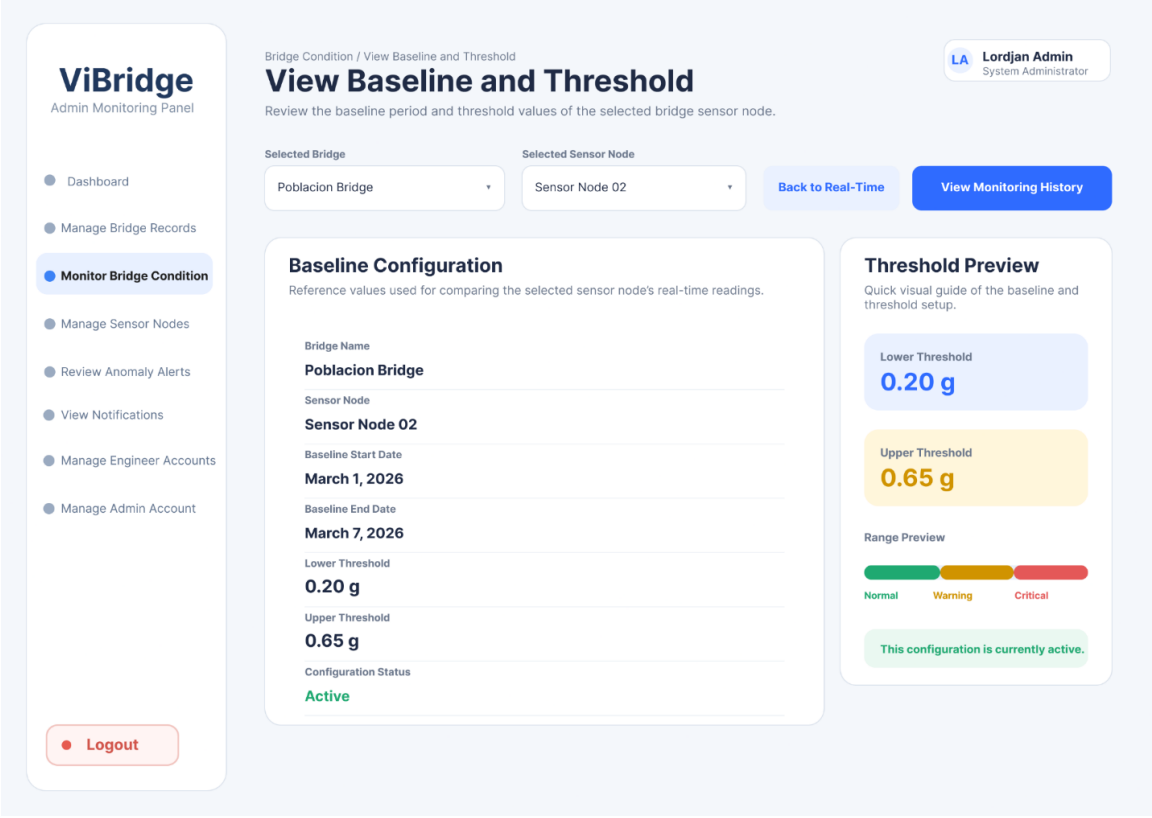


Figure 59: **Web Administration Baseline and Threshold Page**

Figure 59 presents the View Baseline and Threshold screen, where the administrator can review the saved baseline period and threshold values of the selected bridge sensor node. It shows the bridge name, sensor node, baseline start and end dates, lower threshold, upper threshold, and configuration status. The threshold preview panel also provides a quick visual guide of the monitoring range.

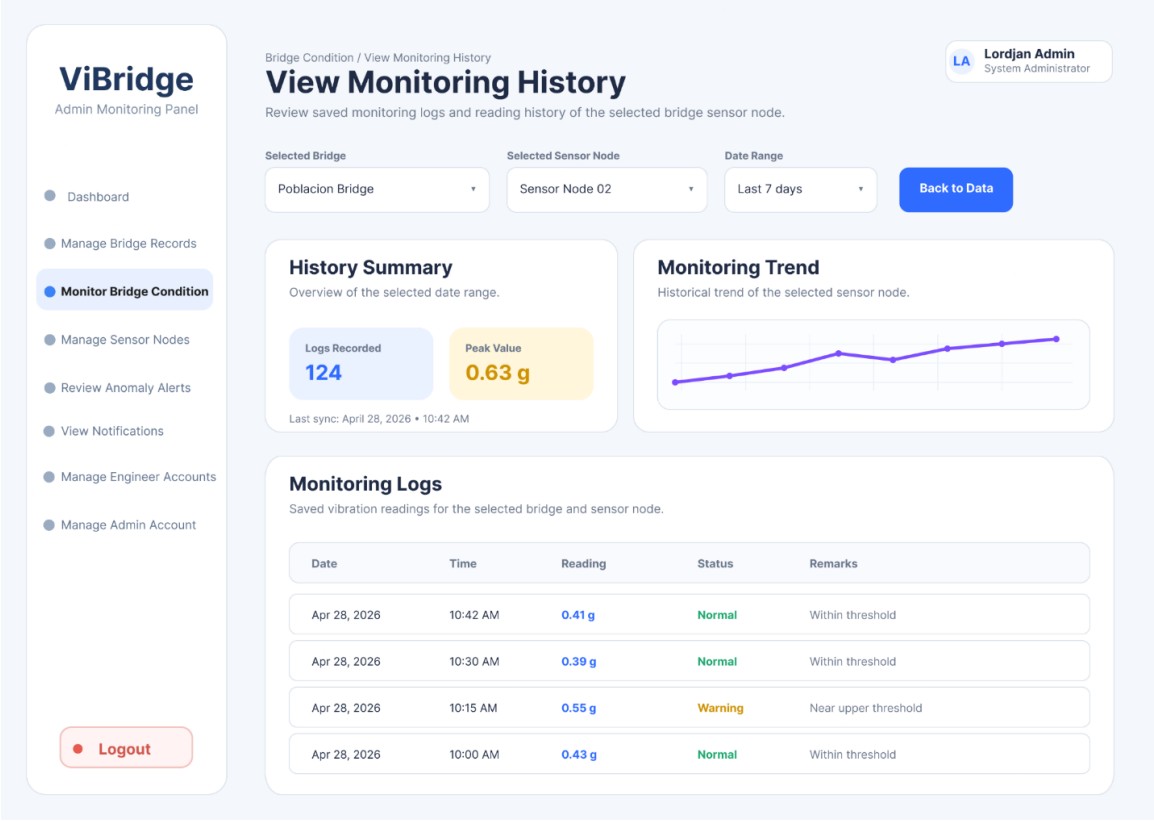


Figure 60: **Web Administration Monitoring History Page**

Figure 60 presents the View Monitoring History screen, where the administrator can review saved monitoring logs and historical vibration readings of the selected bridge sensor node. It shows a summary of recorded logs, peak value, a monitoring trend graph, and a detailed log table with date, time, reading, status, and remarks. This screen helps the administrator assess past bridge condition records for monitoring review.

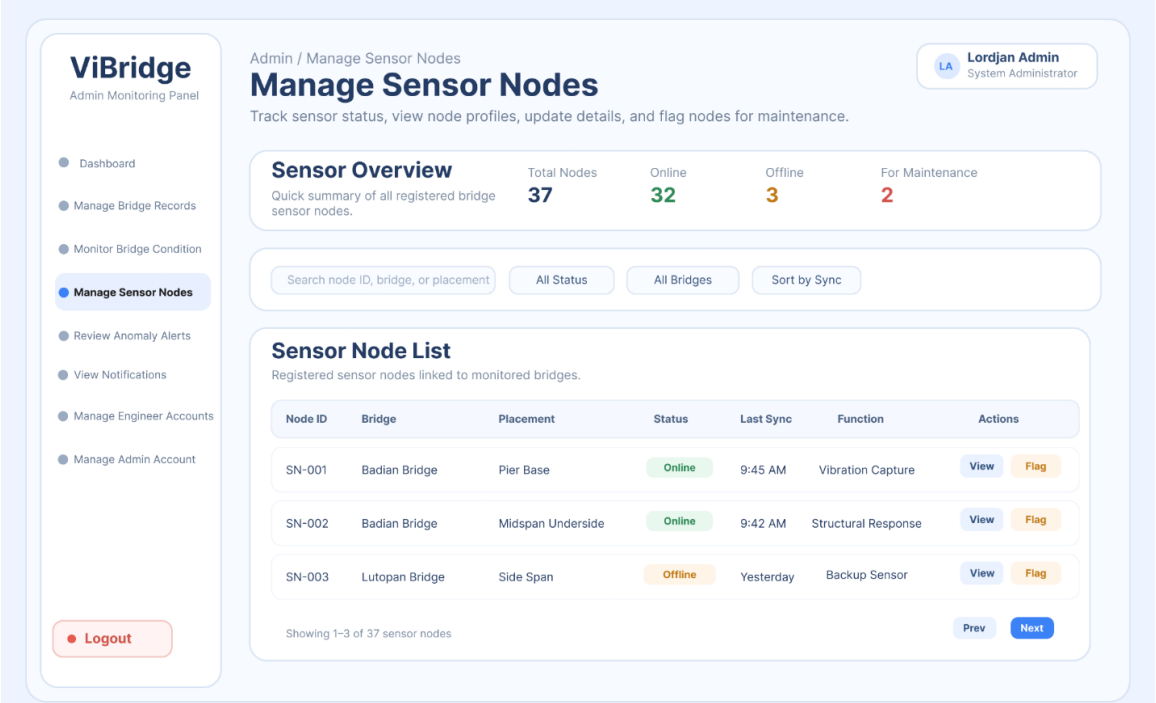


Figure 61: **Web Administration Manage Sensor Nodes Page**

Figure 61 presents the Manage Sensor Nodes screen, where the administrator can monitor and manage all registered sensor nodes linked to bridges. It provides a summary of total, online, offline, and maintenance-related nodes, together with search, filter, and sorting controls. The sensor node list also shows each node's bridge, placement, status, last sync, function, and available actions such as View and Flag.



Figure 62: **Web Administration View Sensor Node Profile Page**

Figure 62 presents the View Sensor Node Profile screen, where the administrator can review the complete information and current condition of the selected sensor node. It shows the node profile details, configuration values, performance snapshot, and notes or remarks. The available actions also allow the administrator to edit the sensor, open its configuration, or flag it for maintenance.

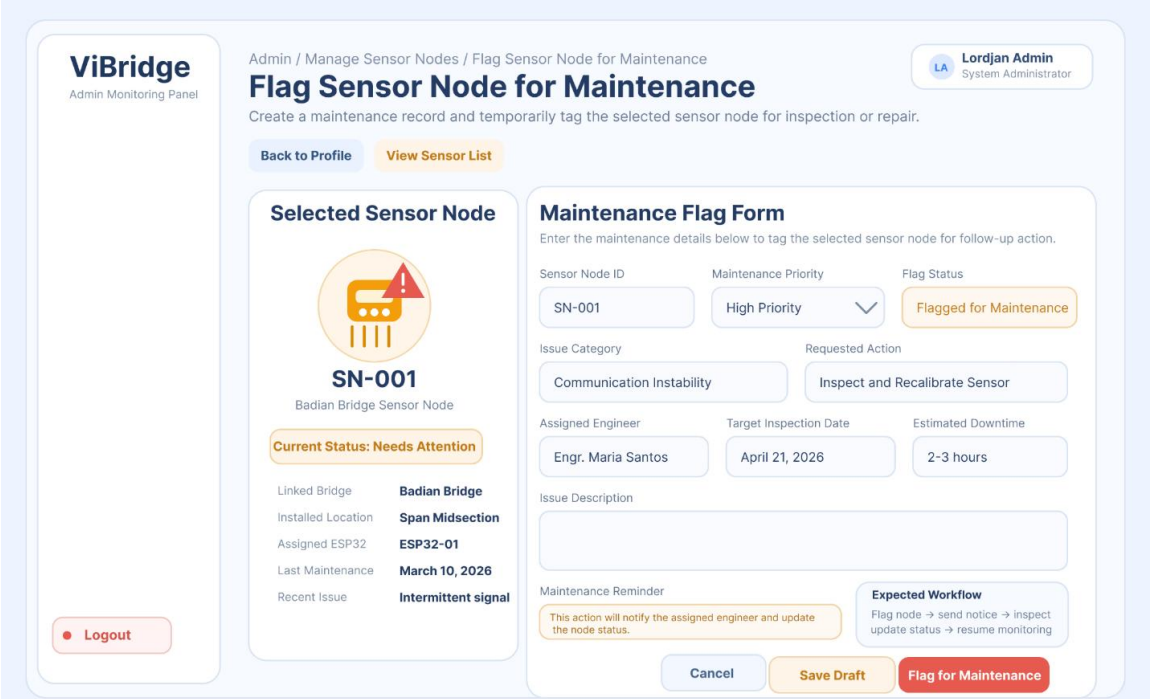


Figure 63: **Web Administration Flag Sensor Node for Maintenance Page**

Figure 63 presents the Flag Sensor Node for Maintenance screen, where the administrator creates a maintenance record for the selected sensor node. It shows the node details and provides fields for priority, issue category, requested action, assigned engineer, target inspection date, estimated downtime, and issue description. The Save Draft button stores the maintenance entry temporarily, while Flag for Maintenance submits it for follow-up action.

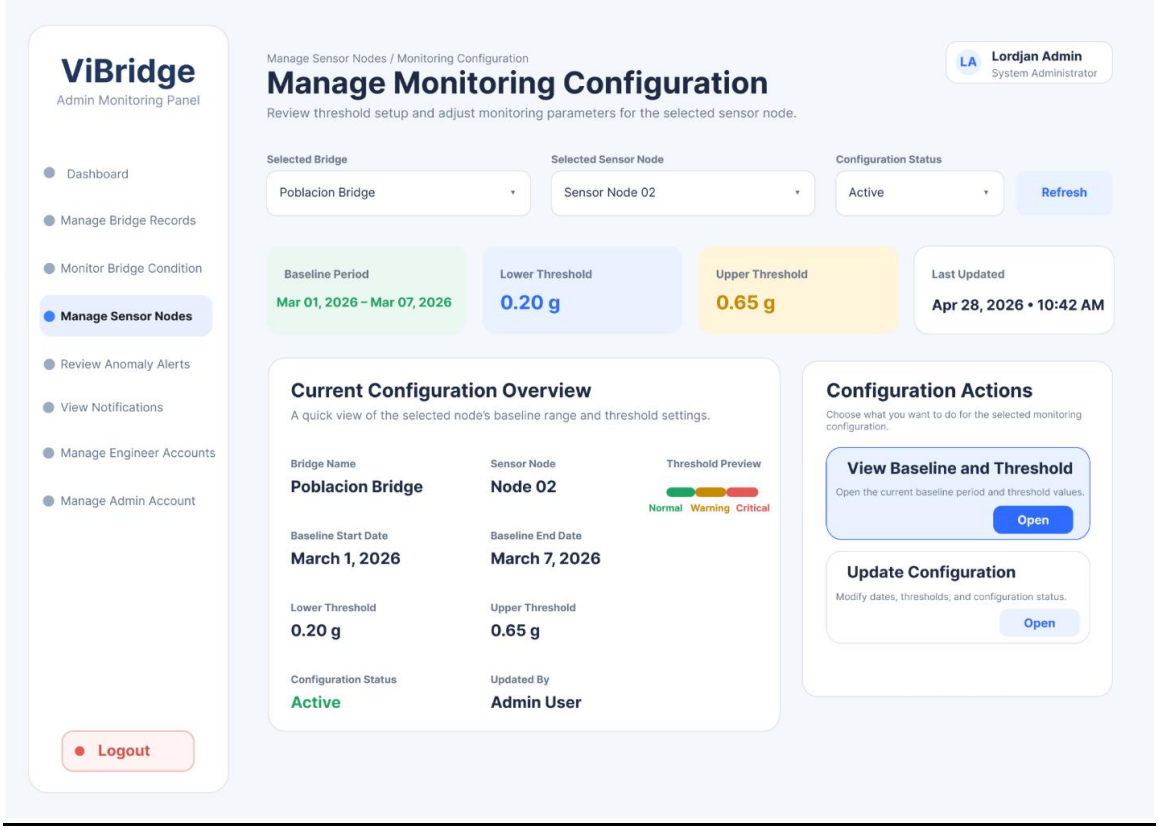


Figure 64: **Web Administration Manage Monitoring Configuration Page**

Figure 64 presents the Manage Monitoring Configuration screen, where the administrator can review and manage the monitoring setup of the selected sensor node. It shows the baseline period, lower and upper threshold values, configuration status, and last updated details. The available actions allow the administrator to open the baseline and threshold view or proceed to update the configuration.

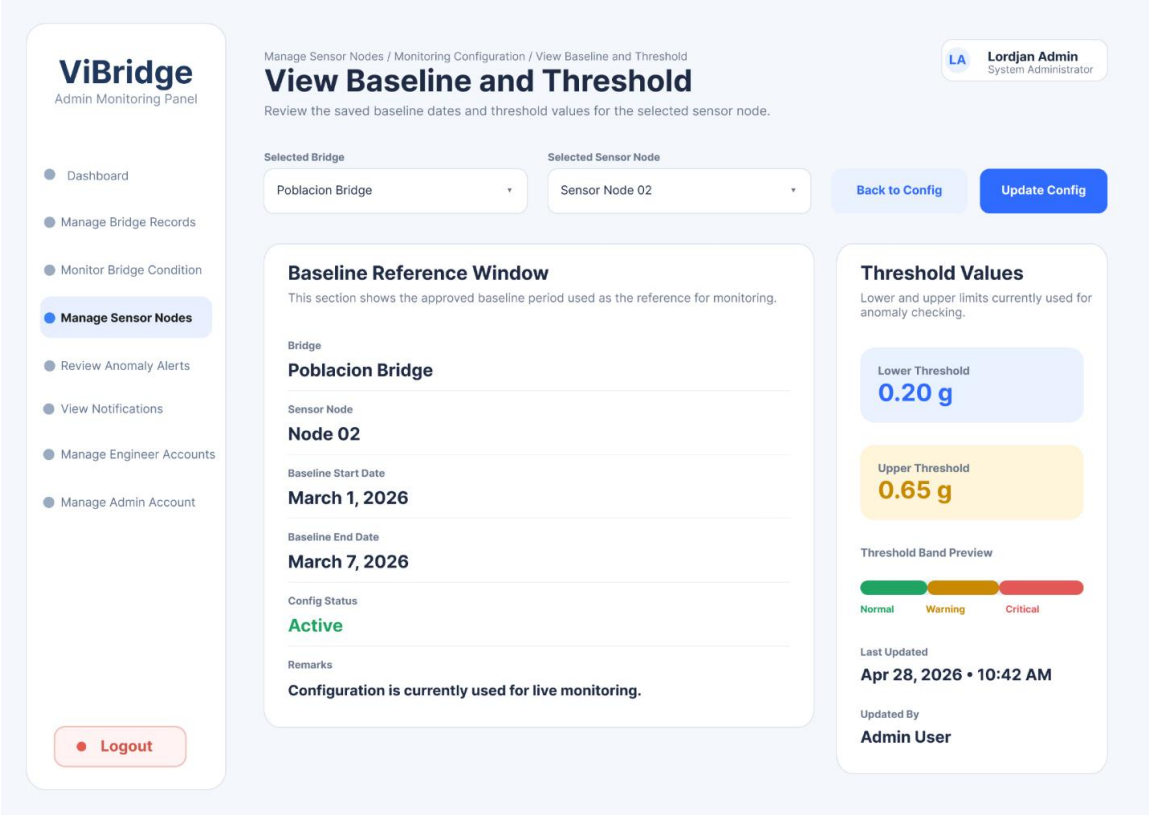


Figure 65: **Web Administration View Baseline and Threshold Page**

Figure 65 presents the View Baseline and Threshold screen, where the administrator can review the saved baseline dates and threshold values of the selected sensor node. It shows the baseline reference window, lower and upper threshold values, configuration status, remarks, and update details. The Back to Config and Update Config buttons allow the administrator to return or proceed to configuration editing.

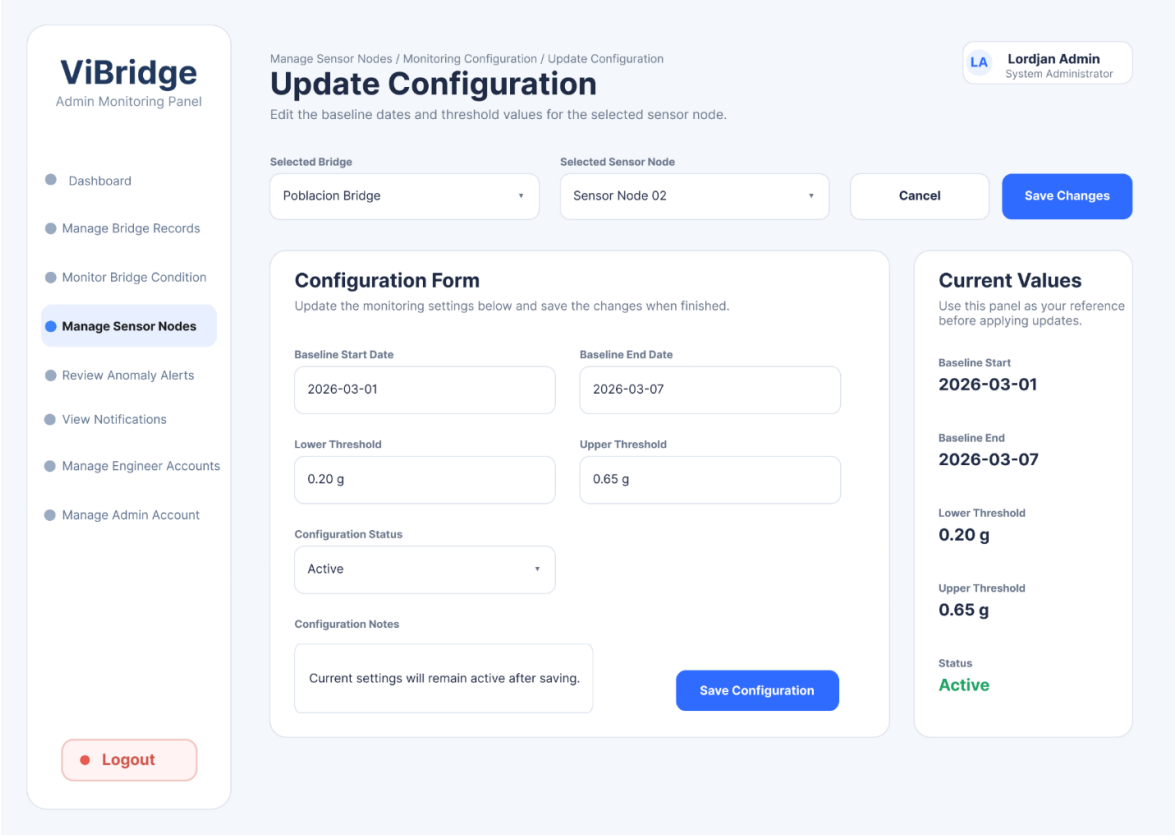


Figure 66: **Web Administration Update Configuration Page**

Figure 66 presents the Update Configuration screen, where the administrator can edit the baseline dates and threshold values of the selected sensor node. It provides a configuration form for updating the baseline period, lower threshold, upper threshold, and configuration status. The Cancel, Save Changes, and Save Configuration actions allow the administrator to discard or apply the updated monitoring settings.

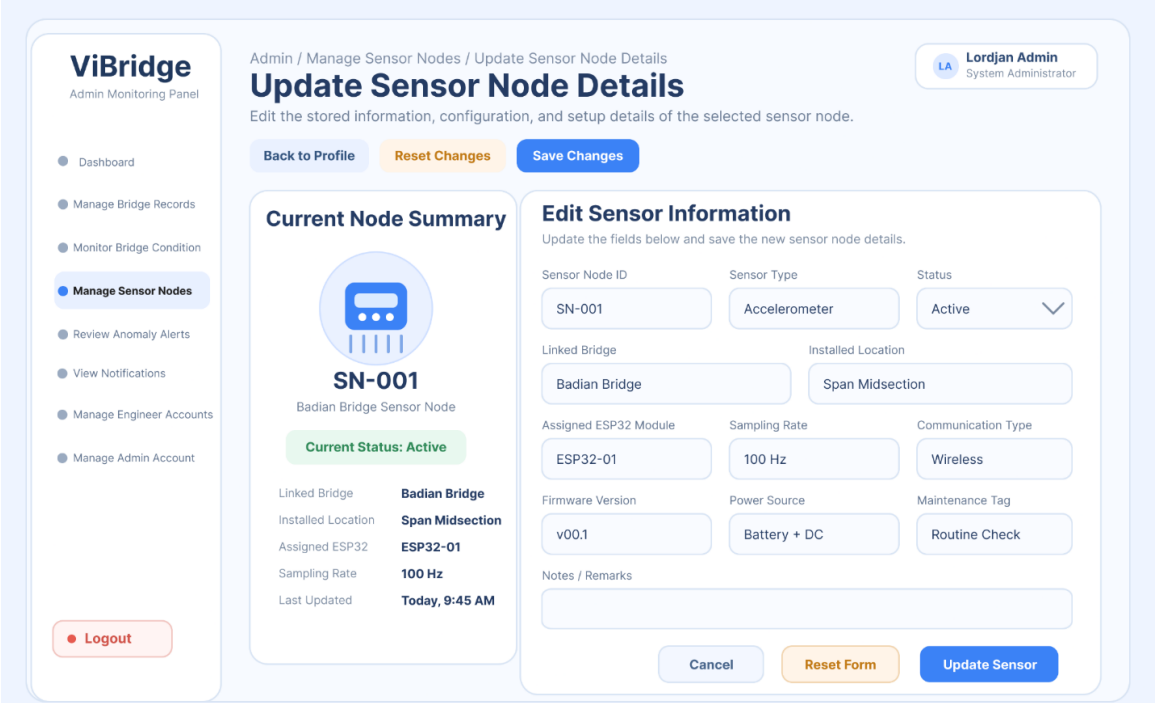


Figure 67: **Web Administration Update Sensor Node Details Page**

Figure 67 presents the Update Sensor Node Details screen, where the administrator can edit the stored information and setup details of the selected sensor node. It provides fields for sensor type, status, linked bridge, installed location, ESP32 module, sampling rate, communication type, firmware version, power source, maintenance tag, and notes. The available actions allow the administrator to cancel, reset, or save the updated sensor node details.

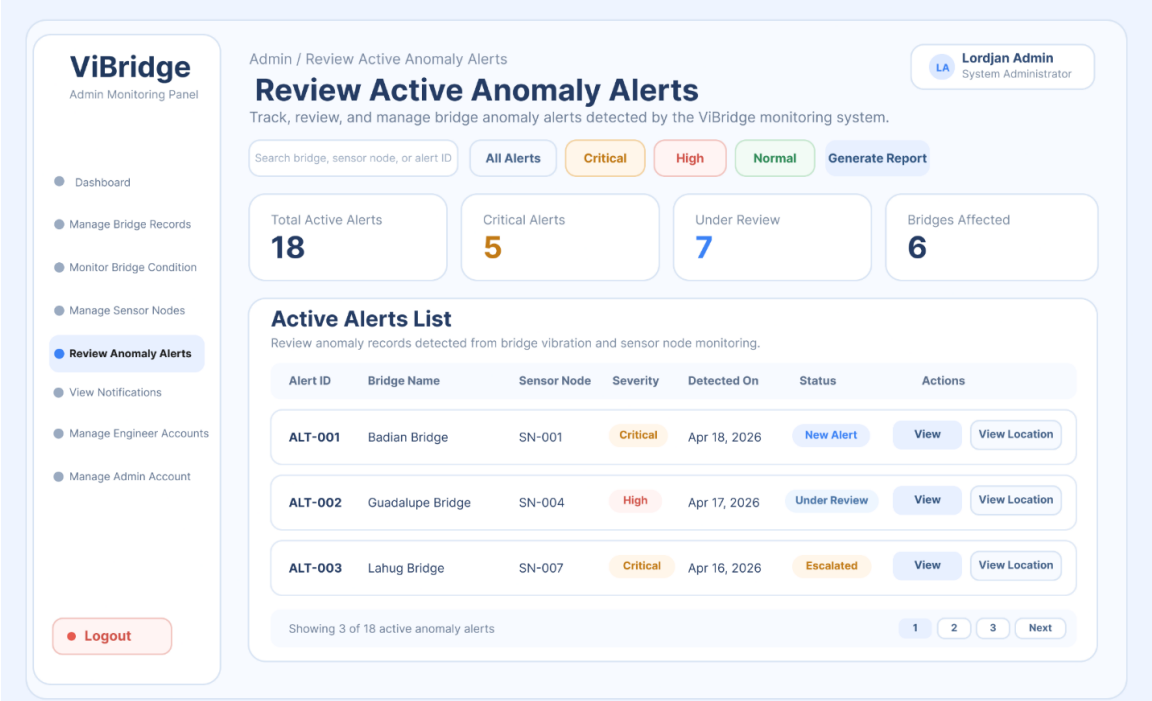


Figure 68: **Web Administration Review Active Anomaly Alerts Page**

Figure 68 presents the Review Active Anomaly Alerts screen, where the administrator can track and manage anomaly alerts detected by the system. It shows alert filters, summary counts, and an active alerts list with severity, status, and affected bridge details. The View and View Location actions allow the administrator to inspect each alert further.

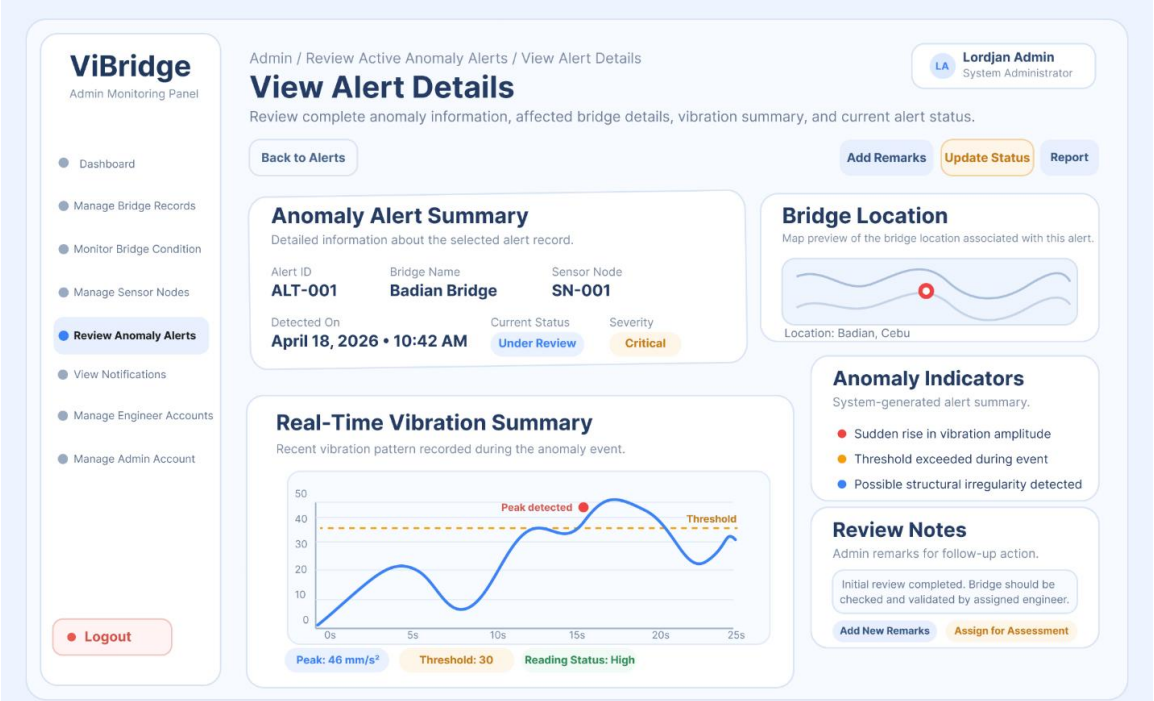


Figure 69: **Web Administration View Alert Details Page**

Figure 69 presents the View Alert Details screen, where the administrator can review the complete information of a selected anomaly alert. It shows the alert summary, bridge location, vibration summary, anomaly indicators, and review notes. The available actions also allow the administrator to add remarks, update the alert status, generate a report, or assign the case for assessment.

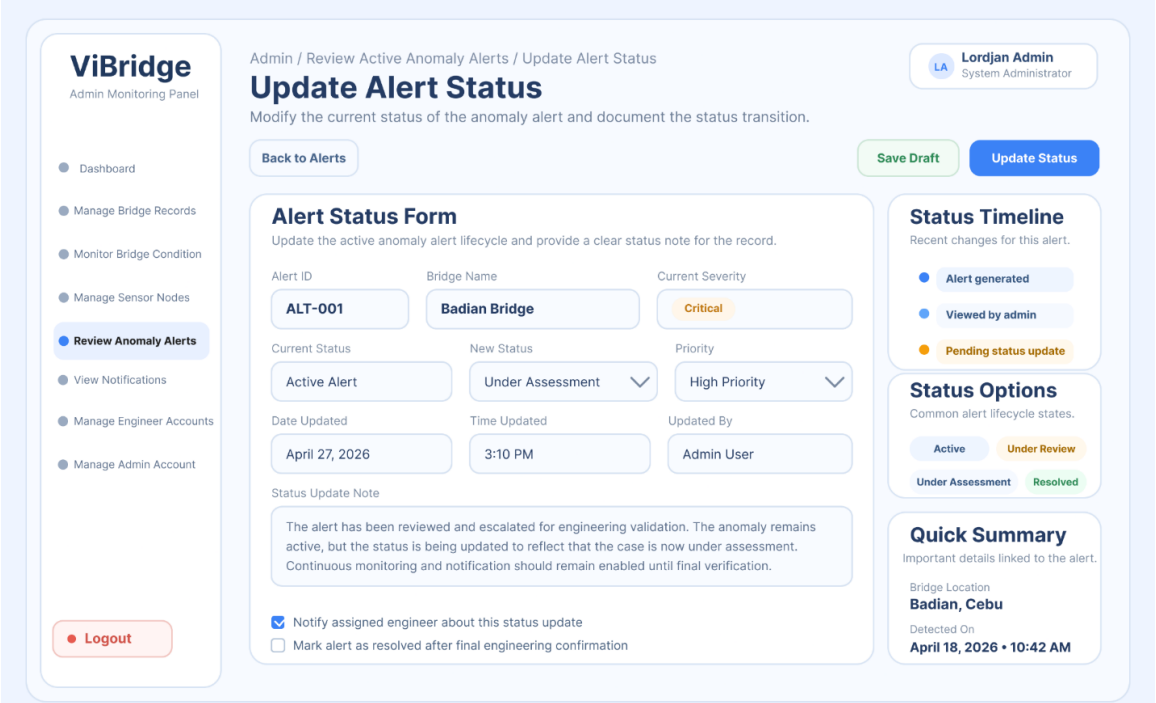


Figure 70: **Web Administration Update Alert Status Page**

Figure 70 presents the Update Alert Status screen, where the administrator can modify the current status of a selected anomaly alert. It provides fields for the new status, priority, update date and time, responsible user, and status note, together with timeline and summary panels. The Save Draft button stores the update temporarily, while Update Status applies the new alert status.

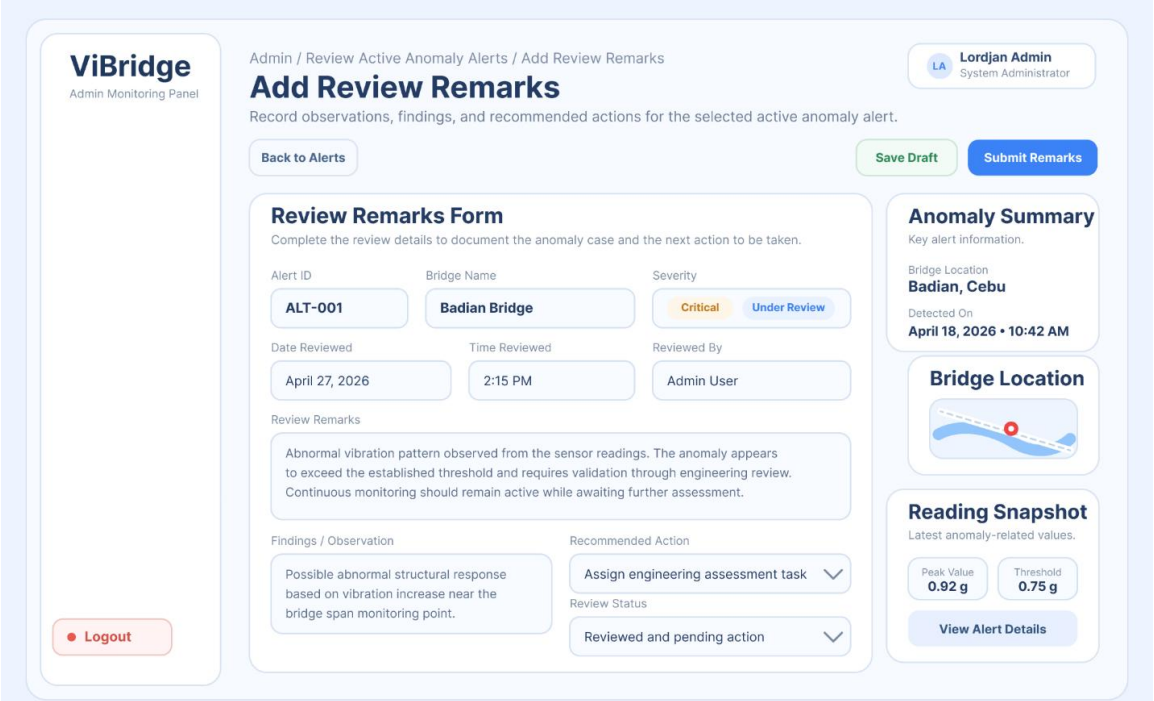


Figure 71: **Web Administration Update Review Remarks Page**

Figure 71 presents the Generate Anomaly Reports screen, where the administrator prepares a formal anomaly report for documentation and monitoring decisions. It provides options for configuring report details, date range, output format, and included report contents. The Save Draft button stores the report temporarily, while Generate Report creates the final anomaly report.

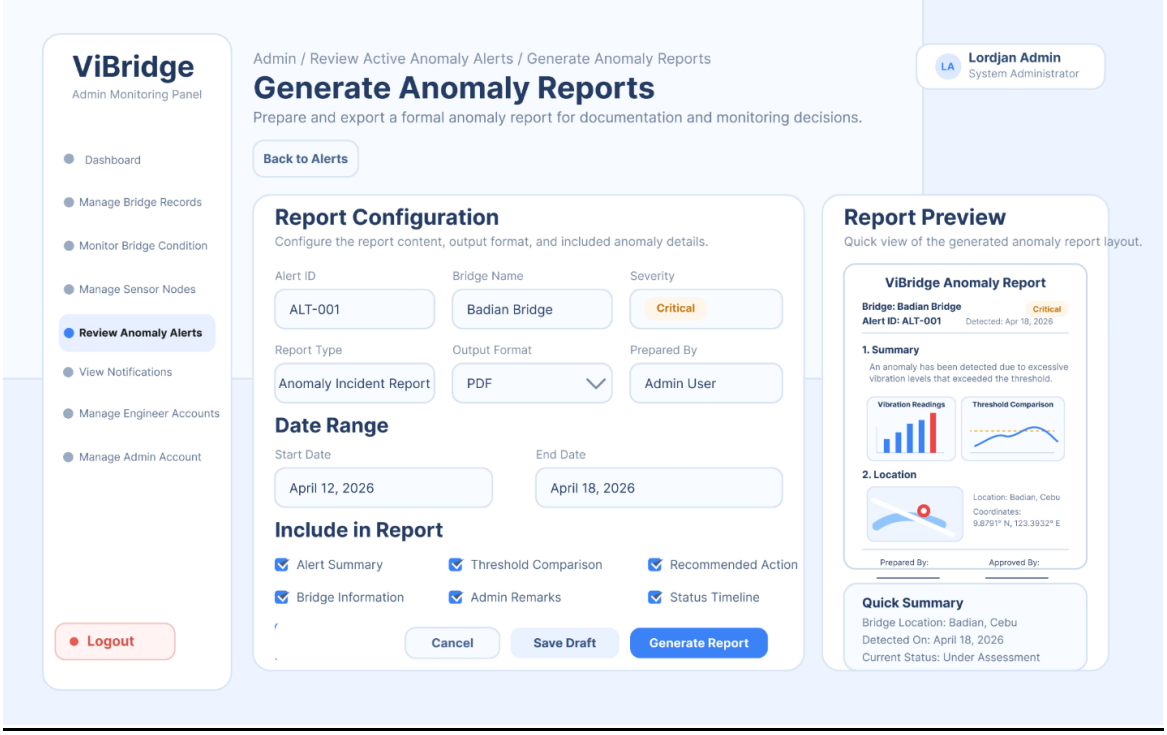


Figure 72: **Web Administration Generate Anomaly Reports Page**

Figure 72 presents the Report Preview section, where the administrator can review the layout and included contents of the anomaly report before final generation. It shows a preview of the summary, charts, location, and quick report details. This section helps confirm that the report is complete before clicking Generate Report.

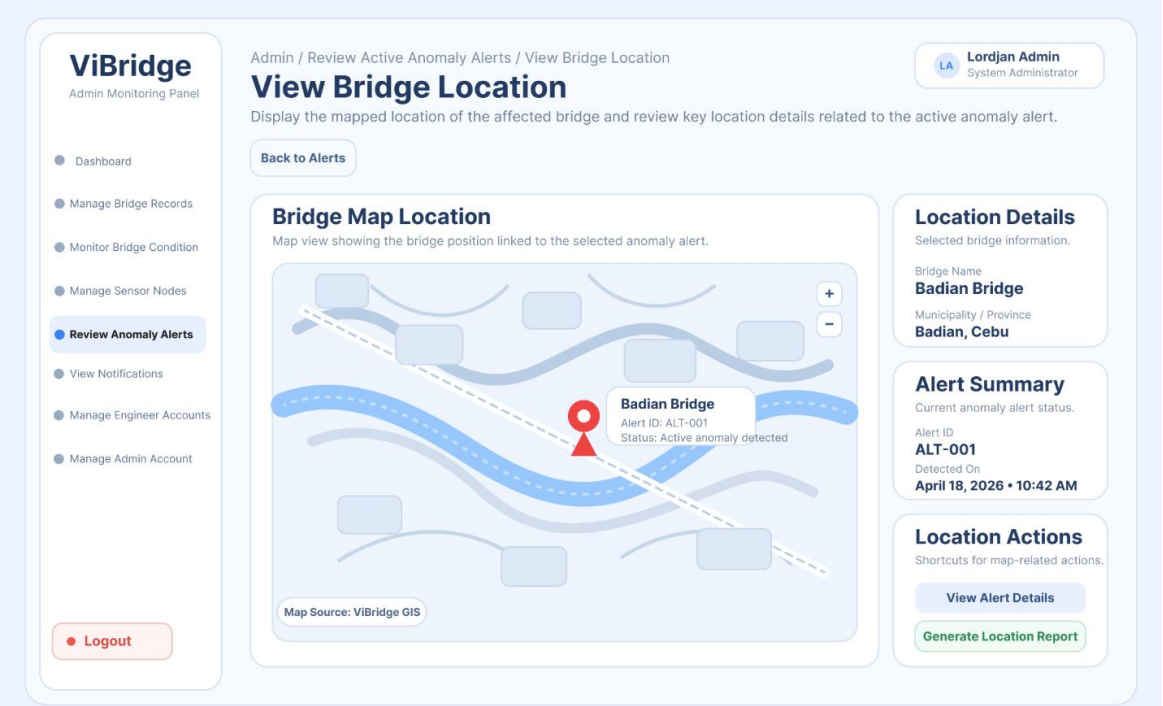


Figure 73: **Web Administration View Bridge Location Page**

Figure 73 presents the View Bridge Location screen, where the administrator can view the mapped location of the affected bridge linked to the anomaly alert. It shows the bridge map, location details, and alert summary for quick reference. The available actions also allow the administrator to return to the alert details or generate a location report.

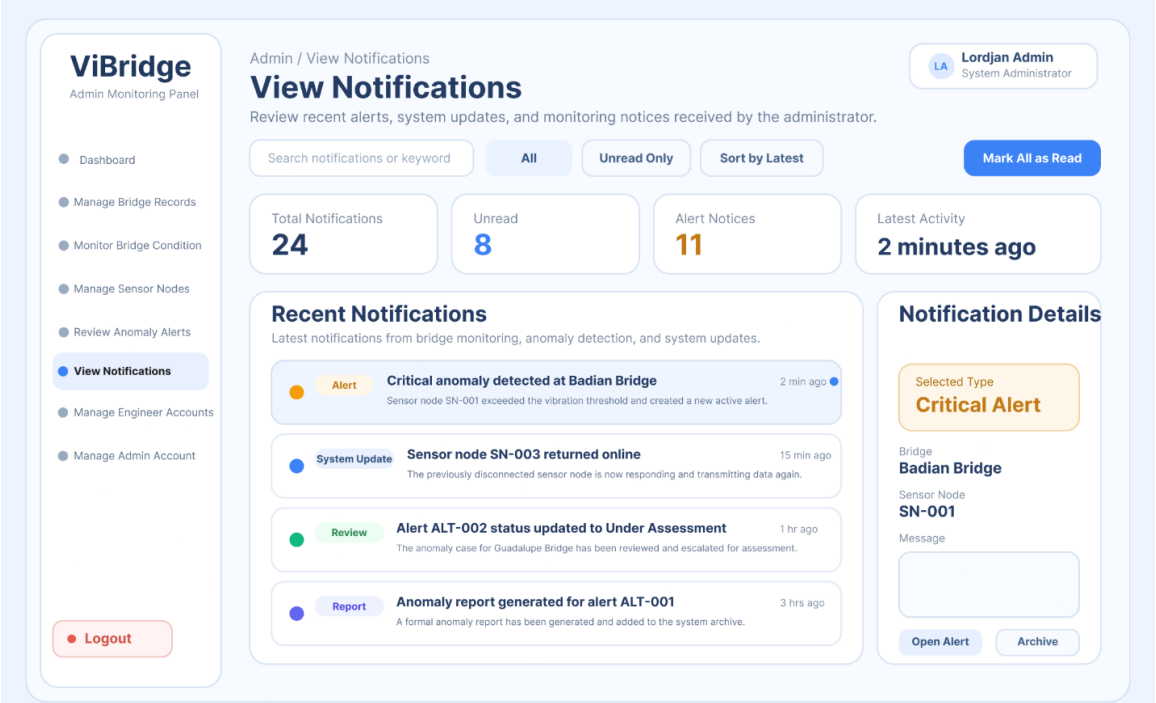


Figure 74: **Web Administration View Notifications Page**

Figure 74 presents the View Notifications screen, where the administrator can review recent alerts, system updates, and monitoring notices. It provides search, filter, and sorting options together with a notification list and selected notification details. The available actions also allow the administrator to mark notifications as read, open the related alert, or archive the notice.

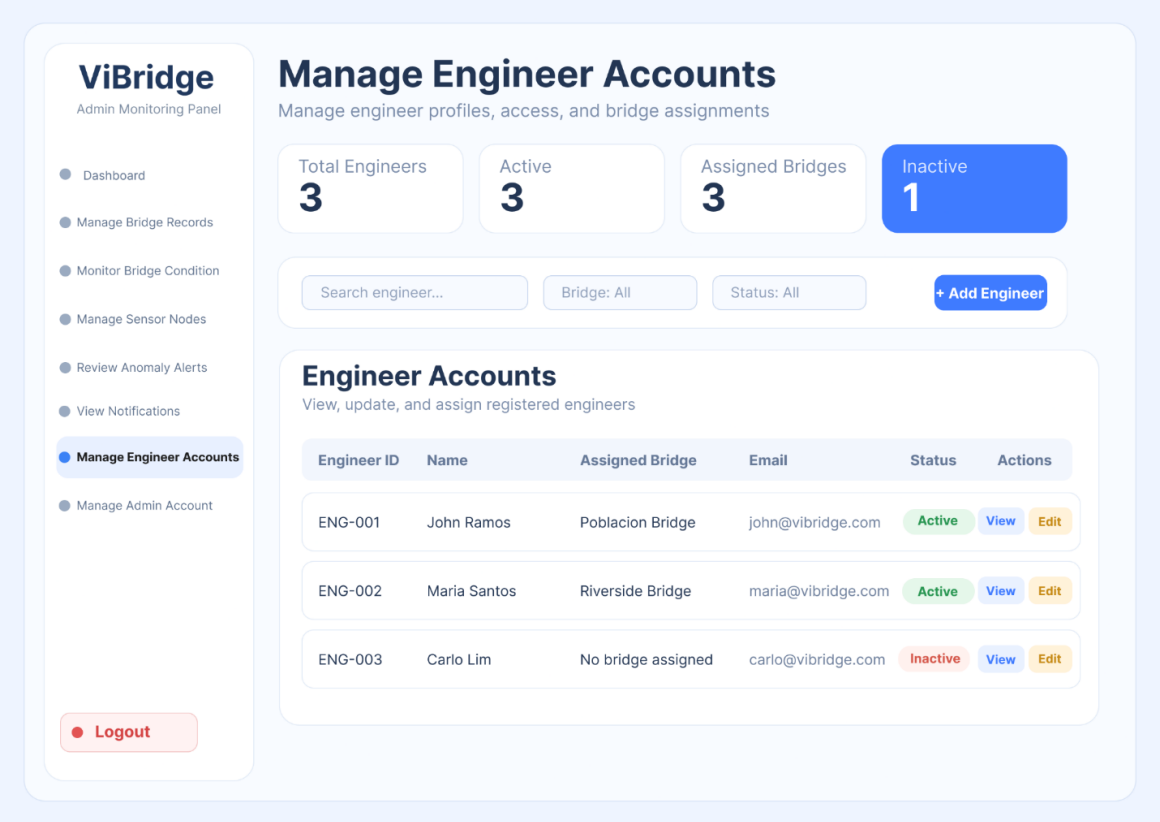


Figure 75: **Web Administration Manage Engineer Accounts Page**

Figure 75 presents the Manage Engineer Accounts screen, where the administrator can manage registered engineer profiles, access, and bridge assignments. It shows engineer account summaries, search and filter controls, and a table of engineer records with status and assigned bridge information. The available actions allow the administrator to add, view, or edit engineer accounts.

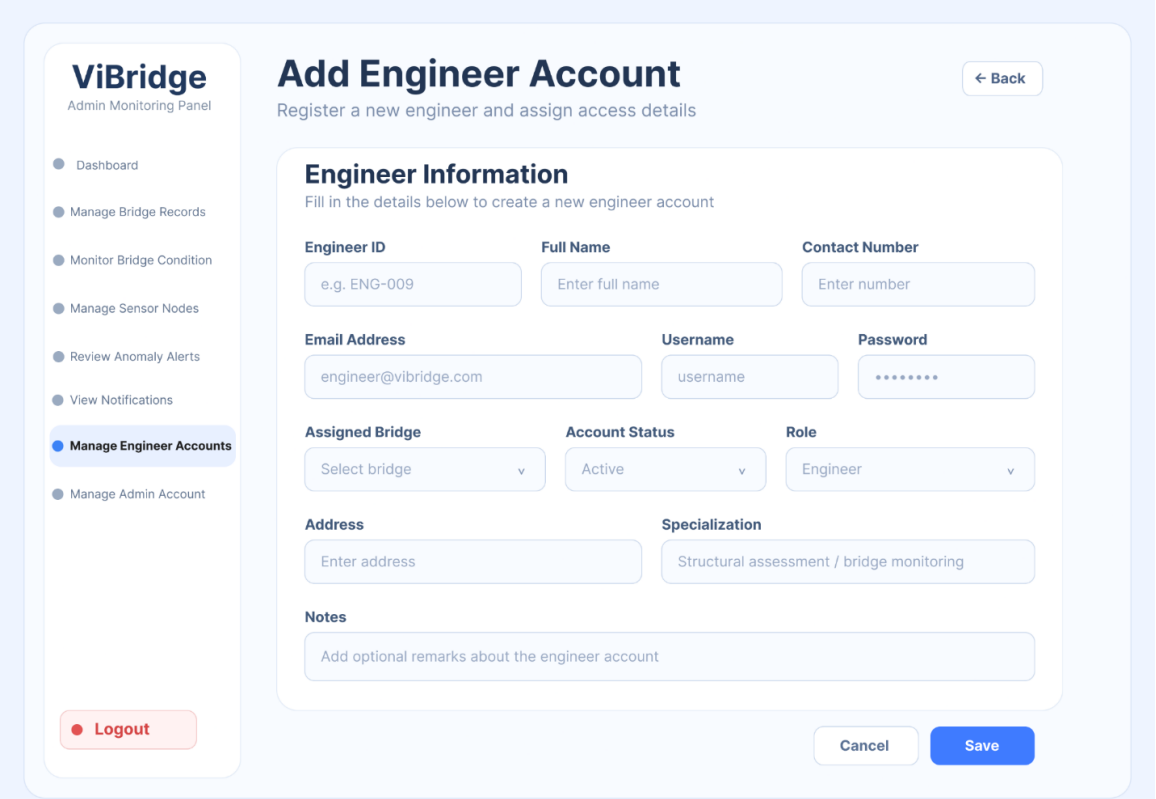


Figure 76: **Web Administration Add Engineer Account Page**

Figure 76 presents the Add Engineer Account screen, where the administrator can register a new engineer and assign access details. It provides input fields for engineer ID, name, contact information, username, password, assigned bridge, role, and other account details. The Save button creates the new engineer account, while Cancel stops the process.

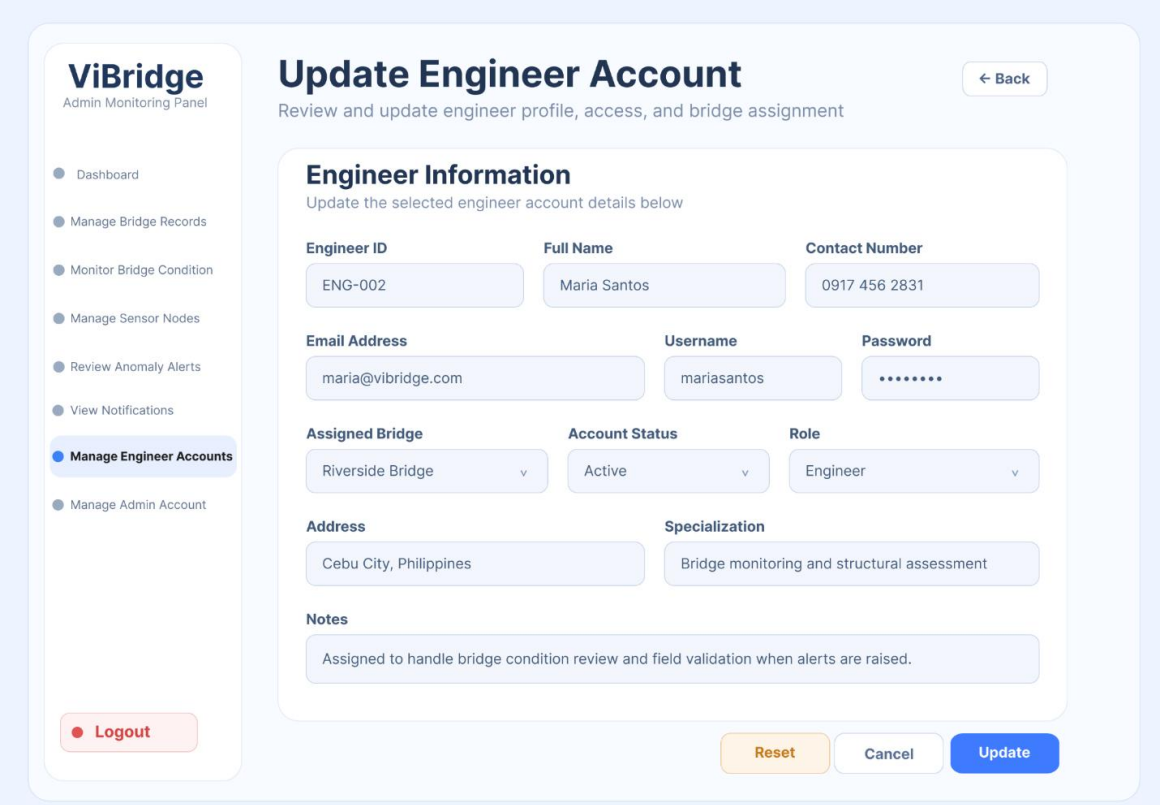


Figure 77: **Web Administration Update Engineer Account Page**

Figure 77 presents the Update Engineer Account screen, where the administrator can review and modify the selected engineer's account details. It includes fields for personal information, contact details, assigned bridge, account status, role, specialization, and notes. The **Reset**, **Cancel**, and **Update** buttons allow the administrator to manage the changes before saving.

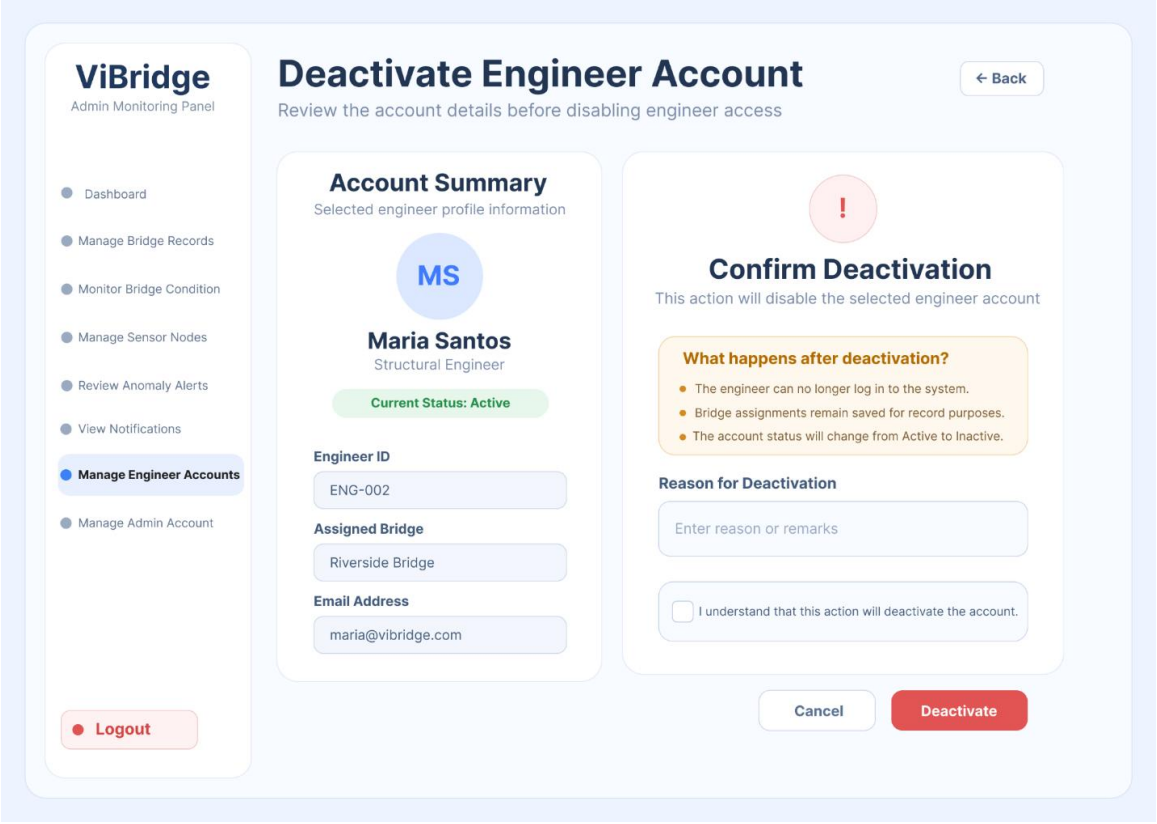


Figure 78: **Web Administration Deactivate Engineer Account Page**

Figure 78 presents the Deactivate Engineer Account screen, where the administrator reviews the selected engineer's account before disabling access. It shows the account summary and a confirmation section explaining the effect of deactivation. The Deactivate button disables the account, while Cancel stops the process.

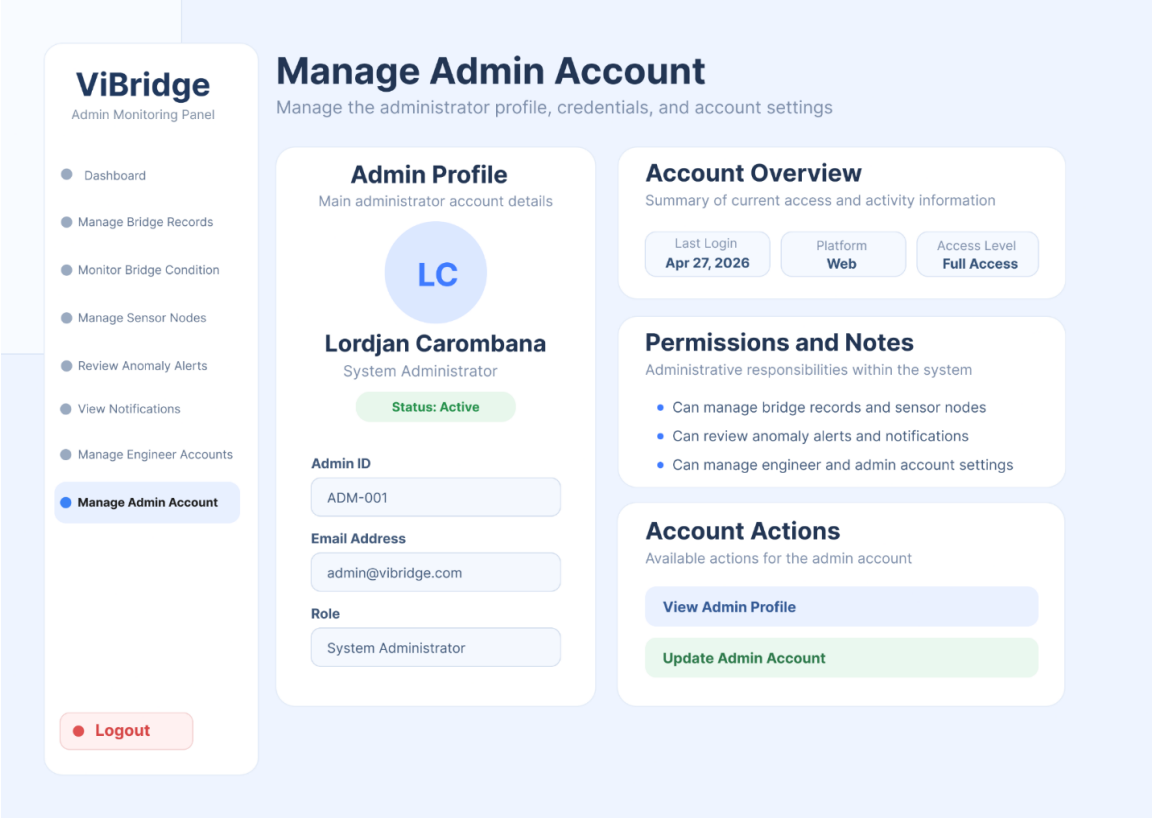


Figure 79: **Web Administration Manage Admin Account Page**

Figure 79 presents the Manage Admin Account screen, where the administrator can view and manage the main admin profile and account settings. It shows the admin profile, account overview, permissions, and available account actions. The provided actions allow the administrator to open the admin profile or proceed to account updating.

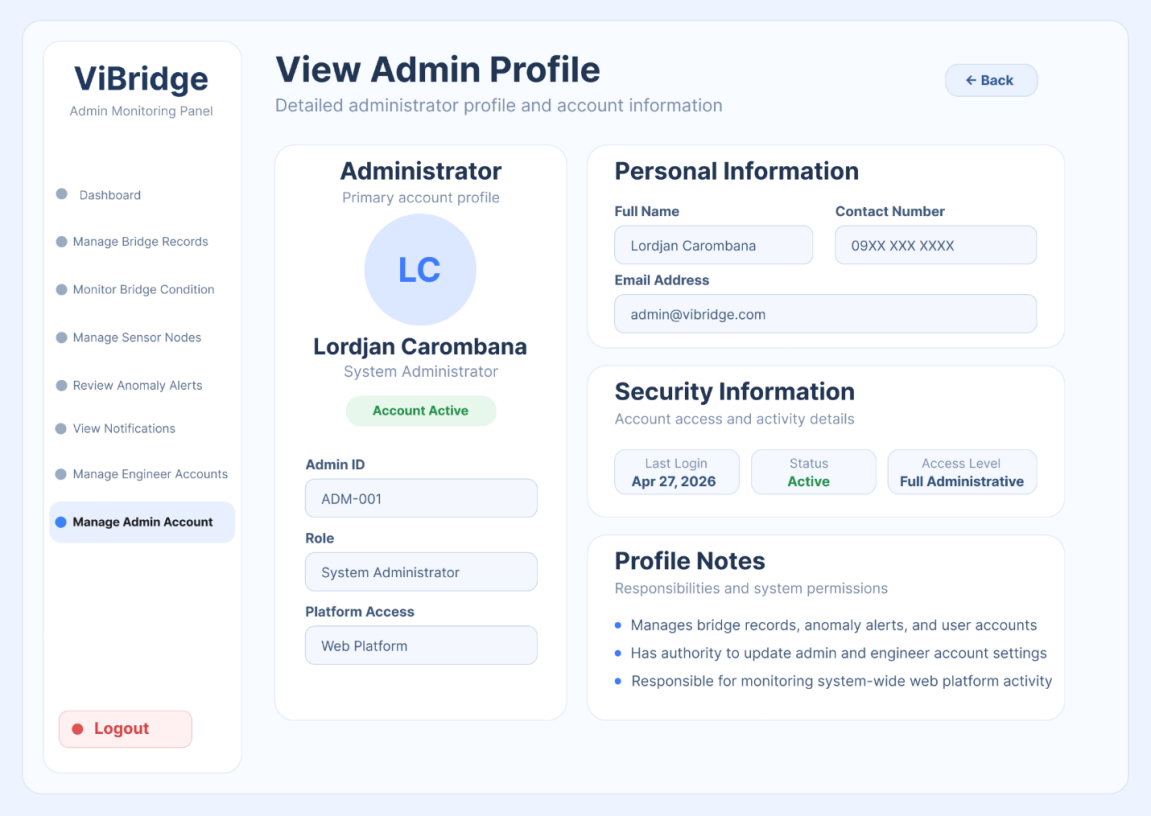


Figure 80: **Web Administration View Admin Profile Page**

Figure 80 presents the View Admin Profile screen, where the administrator can review detailed profile and account information. It shows personal information, security details, account status, access level, and profile notes. This screen provides a complete view of the administrator's account information.

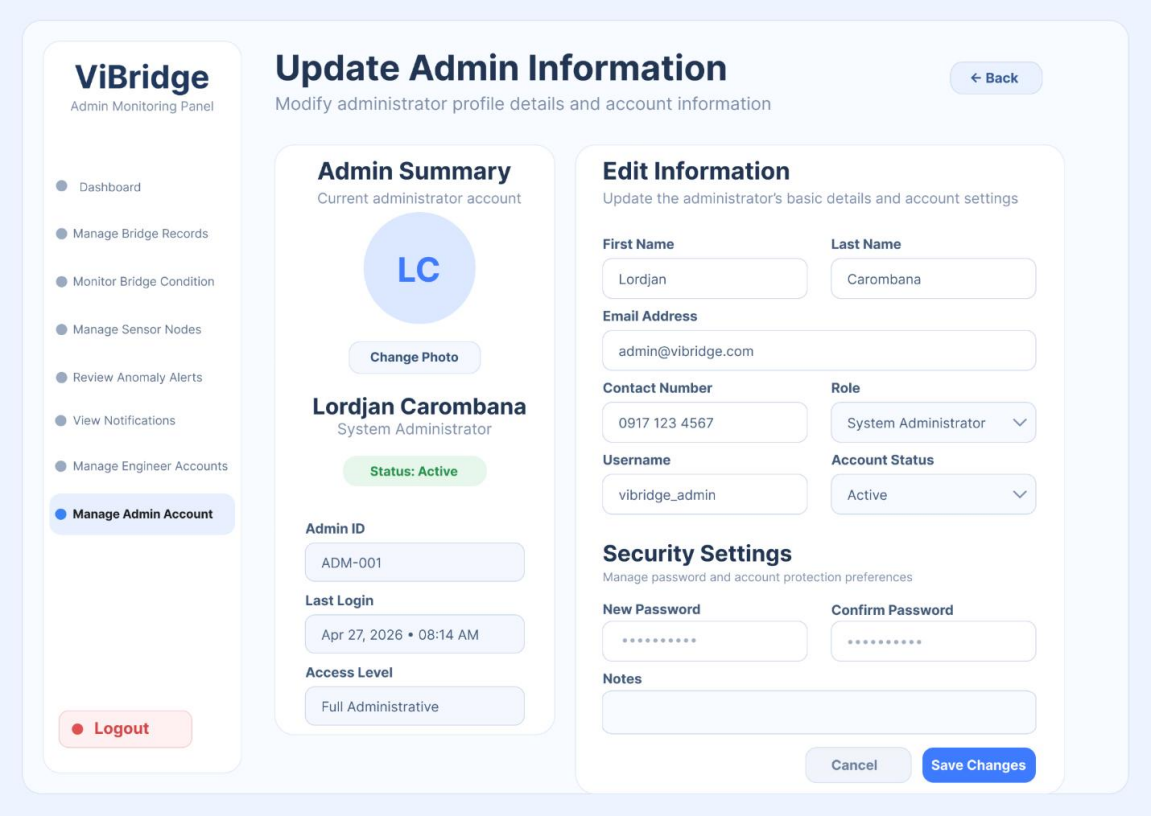


Figure 81: **Web Administration Update Admin Information Page**

Figure 81 presents the Update Admin Information screen, where the administrator can modify profile details and account settings. It includes editable fields for personal information, account credentials, and security settings. The Save Changes button applies updates, while Cancel discards them.

Storyboard

The storyboard illustrates the user journey step by step, visually representing the sequence of user interfaces. It starts from the initial screen and progresses to the final interface, highlighting the system's overall flow.

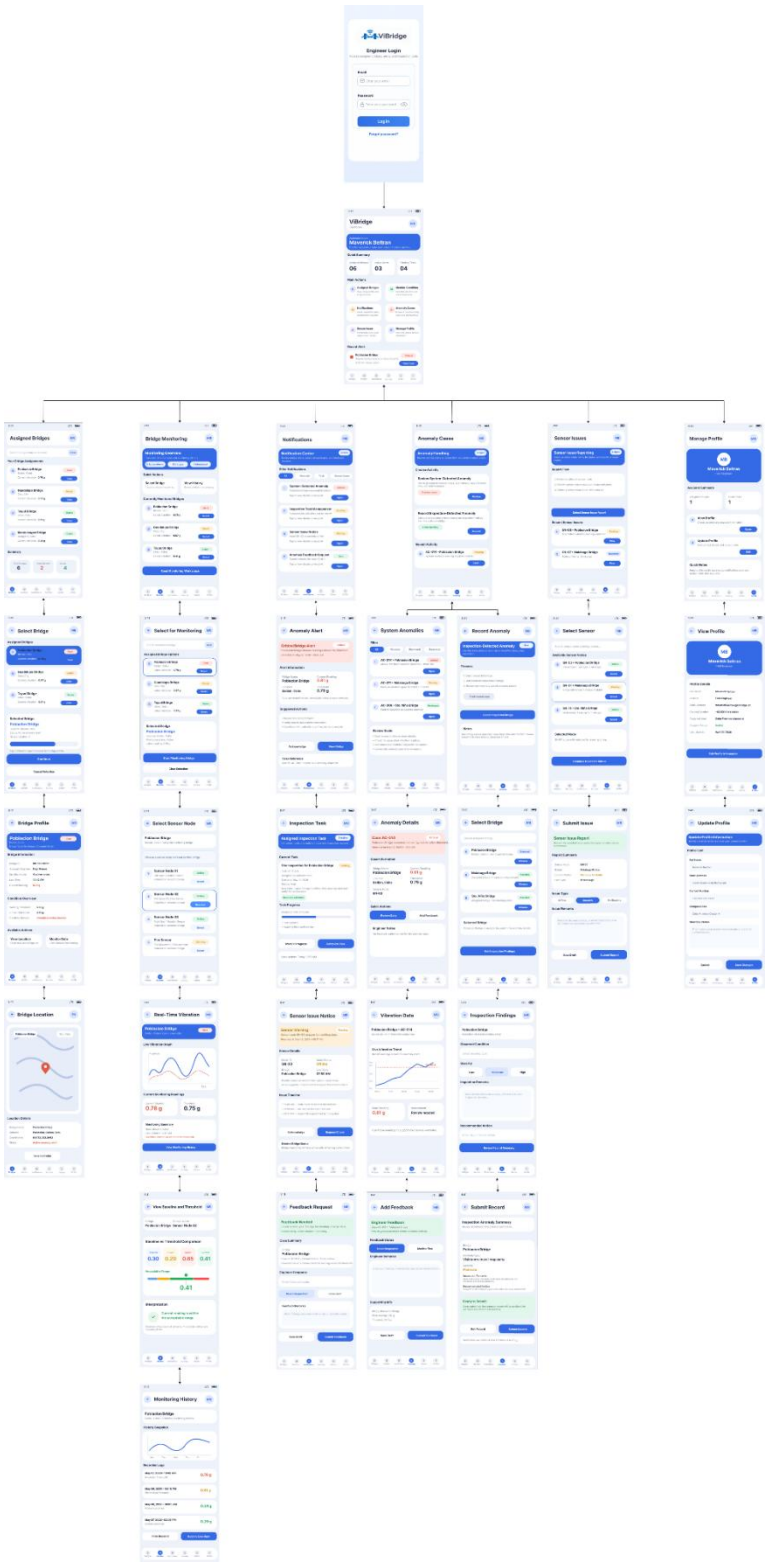


Figure 82: **Storyboard of Mobile Monitoring Application**

Figure 82 illustrates the storyboard of the ViBridge mobile monitoring application, designed for DPWH engineers to monitor bridge conditions and respond to structural anomalies. Upon launching the application, users are directed to the login interface, where they securely access the system using their registered email and password.

After successful authentication, engineers are presented with the dashboard, which serves as the central hub for monitoring assigned bridges, viewing alerts, and tracking recent activities. From this screen, users can navigate to different features through the bottom navigation menu, including assigned bridges, notifications, and account settings.

Within the assigned bridges section, engineers can view a list of bridges under their responsibility along with their current status. Selecting a bridge allows users to access detailed functionalities such as reviewing anomalies, monitoring real-time vibration data, and providing inspection feedback. The anomaly review screen presents detailed information about detected issues, including sensor readings, thresholds, and location mapping, enabling engineers to assess structural conditions accurately.

The real-time monitoring feature provides graphical visualization of vibration data, allowing users to observe trends and evaluate bridge behavior over time. Engineers can also submit inspection feedback through a dedicated form, where they provide observations, assign priorities, and recommend necessary actions for bridge inspection.

The notifications section keeps users informed of system alerts, feedback updates, and important messages, ensuring timely response to potential issues. Additionally, the account section allows engineers to view and update their profile information, maintaining accurate user records within the system.

The storyboard demonstrates the flow of interactions within the ViBridge mobile application, highlighting its role in supporting real-time bridge monitoring, anomaly detection, and efficient engineering response for infrastructure management.

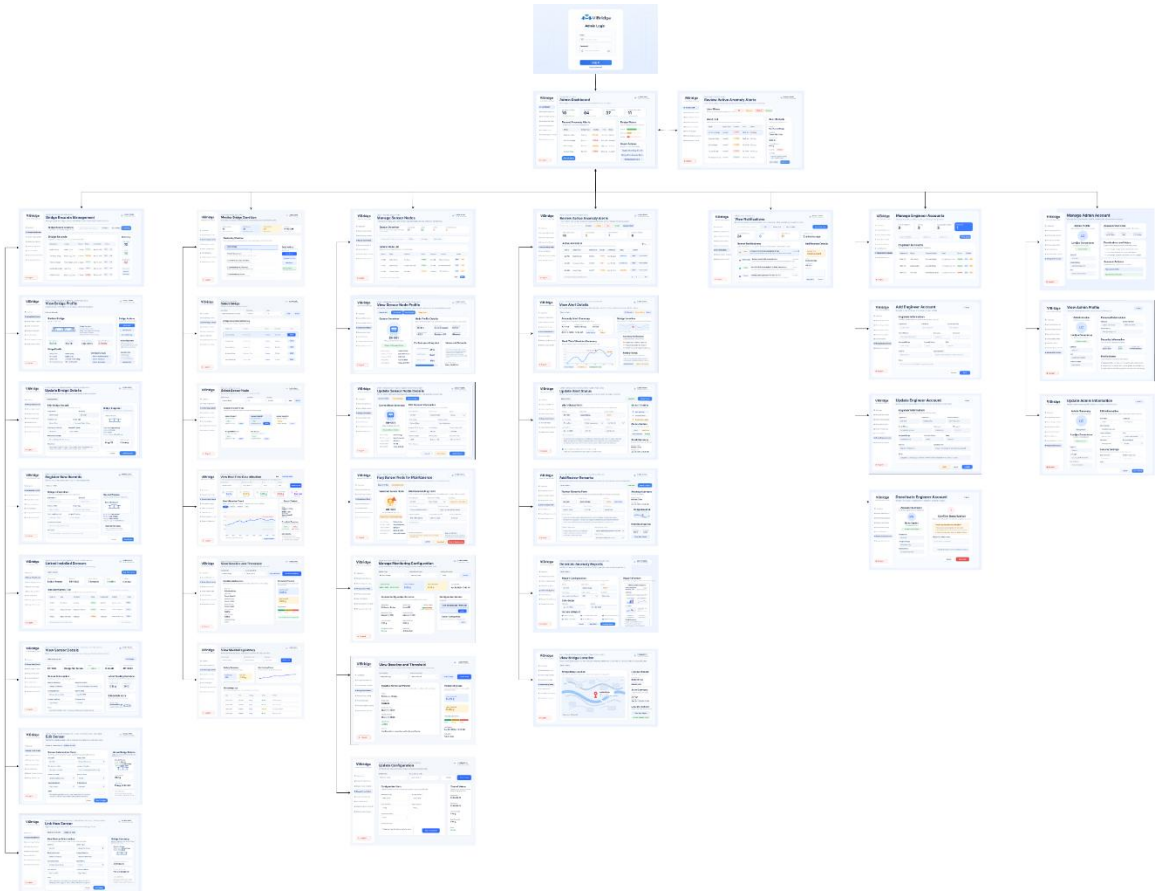


Figure 83: **Storyboard of Web Monitoring Application**

Figure 83 illustrates the user flow of the ViBridge web administration system, designed for DPWH administrators to manage bridge monitoring operations and system data. The process begins at the login page, where administrators input valid credentials to securely access the system. If credentials are valid, the user is directed to the dashboard, which serves as the central interface for monitoring bridge conditions, alerts, and system activities.

Upon successful login, the administrator is presented with the dashboard, which provides an overview of key metrics such as monitored bridges, active alerts, resolved alerts, and maintenance needs. From this central hub, administrators can navigate through different system modules using the sidebar menu, allowing efficient access to core functionalities.

The system includes several major sections for management operations. The Active Alerts section enables administrators to view detected anomalies and access detailed anomaly reports, including sensor data, threshold values, and location mapping. The Manage Bridges section allows administrators to register new bridges, view bridge details, update bridge information, and monitor associated sensors. Within this module, administrators can configure sensors, set baseline values and thresholds, and access real-time vibration monitoring to observe structural behavior.

The Assign Tasks section provides functionality for creating and managing engineering assessment tasks, allowing administrators to assign responsibilities, set priorities, and track task progress. The Notifications section enables administrators to monitor system alerts, messages, and delivery status, ensuring timely communication of critical information. The Account Management section allows administrators to manage user accounts, including adding new users, viewing user details, editing account information, and disabling accounts when necessary.

The system also supports detailed interaction flows, such as adding and editing bridge records, managing sensor configurations, and updating threshold settings, all through structured forms and modal interfaces. These features ensure accurate data management and system configuration.

The storyboard demonstrates a structured and user-friendly flow for managing bridge monitoring operations within the ViBridge system, emphasizing secure access, efficient navigation, real-time monitoring, and effective administrative control.

Database Design

The database design provides a detailed representation of the system's data model. It involves defining logical relationships among data, selecting appropriate physical design options, and specifying storage characteristics, which are then implemented using a data definition language.

Entity Relationship Diagram

The Entity Relationship Diagram (ERD) illustrates the logical structure of the database, showing the key entities, their attributes, and the relationships between them. It serves as the foundation for organizing, storing, and managing data within the system.

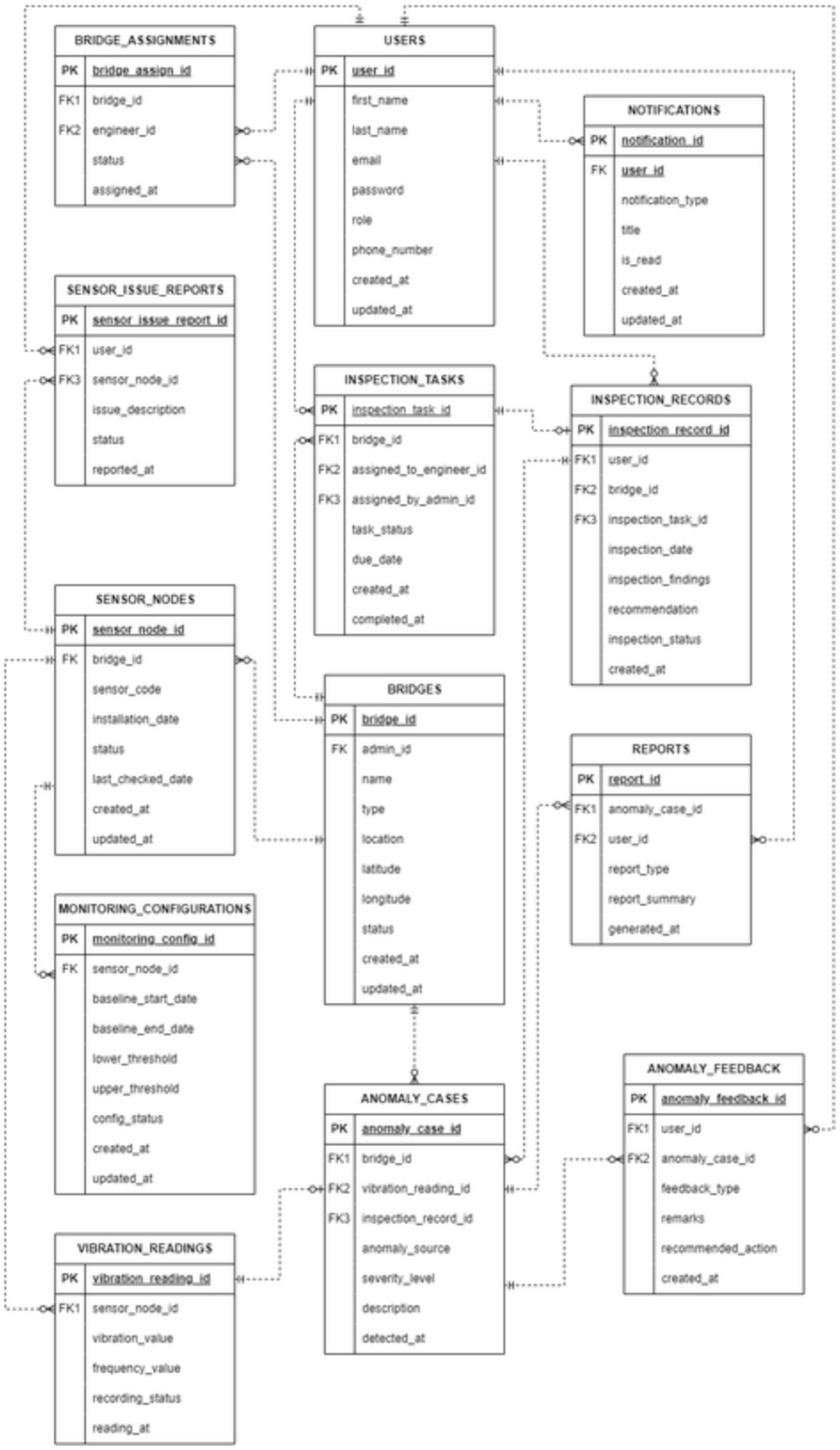


Figure 84: **Entity Relationship Diagram**

Figure 84 presents the Entity Relationship Diagram of the "ViBridge" system. It shows how the different tables of the system are connected to support bridge monitoring, sensor data collection, anomaly handling, inspection recording, notifications, and report generation.

The users table stores the accounts of all authorized users of the system. These users may be administrators or engineers. Administrators can manage bridge records, sensor nodes, monitoring settings, alerts, reports, and user accounts. Engineers can view assigned bridges, monitor bridge data, receive notifications, submit inspection records, give anomaly feedback, and report sensor issues.

The bridges table stores the main information about the bridges monitored by the system. It includes the bridge name, type, location, coordinates, status, and the administrator who manages the bridge record. A user with an administrator role can create and manage many bridge records, while each bridge record is connected to one administrator.

The bridge_assignments table connects engineers to the bridges assigned to them. This table records which engineer is assigned to a specific bridge, the assignment status, and the date when the bridge was assigned. Through this table, one bridge can be assigned to one or more engineers, and one engineer can also be assigned to one or more bridges.

The sensor_nodes table stores the sensor devices installed on bridges. Each sensor node belongs to a specific bridge. It records the sensor code, installation location, installation date, current status, and last checked date. This allows the system to identify which bridge a sensor node belongs to and where it is installed.

The vibration_readings table stores the readings collected from the sensor nodes installed on the bridge. Since the ADXL355 is a triaxial accelerometer, each reading includes values from the X-axis, Y-axis, and Z-axis. These values are stored as acceleration_x, acceleration_y, and acceleration_z. The table also stores the overall vibration_value, the frequency_value, and the date and time when the reading was collected. These records are used for real-time monitoring, monitoring history, threshold checking, and anomaly detection.

The monitoring_configurations table stores the monitoring settings for each sensor node. It includes the baseline start date, baseline end date, lower threshold, upper threshold, configuration status, and the user who created or updated the configuration. This table is where the system keeps the baseline and threshold values used for checking future vibration readings.

The inspection_tasks table stores inspection assignments for bridges. Each inspection task is connected to a bridge, an assigned engineer, and the user who created the task. It also records the task status, due date, creation date, and completion date. This helps the system track which bridge needs inspection and which engineer is responsible for it.

The inspection_records table stores the findings submitted by engineers after inspecting a bridge. Each inspection record is connected to a user, a bridge, and an inspection task. It records the inspection date, findings, recommendation, status, and creation date. If the engineer finds a possible structural concern during inspection, the record may later be connected to an anomaly case.

The anomaly_cases table stores abnormal bridge conditions detected either by the system or by an engineer. If the anomaly is detected by the system, it can be connected to a specific vibration_readings record. If the anomaly is recorded after an inspection, it can be connected to a specific inspection_records record. It also stores the affected bridge, anomaly source, severity level, description, status, and detection date.

The anomaly_feedback table stores comments and recommended actions related to an anomaly case. Each feedback record is connected to a user and an anomaly case. This allows administrators and engineers to add remarks, explain their review, and suggest what action should be taken for the anomaly.

The sensor_issue_reports table stores reported problems related to sensor nodes. Each report is connected to the user who submitted it and the sensor node being reported. It records the issue description, status, and report date. The affected bridge can still be identified through the connected sensor node.

The notifications table stores system messages sent to users. These notifications may be about detected anomalies, inspection tasks, sensor issue notices, feedback requests, or report updates. Each notification is connected to a receiving user and includes the notification type, title, message, related record information, read status, and timestamps.

The reports table stores generated reports related to anomaly cases. Each report is connected to an anomaly case and the user who generated it. It includes the report type, report summary, and generation date. Through the related anomaly case, the system can identify which bridge is affected.

Overall, the ERD shows that "ViBridge" is organized around bridges, sensor nodes, vibration readings, monitoring configurations, inspection tasks, anomaly cases, feedback, notifications, and reports. These tables work together to support real-time bridge monitoring, threshold checking, anomaly recording, inspection follow-up, and report generation.

Data Dictionary

This section provides a summary of each database table, including the table name, attribute names, data types, content descriptions, primary keys (PK), foreign keys (FK), and the tables each FK references.

Table 7.1

USERS

TABLE NAME	ATTRIBUTE NAME	CONTENTS	TYPE	PK or FK	FK REFERENCED TABLE
users	user_id	User's unique identification	INT	PK	
users	first_name	User's first name	VARCHAR(100)		
users	last_name	User's last name	VARCHAR(100)		
users	email	User's email address	VARCHAR(150)		
users	password	Encrypted password of the user	VARCHAR(255)		
users	role	Assigned role of the user, such as Administrator or Engineer	VARCHAR(50)		
users	phone_number	Contact number of the user	VARCHAR(50)		
users	created_at	Date and time the record was created	DATETIME		

Table 7.2

USERS cont'd

TABLE NAME	ATTRIBUTE NAME	CONTENTS	TYPE	PK or FK	FK REFERENCED TABLE
users	updated_at	Date and time the record was last updated	DATETIME		

Overview of Table 7.1 and 7.2, the users table. The users table stores the account information of all authorized ViBridge users. It includes user identity, login credentials, role assignment, contact information, and timestamps. This table is used to identify administrators and engineers who access the web and mobile monitoring platforms.

Table 8.1

BRIDGES

TABLE NAME	ATTRIBUTE NAME	CONTENTS	TYPE	PK or FK	FK REFERENCED TABLE
bridges	bridge_id	Bridge's unique identification	INT	PK	
bridges	admin_id	Administrator who created or manages the bridge record	INT	FK	users
bridges	name	Name of the bridge	VARCHAR(150)		
bridges	type	Type or classification of the bridge	VARCHAR(100)		
bridges	location	Location or address of the bridge	VARCHAR(255)		
bridges	latitude	Latitude coordinate of the bridge	DECIMAL(10,6)		
bridges	longitude	Longitude coordinate of the bridge	DECIMAL(10,6)		

Table 8.2

BRIDGES cont'd

TABLE NAME	ATTRIBUTE NAME	CONTENTS	TYPE	PK or FK	FK REFERENCED TABLE
bridges	status	Current status of the bridge record	VARCHAR(50)		
bridges	created_at	Date and time the record was created	DATETIME		
bridges	updated_at	Date and time the record was last updated	DATETIME		

Overview of Table 8.1 and 8.2, the bridges table. The bridges table stores the master records of bridges monitored by ViBridge. It includes the bridge name, type, location, coordinates, status, and the administrator who created or manages the bridge record.

Table 9.1

BRIDGE_ASSIGNMENTS

TABLE NAME	ATTRIBUTE NAME	CONTENTS	TYPE	PK or FK	FK REFERENCED TABLE
bridge_assignments	bridge_assign_id	Bridge assignment's unique identification	INT	PK	
bridge_assignments	bridge_id	Assigned bridge's unique identification	INT	FK	bridges
bridge_assignments	engineer_id	Engineer assigned to the bridge	INT	FK	users

Table 9.2

BRIDGE_ASSIGNMENTS cont'd

TABLE NAME	ATTRIBUTE NAME	CONTENTS	TYPE	PK or FK	FK REFERENCED TABLE
bridge_assignments	status	Current status of the assignment	VARCHAR(50)		
bridge_assignments	assigned_at	Date and time the bridge was assigned	DATETIME		

Overview of Table 9.1 and 9.2, the bridge_assignments table. The bridge_assignments table serves as the junction table between bridges and engineers. It records which engineer is assigned to a specific bridge and tracks the assignment status and date.

Table 10.1

SENSOR_NODES

TABLE NAME	ATTRIBUTE NAME	CONTENTS	TYPE	PK or FK	FK REFERENCED TABLE
sensor_nodes	sensor_node_id	Sensor node's unique identification	INT	PK	
sensor_nodes	bridge_id	Bridge where the sensor node is installed	INT	FK	bridges
sensor_nodes	sensor_code	Code or label assigned to the sensor node	VARCHAR(100)		sensor_nodes
sensor_nodes	installation_location	Physical placement of the sensor node on the bridge	VARCHAR(150)		sensor_nodes

Table 10.2

SENSOR_NODES cont'd

TABLE NAME	ATTRIBUTE NAME	CONTENTS	TYPE	PK or FK	FK REFERENCED TABLE
sensor_nodes	installation_date	Date when the sensor node was installed	DATE		sensor_nodes
sensor_nodes	status	Current status of the sensor node	VARCHAR(50)		sensor_nodes
sensor_nodes	last_checked_date	Most recent date when the sensor node was checked	DATE		sensor_nodes
sensor_nodes	created_at	Date and time the record was created	DATETIME	sensor_nodes	created_at
sensor_nodes	updated_at	Date and time the record was last updated	DATETIME	sensor_nodes	updated_at

Overview of Table 10.1 and 10.2, the sensor_nodes table. The sensor_nodes table stores information about installed sensor nodes linked to bridges. It includes the assigned bridge, sensor code, installation details, physical installation location, status, and the latest date the sensor was checked.

Table 11.1

VIBRATION_READINGS

TABLE NAME	ATTRIBUTE NAME	CONTENTS	TYPE	PK or FK	FK REFERENCED TABLE
vibration_readings	vibration_reading_id	Vibration reading's unique identification	INT	PK	
vibration_readings	sensor_node_id	Sensor node that produced the reading	INT	FK	sensor_nodes

Table 11.2

VIBRATION_READINGS cont'd

TABLE NAME	ATTRIBUTE NAME	CONTENTS	TYPE	PK or FK	FK REFERENCED TABLE
vibration_readings	vibration_value	overall vibration value computed from X, Y, and Z	FLOAT		
vibration_readings	frequency_value	Measured or processed vibration frequency value	FLOAT		
vibration_readings	recording_status	Recording state of the reading, such as active or completed	VARCHAR(50)		
vibration_readings	acceleration_x	Acceleration reading from the X-axis of the sensor	FLOAT		
vibration_readings	acceleration_y	Acceleration reading from the Y-axis of the sensor	FLOAT		
vibration_readings	acceleration_z	Acceleration reading from the Z-axis of the sensor	FLOAT		
vibration_readings	reading_at	Date and time the reading was recorded	DATETIME		

Overview of Table 11.1 and 11.2, the vibration_readings table. The vibration_readings table stores the actual readings collected from sensor nodes. These records are used for real-time

monitoring, monitoring history, baseline computation, threshold comparison, and possible anomaly detection.

Table 12

MONITORING_CONFIGURATIONS

TABLE NAME	ATTRIBUTE NAME	CONTENTS	TYPE	PK or FK	FK REFERENCED TABLE
monitoring_configurations	monitoring_config_id	Monitoring configuration's unique identification	INT	PK	
monitoring_configurations	sensor_node_id	Sensor node where the configuration applies	INT	FK	sensor_nodes
monitoring_configurations	set_by_user_id	User who created or updated the configuration	INT	FK	users
monitoring_configurations	baseline_start_date	Start date of the baseline data collection period	DATE		
monitoring_configurations	baseline_end_date	End date of the baseline data collection period	DATE		
monitoring_configurations	lower_threshold	Lowest acceptable frequency or vibration limit	FLOAT		
monitoring_configurations	upper_threshold	Highest acceptable frequency or vibration limit	FLOAT		

Overview of Table 12, the monitoring_configurations table. The monitoring_configurations table stores the baseline period and threshold limits assigned to each sensor node. The lower and upper thresholds are generated after analyzing vibration readings collected during the baseline period and are used as references for future anomaly checking.

Table 13

INSPECTION_TASKS

TABLE NAME	ATTRIBUTE NAME	CONTENTS	TYPE	PK or FK	FK REFERENCED TABLE
inspection_tasks	inspection_task_id	Inspection task's unique identification	INT	PK	
inspection_tasks	bridge_id	Bridge that needs to be inspected	INT	FK	bridges
inspection_tasks	assigned_to_engineer_id	Engineer assigned to the inspection task	INT	FK	users
inspection_tasks	assigned_by_id	Administrator or user who assigned the inspection task	INT	FK	users
inspection_tasks	task_status	Current status of the inspection task	VARCHAR (50)		
inspection_tasks	due_date	Target completion date of the task	DATE		
inspection_tasks	created_at	Date and time the task was created	DATETIME		
inspection_tasks	completed_at	Date and time the task was completed	DATETIME		

Overview of Table 13, the inspection_tasks table. The inspection_tasks table stores inspection assignments created for specific bridges. It records the bridge to be inspected, the engineer assigned to perform the task, the user who assigned the task, the task status, due date, and completion date.

Table 14

INSPECTION_RECORDS

TABLE NAME	ATTRIBUTE NAME	CONTENTS	TYPE	PK or FK	FK REFERENCED TABLE
inspection_records	inspection_record_id	Inspection record's unique identification	INT	PK	
inspection_records	user_id	Engineer who submitted the inspection record	INT	FK	users
inspection_records	bridge_id	Bridge inspected by the engineer	INT	FK	bridges
inspection_records	inspection_task_id	Inspection task related to the record	INT	FK	inspection_tasks
inspection_records	inspection_date	Date when the inspection was conducted	DATE		
inspection_records	inspection_findings	Observed findings during the inspection	TEXT		
inspection_records	recommendation	Recommended action based on the inspection result	TEXT		
inspection_records	inspection_status	Status of the inspection record	VARCHAR (50)		
inspection_records	created_at	Date and time the record was created	DATETIME		

Overview of Table 14, the inspection_records table. The inspection_records table stores the findings submitted after an engineer inspects a bridge. It records the engineer, related bridge, related inspection task, findings, recommendations, status, and the inspection date. An inspection record may later create one or more inspection-detected anomaly cases.

Table 15.1

ANOMALY_CASES

TABLE NAME	ATTRIBUTE NAME	CONTENTS	TYPE	PK or FK	FK REFERENCED TABLE
anomaly_cases	anomaly_case_id	Anomaly case's unique identification	INT	PK	
anomaly_cases	bridge_id	Bridge affected by the anomaly	INT	FK	bridges
anomaly_cases	vibration_reading_id	Vibration reading that triggered the anomaly, if system-detected	INT	FK	vibration_readings
anomaly_cases	inspection_record_id	Inspection record that created the anomaly, if inspection-detected	INT	FK	inspection_records
anomaly_cases	anomaly_source	Source of the anomaly, such as system-detected or inspection-detected	VARCHAR (50)		
anomaly_cases	severity_level	Severity level of the anomaly	VARCHAR (50)		
anomaly_cases	description	Description of the anomaly case	TEXT		
anomaly_cases	status	Current status of the anomaly case	VARCHAR (50)		
anomaly_cases	detected_at	Date and time the anomaly was detected or recorded	DATETIME		

Table 15.2

ANOMALY_CASES cont'd

TABLE NAME	ATTRIBUTE NAME	CONTENTS	TYPE	PK or FK	FK REFERENCED TABLE
anomaly_cases	created_at	Date and time the record was created	DATETIME		
anomaly_cases	updated_at	Date and time the record was last updated	DATETIME		

Overview of Table 15.1 and 15.2, the anomaly_cases table. The anomaly_cases table stores abnormal bridge conditions detected either by the system through vibration readings or by engineers through inspection records. It records the affected bridge, the possible source record, anomaly source, severity level, description, status, and detection time.

Table 16.1

ANOMALY FEEDBACK

TABLE NAME	ATTRIBUTE NAME	CONTENTS	TYPE	PK or FK	FK REFERENCED TABLE
anomaly_feedback	anomaly_feedback_id	Anomaly feedback's unique identification	INT	PK	
anomaly_feedback	user_id	User who submitted the feedback	INT	FK	users
anomaly_feedback	anomaly_case_id	Anomaly case related to the feedback	INT	FK	anomaly_cases
anomaly_feedback	feedback_type	Type of feedback submitted	VARCHAR (100)		

Table 16.2

ANOMALY FEEDBACK cont'd

TABLE NAME	ATTRIBUTE NAME	CONTENTS	TYPE	PK or FK	FK REFERENCED TABLE
anomaly_feedback	remarks	Remarks or comments about the anomaly case	TEXT		
anomaly_feedback	recommended_action	Recommended action for the anomaly case	TEXT		
anomaly_feedback	created_at	Date and time the feedback was created	DATETIME		

Overview of Table 16.1 and 16.2, the anomaly_feedback table. The anomaly_feedback table stores remarks, feedback, and recommended actions submitted by administrators or engineers for a specific anomaly case. It supports review history and accountability during anomaly handling.

Table 17.1

SENSOR_ISSUE_REPORTS

TABLE NAME	ATTRIBUTE NAME	CONTENTS	TYPE	PK or FK	FK REFERENCED TABLE
sensor_issue_reports	sensor_issue_report_id	Sensor issue report's unique identification	INT	PK	
sensor_issue_reports	user_id	User who reported the sensor issue	INT	FK	users
sensor_issue_reports	sensor_node_id	Sensor node with the reported issue	INT	FK	sensor_nodes

Table 17.2

SENSOR_ISSUE_REPORTS cont'd

TABLE NAME	ATTRIBUTE NAME	CONTENTS	TYPE	PK or FK	FK REFERENCED TABLE
sensor_issue_reports	issue_description	Description of the sensor issue	TEXT		
sensor_issue_reports	status	Current status of the issue report	VARCHAR (50)		
sensor_issue_reports	reported_at	Date and time the issue was reported	DATETIME		

Overview of Table 17.1 and 17.2, the sensor_issue_reports table. The sensor_issue_reports table stores reported problems related to installed sensor nodes. It records the user who reported the issue, the affected sensor node, issue description, status, and report date. The related bridge can be identified through the sensor node record.

Table 18.1

NOTIFICATIONS

TABLE NAME	ATTRIBUTE NAME	CONTENTS	TYPE	PK or FK	FK REFERENCED TABLE
notifications	notification_id	Notification's unique identification	INT	PK	
notifications	user_id	User who receives the notification	INT	FK	users
notifications	notification_type	Type of notification sent by the system	VARCHAR (100)		

Table 18.2

NOTIFICATIONS cont'd

TABLE NAME	ATTRIBUTE NAME	CONTENTS	TYPE	PK or FK	FK REFERENCED TABLE
notifications	title	Short title of the notification	VARCHAR (150)		
notifications	message	Content or message of the notification	TEXT		
notifications	related_type	Type of related record, such as anomaly case or inspection task	VARCHAR (100)		
notifications	related_id	Identification number of the related record	INT		
notifications	is_read	Read status of the notification	BOOLEAN		
notifications	created_at	Date and time the notification was created	DATETIME		
notifications	updated_at	Date and time the notification was last updated	DATETIME		

Overview of Table 18.1 and 18.2, the notifications table. The notifications table stores system messages sent to users. It includes the receiving user, notification type, title, message, optional related record information, read status, and timestamps.

Table 19.1

REPORTS

TABLE NAME	ATTRIBUTE NAME	CONTENTS	TYPE	PK or FK	FK REFERENCED TABLE
reports	report_id	Report's unique identification	INT	PK	

Table 19.2

REPORTS cont'd

TABLE NAME	ATTRIBUTE NAME	CONTENTS	TYPE	PK or FK	FK REFERENCED TABLE
reports	anomaly_case_id	Anomaly case related to the report	INT	FK	anomaly_cases
reports	generated_by_user_id	User who generated the report	INT	FK	users
reports	report_type	Type or classification of the generated report	VARCHAR (100)		
reports	report_summary	Summary or content overview of the report	TEXT		
reports	generated_at	Date and time the report was generated	DATETIME		

Overview of Table 19.1 and 19.2, the reports table. The reports table stores generated reports related to anomaly cases. It records the anomaly case, the user who generated the report, the report type, summary, and generation date. The affected bridge can be traced through the related anomaly case.

Network Design

This section presents the planning and implementation of "ViBridge's" network infrastructure. It includes evaluating, analyzing, and sizing the network to ensure that it can support the continuous collection, transmission, storage, and viewing of bridge monitoring data. The network is designed to connect the bridge-mounted devices, communication modules, database, server, and Android and web-based monitoring platforms so that real-time readings and alerts can be accessed by authorized users. A well-structured network diagram representing the overall network architecture serves as a guide for the proper setup of communication between the hardware and software components of the system.

Network Model

The ViBridge application has two users in the database: the Administrator and the Engineer. The Engineer accesses the system through the Android application and web application to view bridge monitoring data, alerts, and assigned bridge information. The Administrator accesses the system through the web application to manage user accounts, monitor bridge conditions, review alerts, and supervise the overall operation of the system. The ViBridge IoT device collects bridge vibration data and sends it through the network for storage and access in the database. Through this network model, the Administrator oversees the registered Engineer accounts and system records, while both users are able to access monitoring information according to their assigned roles in the ViBridge system.

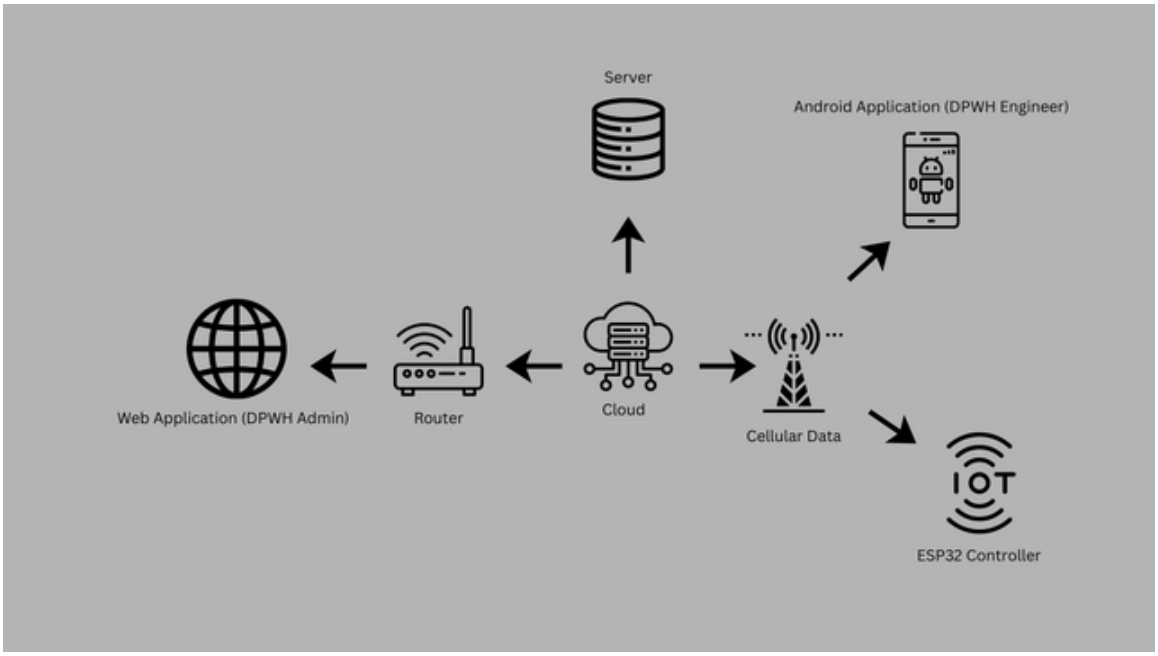


Figure 85: **Network Models**

Figure 85 shows the network model of the ViBridge system. The figure illustrates how the two authorized users, the Administrator and the Engineer, are connected within the system through the available platforms and network structure. The Engineer can access the system through the Android application and web application, while the Administrator uses the web application for monitoring and management tasks. The figure also shows that the ViBridge IoT device is included in the network structure as the hardware responsible for collecting bridge monitoring data. These system components are connected through the network so that monitoring information can be stored in the database and made accessible to the authorized users. Through this setup, both the

Administrator and the Engineer are able to access bridge monitoring information according to their assigned roles in the system.

Network Topology

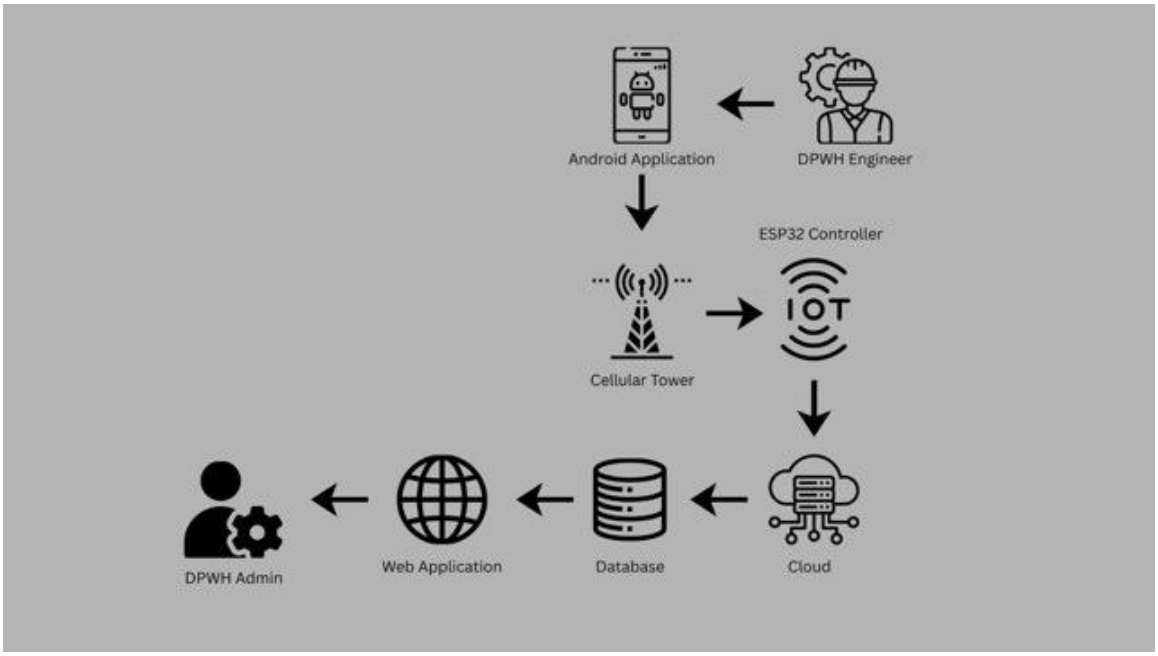


Figure 86: **Network Topology**

Figure 86 shows the network topology of the ViBridge system. The figure illustrates how the ViBridge Sensor Node (ESP32 microcontroller) transmits bridge monitoring data through available communication paths, specifically the router and cellular data, which then send the information to the cloud and Firebase server for storage and system access. It also shows the two authorized users of the system: the Engineer and the Administrator. The Engineer can access the system through both the Android application and the web platform, while the Administrator accesses the system through the web platform only. Through this topology, real-time monitoring data collected by the sensor node can be transmitted, stored, and viewed by the users according to their assigned roles in the ViBridge system.

Development/Construction/Build Phase

During the development phase, all resources and requirements needed to implement the ViBridge system are organized and prepared. This includes coordinating the needed hardware and software components, identifying possible suppliers or technical support sources, creating a development timeline, preparing the required tools and materials, and giving clear instructions for

the implementation of the system. This phase ensures that the development of the web platform, android application, database, and bridge monitoring components will proceed in an organized and systematic manner.

Technology Stack Diagram

Table 20

TECHNOLOGY STACK

<table>
<tr><th>ANDROID DEVELOPMENT</th><th>IOT DEVELOPMENT</th><th>WEB DEVELOPMENT</th></tr>
<tr><td>Kotlin
Android Studio
Android SDK</td><td rowspan="3">ESP32 Microcontroller
ADXL355 Accelerometer
Wi-Fi Communication
Power Supply (Battery/Solar)</td><td>React (JavaScript)
Node.js / Express
Vite</td></tr>
<tr><td>CLOUD SERVICES</td><td>CLOUD SERVICES</td></tr>
<tr><td>Firebase Authentication
Firebase Cloud Messaging
GitHub</td><td>Firebase Admin SDK
Firebase Cloud Functions
Vercel / Render</td></tr>
<tr><td colspan="3">SERVER</td></tr>
<tr><td colspan="3">Firebase Realtime Database
Firebase Storage</td></tr>
</table>

ViBridge, an IoT-based bridge vibration and structural condition monitoring system for small bridges, is developed using a technology stack that supports Android monitoring, web administration, IoT data collection, cloud services, and server-side data handling. The Android application is developed using Kotlin in Android Studio, supported by the Android SDK. This mobile application allows authorized engineers to view assigned bridges, monitor vibration data, receive alerts, review anomaly details, and report sensor issues.

For the IoT component, ViBridge uses an ESP32 microcontroller connected to an ADXL355 triaxial accelerometer. The ADXL355 collects bridge vibration readings, while the ESP32 transmits the collected data through Wi-Fi communication. The sensor node is powered using a battery or solar-based power supply, making it suitable for bridge monitoring deployment. These components support real-time vibration data collection from selected small single-span bridges.

The web platform is developed using React JavaScript, with Node.js and Express supporting backend communication and data handling. Vite is used to improve the development and build process of the web application. The web platform allows administrators to manage

bridges, sensor nodes, users, monitoring configurations, alerts, and system records. For cloud services, ViBridge uses Firebase Authentication for user access control, Firebase Cloud Messaging for notifications, and GitHub for source code management.

The server and database components use Firebase Admin SDK, Firebase Cloud Functions, and deployment platforms such as Vercel or Render. System data such as vibration readings, baseline and threshold records, anomaly alerts, users, and bridge information are stored using Firebase Realtime Database, while Firebase Storage is used for storing related files and system resources. Together, these technologies support ViBridge's main functions, including vibration monitoring, wireless data transmission, anomaly detection support, alert generation, and map-based bridge location viewing.

Software Specification

This section presents the software specifications that will be used in developing the ViBridge system. It includes the operating system, programming languages, development tools, frameworks, platforms, database, authentication, and cloud services needed for the Android application, web-based monitoring platform, server, and IoT data handling. These specifications serve as a technical guide for the researchers in building the system correctly and ensuring that all software components support vibration data monitoring, baseline and threshold checking, anomaly detection, alert notifications, and bridge location viewing.

Table 21.1

SOFTWARE SPECIFICATION

Component	**Specification**
Operating System	Windows, Android OS Version 5 and above
Component	**Specification**
Text Editor/IDE	Visual Studio Code and Android Studio
Programming Language	Kotlin, JavaScript
Web Browser	Google Chrome, Microsoft Edge, Opera
Mobile	Android

Table 21.2

SOFTWARE SPECIFICATION cont'd

Component	**Specification**
Database	Firebase Realtime Database

Tables 21.1 and 21.2 present the software specifications required for the development of the ViBridge system. The table identifies the operating system, development tools, programming languages, web browsers, mobile platform, and database that will be used in building the system. These software components support the development of ViBridge's Android application, web-based monitoring platform, and database operations.

The system will run on Windows for development and will support Android OS version 5 and above for the mobile application. Visual Studio Code and Android Studio will be used as the main development environments. The programming languages include Kotlin for the Android application and JavaScript for the web-based platform. The system can be accessed through common web browsers such as Google Chrome, Microsoft Edge, and Opera. For data storage, Firebase Realtime Database will be used to store and manage system data such as vibration readings, bridge records, monitoring data, alerts, and user-related information.

Overall, the software specifications provide the technical foundation needed to develop and operate ViBridge. These tools and platforms support real-time vibration monitoring, baseline and threshold-based checking, anomaly detection, alert notifications, and web and mobile access for authorized users.

Hardware Specification

This section presents the hardware specifications required for the development and operation of the ViBridge system. It identifies the necessary devices and components, including the computer, Android mobile device, ESP32 microcontroller, ADXL355 triaxial MEMS accelerometer, power supply, and sensor-node enclosure. These hardware specifications serve as a guide to ensure that the system can collect, transmit, and display bridge vibration data properly for real-time monitoring, baseline and threshold checking, anomaly detection, and alert generation.

Table 22.1

HARDWARE SPECIFICATION

PC	
CPU	64-bit processor support; Intel Core i3 10th Gen / AMD Ryzen 3 or higher
Memory	At least 8 GB RAM or above
Storage	10 GB Available space or more
GPU	Integrated or dedicated graphics; Intel UHD Graphics / AMD Graphics / NVIDIA Graphics
Mobile	
CPU	Quad-core processor or higher
Memory	At least 4 GB RAM or above
Storage	6 GB available space or more
Connectivity	Wi-Fi or mobile data connection
IOT Sensors	
Microcontroller	ESP32 microcontroller
Vibration Sensor	ADXL355 triaxial MEMS accelerometer
Sensor Reading Types	X-axis, Y-axis, and Z-axis acceleration readings for bridge vibration monitoring

Table 22.2

HARDWARE SPECIFICATION cont'd

Wireless Communication	Built-in ESP32 Wi-Fi module
Data Transmission	Wireless transmission of vibration readings from the sensor node to the cloud database or server
Power Supply	Regulated power supply, rechargeable battery pack, or suitable external power source for ESP32 and sensor operation
Protection Casing	Weather-resistant enclosure or protection box for outdoor bridge installation
Mounting Materials	Brackets, screws, adhesive mount, or other secure fastening materials for attaching the sensor node to selected bridge locations
Deployment Structure	Selected small single-span bridge for pilot implementation

Table 22.1 and 22.2 presents the hardware specifications required for the ViBridge system. The listed components support the operation of the web platform, Android mobile application, and IoT sensor node. A PC or laptop is required for web-based monitoring and system administration, while an Android device is used for mobile monitoring by authorized users. The IoT sensor node consists of an ESP32 microcontroller and an ADXL355 triaxial MEMS accelerometer, which collect bridge vibration readings and transmit them wirelessly to the cloud database or server. These hardware components are selected to support real-time vibration monitoring, baseline and threshold checking, anomaly alert generation, and location-based monitoring for selected small single-span bridges.

Program Specification

The program specification consists of the gathered information, organized into procedures that define the required functions and expected outputs of the ViBridge system. It also identifies the modules assigned to each team member and the system access of the DPWH Administrator and DPWH Engineer.

List of Modules

Table 23.1

Programmers	Modules	DPWH Administrator	DPWH Engineer
Lord Jan Rolmar Carombana Paul Vincent Lepatan	**Account Management**		
	1. Login	*	*
	2. Logout	*	*
	3. Forgot and Reset Password	*	*
	4. Edit Account Information	*	*
	5. Manage Engineer Accounts	*	
	6. Activate or Deactivate User Account	*	
	No. of Points	**1**	**1**
	Bridge Management	**DPWH Administrator**	**DPWH Engineer**
Maverick Beltran Christian Rosales	1. View Bridge Records	*	*
	2. Register New Bridge Record	*	
	3. View Bridge Profile	*	*
	4. Update Bridge Details	*	
	5. Link Installed Sensor Node	*	
	6. Update Bridge Status	*	
	No. of Points	**1**	**1**

LIST OF MODULES cont'd

Table 23.2

Programmers	Modules	DPWH Administrator	DPWH Engineer
Lord Jan Rolmar Carombana Maverick Beltran	**Bridge Monitoring and Vibration Data Management**		
	1. View Real-Time Vibration Data	*	*
	2. View Frequency and PSD Visualization	*	*
	3. View Vibration Reading History	*	*
	4. View Baseline and Threshold Values	*	*
	5. Configure Baseline Period	*	
	6. Set or Update Threshold Values	*	
	7. Export Monitoring Data	*	
	No. of Points	**1**	**1**
Paul Vincent Lepatan Christian Rosales	**Sensor Node Management**	**DPWH Administrator**	**DPWH Engineer**
	1. Register Sensor Node	*	
	2. View Sensor Node List	*	*
	3. View Sensor Node Status	*	*
	4. Update Sensor Node Details	*	
	5. Mark Sensor Node for Maintenance or Replacement	*	
	No. of Points	**1**	**1**

LIST OF MODULES cont'd

Table 23.3

Programmers	Modules	DPWH Administrator	DPWH Engineer
Lord Jan Rolmar Carombana Christian Rosales	**Anomaly and Alert Management**		
	1. Receive Threshold Alerts	*	*
	2. View Active Alerts	*	*
	3. Review Anomaly Details	*	*
	4. Update Alert Status	*	
	5. Generate Anomaly Report	*	
	6. Notify Assigned Engineer	*	*
	No. of Points	**1**	**1**
Paul Vincent Lepatan Maverick Beltran	**Inspection Task and Engineer Feedback Management**	**DPWH Administrator**	**DPWH Engineer**
	1. View Assigned Bridges		*
	2. View Suggested Bridge for Inspection		*
	3. Create Inspection Task	*	
	4. View Inspection Task Details	*	*
	5. Update Inspection Task Status		*
	6. Submit Inspection Feedback		*
	7. Review Engineer Feedback	*	
	No. of Points	**1**	**1**

LIST OF MODULES cont'd

Table 23.4

Programmers	Modules	DPWH Administrator	DPWH Engineer
Maverick Beltran Christian Rosales	**Notification, Logs, and Report Management**		
	1. View Notifications	*	*
	2. View Notification History	*	*
	3. View Data Logs	*	*
	4. Generate Monitoring Reports	*	
	5. Export Reports	*	
	6. View System Activity Logs	*	
	No. of Points	**1**	**1**
Lord Jan Rolmar Carombana Paul Vincent Lepatan	**Profile and Support Management**	**DPWH Administrator**	**DPWH Engineer**
	1. View Profile	*	*
	2. Edit Profile	*	*
	3. View About System Information	*	*
	4. View Technical Support Information	*	*
	No. of Points	**1**	**1**

References

Books

- Baylon, M. B. *Developing Fragility Curves in Seismic Assessment of Pier.* LAP Lambert Academic Publishing, 2020.
- Cao, J., & Liu, X. *Wireless Sensor Networks for Structural Health Monitoring.* Springer, 2020.
- Gonzalez, A., Schorr, M., Valdez, B., & Mungaray, A. *Bridges: Structures and Materials, Ancient and Modern.* IntechOpen, 2020.
- Rao, K. R., Kim, D. N., & Hwang, J. J. *Fast Fourier Transform: Algorithms and Applications.* Springer Science & Business Media, 2020.
- Shankar, P. M. *Introduction to Wireless Systems.* Wiley and Sons, 2020.

Reports

- Couture, Z. *Structural Health Monitoring.* Worcester Polytechnic Institute, 2020.
- Cruz, F. R. G., Garcia, R. G., Faicol, J. R. C., Gica, Z. L. D., Pancho, R. M. P., & Tuyay, M. A. L. *Wireless Sensor Nodes with MEMS Accelerometer for Structural Health Monitoring System of Jones Bridge.* B.S. Thesis, School of EECE, Mapua University, 2020.
- Gawthrop, P. J. "Physical Model Based Predictive Control," 2020.
- Guan, H. "Vibration-Based Structural Health Monitoring for Highway Bridges," 2020.
- James, H., & Doroteo, E. *Monographic Report: Philippines Disaster Risk Profile and Disaster Risk Reduction (DRR) Framework: Natural Calamities.* 2020.
- Miranville, A. "Annual Report 2018." *AIMS Mathematics,* 2020.
- Peng, Z. K., Lang, Z. Q., & Billings, S. A. "Crack Detection Using Nonlinear Output Frequency Response Functions – An Experimental Study." University of Sheffield, 2020.
- Rice, J. A., & Spencer Jr., B. F. "Flexible Smart Sensor Framework for Autonomous Full-Scale Structural Health Monitoring." *NSEL Report Series,* No. 18, University of Illinois at UrbanaChampaign, 2020.
- Shinozuka, M., et al. *Statistical Analysis of Fragility Curves.* University of Southern California, 2020.

- Teng, C. *Structural Health Monitoring of a Bridge Structure Using Wireless Sensor Network.* Master's Thesis, 2020.
- The Project for Study on Improvement of Bridges through Disaster Mitigating Measures for Large Scale Earthquakes in the Republic of the Philippines. 2020.
- The World Bank East Asia and Pacific Region Rural Development. *Disaster Risk Reduction and Management in the Philippines: Enhancing Poverty Alleviation through Disaster Reduction.* 2020.

Journals/Periodicals/Manuals

- AASHTO. *Standard Specifications for Highway Bridges.* 14th ed., 2020.
- Abdrabo, A. (2024). Application of online anomaly detection using one-class classification to the Z24 Bridge. *Sensors, 24*(23), 7866. https://doi.org/10.3390/s24237866
- ACI 318M-11. *Building Code Requirements for Structural Concrete.* American Concrete Institute, 2020.
- Albarbar, A., Badri, A., Sinha, J. K., & Starr, A. "Performance Evaluation of MEMS Accelerometers." *Measurement,* 2020.
- Anaissi, A., Suleiman, B., & Alyassine, W. "Personalised Federated Learning Framework for Damage Detection in Structural Health Monitoring." *Journal of Civil Structural Health Monitoring,* 2023.
- Armijo, A., & Zamora-Sánchez, D. (2024). Integration of railway bridge structural health monitoring into the Internet of Things with a digital twin: A case study. *Sensors, 24*(7), 2115. https://doi.org/10.3390/s24072115
- Association of Structural Engineers of the Philippines. *National Structural Code of the Philippines, Volume 1.* 2020.
- Bao, Y., & Li, H. "Machine Learning Paradigm for Structural Health Monitoring." *Structural Health Monitoring,* 2021.
- Bao, Y., Sun, H., Xu, Y., Guan, X., Pan, Q., & Liu, D. (2025). Recent advances in structural health diagnosis: A machine learning perspective. *Advances in Bridge Engineering, 6,* 7. https://doi.org/10.1186/s43251-024-00155-z
- Beltran, A., Zeny, J., Conde, B., & Serfa, R. *World Congress on Engineering and Technology:*
- *Innovation and Its Sustainability 2018.* 2018.

- Beskhyroun, S., Wegner, L. D., & Sparling, B. F. “Integral Resonant Control Scheme for Cancelling Human-Induced Vibrations in Light-Weight Pedestrian Structures.” *Structural Control and Health Monitoring,* 2020.
- Bhatta, S., & Dang, J. (2024). Use of IoT for structural health monitoring of civil engineering structures: A state-of-the-art review. *Urban Lifeline, 2,* 17. https://doi.org/10.1007/s44285-02400031-2
- “Bridges.” In *National Structural Code of the Philippines, Volume 2,* 2nd ed. Association of Structural Engineers of the Philippines, 2021.
- Brunning, R., Hogan, D., Jones, J., Jones, M., Maltby, E., Robinson, M., & Straker, V. “Saving the Sweet Track: The In-Situ Preservation of a Neolithic Wooden Trackway, Somerset, UK.” *Conservation and Management of Archaeological Sites,* 2021.
- Canuto, E., Ospina, J., & Acuna-Bravo, W. “Dynamic and Relative Calibration of Temperature Sensors under Uncertain Parameters.” *IEEE Control Conference,* 2022.
- Carpio, J. N., Cruz, F. R. G., & Chung, W. Y. “An Earthquake Activated Power Interrupting Device Using a Triaxis Accelerometer.” *IEEE Region 10 Conference (TENCON),* 2020.
- Chen, S. Y., Hsieh, Y. J., Yang, C. C., Wu, C. M., & Huang, C. M. “A Real-Time Bridge Scour Sensor System with Accelerometers.” *IEEE International Instrumentation and Measurement Technology Conference,* 2022.
- Cho, S., Yun, C.-B., Lynch, J. P., Zimmerman, A. T., Spencer, B. F., & Nagayama, T. “Smart Wireless Sensor Technology for Structural Health Monitoring of Civil Structures.” *Steel Structures,* 2023.
- Clemente, P., Bongiovanni, G., Buffarini, G., & Saitta, F. “Structural Health Status Assessment of a Cable-Stayed Bridge by Means of Experimental Vibration Analysis.” *Journal of Civil Structural Health Monitoring,* 2022.
- Concepcion, R. S., & Ilagan, L. C. “Application of Hybrid Soft Computing for Classification of Reinforced Concrete Bridge Structural Health Based on Thermal-Vibration Intelligent System Parameters.” *IEEE International Colloquium on Signal Processing and Its Applications,* 2024.
- Concepcion, R. S., Cruz, F. R. G., Uy, F. A. A., Baltazar, J. M. E., Carpio, J. N., & Tolentino, K. G. “Triaxial MEMS Digital Accelerometer and Temperature Sensor Calibration Techniques for Structural Health Monitoring of Reinforced Concrete Bridge Laboratory Test Platform.” *HNICEM,* 2020.

- De La Torre, R. D., Pasobillo, G. A. E., Rebueno, M. F., Sunga, D. P., Esguerra, B. J. J., &
- Concepcion, R. "Vibration-Based Structural Health Monitoring System for Bridges Using ADXL345 Accelerometer with MATLAB Standalone Application." *IEEE 12th International Conference on Humanoid, Nanotechnology, Information Technology, Communication and Control, Environment, and Management,* 2020.
- Deng, C., Sheng, Y., Wang, S., Hu, W., Diao, S., & Qian, D. "A CMOS Smart Temperature Sensor with Single-Point Calibration Method for Clinical Use." *IEEE Transactions on Circuits and Systems II: Express Briefs,* 2025.
- Deng, Z., Huang, M., Wan, N., & Zhang, J. (2023). The current development of structural health monitoring for bridges: A review. *Buildings, 13*(6), 1360. https://doi.org/10.3390/buildings13061360
- Design Manual for Roads and Bridges. *Design Criteria for Footbridges, Volume 2, Section 2, Part 8,* 2024.
- Diamanti, N., Annan, A. P., & Redman, J. D. "Concrete Bridge Deck Deterioration Assessment Using Ground Penetrating Radar (GPR)." *Journal of Environmental and Engineering Geophysics,* 2021.
- Dinesh, P. B., Lakshmi, T., & KishoreRaja, D. C. "Monitoring of Strength of a Bridge Using Accelerometer." *International Journal of Pure and Applied Mathematics,* 2022.
- Dong, Y., Frangopol, D. M., & Saydam, D. "Pre-Earthquake Multi-Objective Probabilistic Retrofit Optimization of Bridge Networks Based on Sustainability." *Journal of Bridge Engineering,* 2024.
- Duan, J. C., & Chung, F. L. "A Mamdani Type Multistage Fuzzy Neural Network Model." *IEEE International Conference on Fuzzy Systems,* 2020.
- Eberle, R., & Oberguggenberger, M. "Vibrations of a Bridge with Random Structural Irregularities under Random Traffic Load and a Probabilistic Structural Degradation Assessment Approach." *Journal of Vibration Engineering Technologies,* 2023.
- Engineering, C. "Structural Health Monitoring of Bridge Using Wireless Sensor Network with Temperature Compensation via Principal Component Analysis by," 2021.
- Eshkevari, S., Matarazzo, T., & Pakzad, S. "Bridge Modal Identification Using Acceleration Measurements within Moving Vehicles." *Mechanical Systems and Signal Processing,* 2020.

- Farhidzadeh, A., Dehghan-Niri, E., Moustafa, A., Salamone, S., & Whittaker, A. "Damage Assessment of Reinforced Concrete Structures Using Fractal Analysis of Residual Crack Patterns." *Experimental Mechanics,* 2023.
- Farrar, C. R., & Worden, K. "An Introduction to Structural Health Monitoring." *Philosophical Transactions of the Royal Society A,* 2020.
- Fascista, A. "Toward Integrated Large-Scale Environmental Monitoring Using WSN/UAV/Crowdsensing: A Review of Applications, Signal Processing, and Future Perspectives." *Sensors,* 2022.
- Gaviña, J. R., Uy, F. A., & Carreon, J. P. D. "Structural Health Monitoring of Existing Concrete Bridges with AASHTO Type IV Girder Using SmartBridge Sensor Nodes." *IEEE Conference on Technologies for Sustainability (SusTech),* 2020.
- Gaviña, J. R., Uy, F. A., & Carreon, J. P. D. "Wireless Smart Sensor Network System Using SmartBridge Sensor Nodes for Structural Health Monitoring of Existing Concrete Bridges." *IOP Conference Series: Materials Science and Engineering,* 2020.
- Hidalgo-Fort, E., Blanco-Carmona, P., Muñoz-Chavero, F., Torralba, A., & Castro-Triguero, R. (2024). Low-cost, low-power edge computing system for structural health monitoring in an IoT framework. *Sensors, 24*(15), 5078. https://doi.org/10.3390/s24155078
- Hoang, T., Fu, Y., Mechitov, K., Kim, J. R., Lee, D., & Spencer, B. F., Jr. (2020). Autonomous end-to-end wireless monitoring system for railroad bridges. *Advances in Bridge Engineering, 1,* 17. https://doi.org/10.1186/s43251-020-00014-7
- Hoque, R., Hassan, S., Sadaf, M. D. A., Galib, A., & Karim, T. F. "Earthquake Monitoring and Warning System." *International Conference on Advances in Electrical Engineering,* 2020.
- Hoult, N. A., Fiddler, P. R. A., Hill, P. G., & Middleton, C. R. "Long-Term Wireless Structural Health Monitoring of the Ferriby Road Bridge." *Journal of Bridge Engineering,* 2021.
- Huiskes, R., Rulmerman, R., Van Lenthe, G. H., & Janssen, J. D. "Effects of Mechanical Forces on Maintenance and Adaptation of Form in Trabecular Bone." *Nature,* 2020.
- Ibrahim, A., Abdelkhalek, S., Zayed, T., Qureshi, A. H., & Abdelkader, E. M. (2024). A comprehensive review of the key deterioration factors of concrete bridge decks. *Buildings, 14*(11), 3425. https://doi.org/10.3390/buildings14113425 ISO 16063. *A Comprehensive Set of Vibration and Shock Calibration Standards,* 2021.

- Jeong, S. H., Jang, W. S., Nam, J. W., An, H., & Kim, D. J. "Development of a Structural Monitoring System for Cable Bridges by Using Seismic Accelerometers." *Applied Sciences,* 2020.
- Ji, B., & Qu, W. "The Research of Acoustic Emission Techniques for Non-Destructive Testing and Health Monitoring on Civil Engineering Structures." *International Conference on Condition Monitoring and Diagnosis,* 2021.
- Jessica, P., Roque, C., Violanda, R. R., Bernido, C. C., Lea, J., & Soria, A. "Earthquake Occurrences in the Pacific Ring of Fire Exhibit a Collective Stochastic Memory for Magnitudes, Depths, and Relative Distances of Events." JICA. *Proposed DPWH Bridges Seismic Design Specifications, Appendix 1-A,* 2021.
- Kang, X., Zhu, B., Cai, Y., Xiao, Y., Liu, N., Guo, Z., Wang, Q.-A., & Luo, Y. (2025). A concise review of state-of-the-art sensing technologies for bridge structural health monitoring. *Sensors, 25*(17), 5460. https://doi.org/10.3390/s25175460
- Kim, H., Sim, S.-H., Lee, J., Lee, Y.-J., & Kim, J.-M. "Flood Fragility Analysis for Bridges with Multiple Failure Modes." *Advances in Mechanical Engineering,* 2022.
- Li, F. X., Islam, A. A., Jaroo, A. S., Hamid, H., Jalali, J., & Sammartino, M. "Urban Highway Bridge Structure Health Assessments Using Wireless Sensor Network." *IEEE Topical Conference on Wireless Sensors and Sensor Networks (WiSNet),* 2023.
- Liang, Z., Wang, A., Yu, Y., & Yang, P. "Research on Early Weak Structural Damage Detection of Aeroengine Intershaft Bearing Based on Acoustic Emission Technology." *Structural Health Monitoring,* 2020.
- Litwhiler, D. H. "A Custom Vibration Test Fixture Using a Subwoofer." *IAJC-ASEE International Conference,* 2020.
- Locke, W., Sybrandt, J., Redmond, L., Safro, I., & Atamturktur, S. "Using Drive-by Health Monitoring to Detect Bridge Damage Considering Environmental and Operational Effects." *Journal of Sound and Vibration,* 2020.
- Lu, X., Liu, Z., & He, J. "Maximum Likelihood Approach for Low-Cost MEMS Triaxial Accelerometer Calibration." *8th International Conference on Intelligent Human-Machine Systems and Cybernetics,* 2022.
- Ma, H., Feng, M., Feng, R., Zeng, J., & Wen, B. "Mesh Characteristic and Vibration Response Comparison for Geared Systems with Crack and Spalling Faults Using Different Modelling Method." *Prognostics and System Health Management Conference,* 2021.

- Ma, J., & Ge, J. “Ambient Vibration Test of an 18-Story Reinforced Concrete Building.” *E3S Web of Conferences,* 2020.
- Madhukumar, N. “MEMS Based Wireless Sensor Network for Structural Health Monitoring.” *International Journal of Advanced Research in Electronics and Communications Engineering,* 2021.
- Magalhaes, F., Cunha, A., & Caetano, E. “Dynamic Monitoring of a Long Span Arch Bridge.” *Engineering Structures,* 2020.
- Mair, D., Fischer, M., Konzilia, J., Renzler, M., & Ussmueller, T. “Evolutionary Optimization of Antennas for Structural Health Monitoring.” *IEEE Access,* 2023.
- Mamaghani, A. S., & Pedrycz, W. “Structural Optimization of Fuzzy Rule-Based Models: Towards Efficient Complexity Management.” *Expert Systems with Applications,* 2020.
- Mitoulis, S., Domaneschi, M., Cimellaro, G., & Casas, J. “Bridge and Transport Network Resilience—A Perspective.” *Proceedings of the Institution of Civil Engineers – Bridge Engineering,* 2022.
- Moschonas, I. F., Kappos, A. J., Panetsos, P., Papadopoulos, V., Makarios, T., & Thanopoulos, P. “Seismic Fragility Curves for Greek Bridges: Methodology and Case Studies.” *Bulletin of Earthquake Engineering,* 2020.
- Muller, M., Maiwald, V., Kach, M., Hierold, C., & Roman, C. “A Passive Micromechanical Broadband Amplifier for Acoustic Emission Sensing.” *18th International Conference on SolidState Sensors, Actuators and Microsystems (TRANSDUCERS),* 2020.
- National Structural Code of the Philippines (NSCP 2015), Chapter 2: *Instrumentation.*
- Otero, L. D., Moyou, M., Peter, A., & Otero, C. E. “Towards a Remote Sensing System for Railroad Bridge Inspections: A Concrete Crack Detection Component.” *IEEE SOUTHEASTCON,* 2020
- Pan, N. F., Lin, T. C., & Pan, N. H. “Estimating Bridge Performance Based on a Matrix-Driven Fuzzy Linear Regression Model.” *Automation in Construction,* 2021.
- Payawal, J. M. G., Uy, F. A., & Carreon, J. P. D. “Data Calibration of the Actual Versus the Theoretical Micro Electro Mechanical Systems (MEMS) Based Accelerometer Reading through Remote Monitoring of Padre Jacinto Zamora Flyover.” *IEEE Conference on Technologies for Sustainability (SusTech),* 2021.
- Poddaeva, O., Fedosova, A., & Gribach, J. “The Study of Wind Effects on the Bridge Constructions.” *E3S Web of Conferences,* 2020.

- Prendergast, L., Limongelli, M., Ademovic, N., Anžlin, A., Gavin, K., & Zanini, M. "Structural Health Monitoring for Performance Assessment of Bridges under Flooding and Seismic Actions." *Structural Engineering International,* 2021.
- Ribeiro, D., Rakoczy, A. M., Cabral, R., Olaszek, P., Casas, J. R., Figueiras, H., Migliaccio, G. C., & Silvestre, N. (2025). Methodologies for remote bridge inspection—Review. *Sensors, 25*(18), 5708. https://doi.org/10.3390/s25185708
- Risodkar, Y. R., & Pawar, A. S. "A Survey: Structural Health Monitoring of Bridge Using WSN." *International Conference on Global Trends in Signal Processing, Information Computing and Communication,* 2020.
- Ruiz-Jaramillo, J., Montiel-Vega, L., García-Pulido, L. J., Muñoz-González, C., & Blanca-Hoyos, A. "Ambient Vibration as a Basis for Determining the Structural Behaviour of Watchtowers Against Horizontal Loads in Southeast Spain." *Applied Sciences,* 2020.
- Sabato, A. "Pedestrian Bridge Vibration Monitoring Using a Wireless MEMS Accelerometer Board." *IEEE 19th International Conference on Computer Supported Cooperative Work in Design,* 2020.
- Sarhosis, V., De Santis, S., & Felice, G. "A Review of Experimental Investigations and Assessment Methods for Masonry Arch Bridges." *Structure and Infrastructure Engineering,* 2021.
- Sonbul, O. S., & Rashid, M. (2023). Towards the structural health monitoring of bridges using wireless sensor networks: A systematic study. *Sensors, 23*(20), 8468. https://doi.org/10.3390/s23208468
- Strang, G. "Wavelets." *American Scientist,* 2020.
- Sunchikala, S. K., Mohan, S. C., Gopala, S. D., Jayalekshmi, B. R., & Lincy, A. J. (2025). Structural health monitoring of railway truss bridge under moving train load using decision tree models and residual neural networks. *Discover Civil Engineering, 2,* 35. https://doi.org/10.1007/s44290-02500199-8
- Sunjie. "Discussion on Health Monitoring and Damage Detection of a Large-Span Bridge." *International Conference on Electrical Technology and Civil Engineering,* 2021.
- Sweet, M. "Does Traffic Congestion Slow the Economy?" *Journal of Planning Literature,* 2021.
- Tan, T. D., Anh, N. T., & Anh, G. Q. "Low-Cost Structural Health Monitoring Scheme Using MEMS-Based Accelerometers." *Second International Conference on Intelligent Systems, Modelling and Simulation,* 2021.

- Tarighat, A., & Miyamoto, A. “Fuzzy Concrete Bridge Deck Condition Rating Method for Practical Bridge Management System.” *Expert Systems with Applications,* 2020.
- Tsapanos, T. M., & Burton, P. W. “Seismic Hazard Evaluation for Specific Seismic Regions of the World.” *Tectonophysics,* 2020.
- Van der Wielen, A., Courard, L., & Nguyen, F. “Nondestructive Methods for the Detection of Delaminations in Concrete Bridge Decks.” *13th International Conference on Ground Penetrating Radar,* 2021.
- Veldman, C. S. *ISO 16063: A Comprehensive Set of Vibration and Shock Calibration Standards.* 2020.
- Wang, L. “Analysis of Temperature Effect for Horizontal Rotation of Ten Thousand Tons CableStayed Bridge in Construction Monitoring.” *International Conference on Remote Sensing, Environment and Transportation Engineering,* 2020.
- Wang, L., Tang, A., Cui, Y., Yang, S., & Zhan, Z. “Study on the Development of Sihui Bridge Management System.” *WRI World Congress on Software Engineering,* 2022.
- Wang, Q., Zheng, Z., Qiao, H., & De Domenico, D. “Seismic Protection of Reinforced Concrete Continuous Girder Bridges with Inerter-Based Vibration Absorbers.” *Soil Dynamics and Earthquake Engineering,* 2023.
- Wang, Y., Tang, Z., Qian, G., Xu, W., Huang, X., & Fang, H. (2025). A prototype of a lightweight structural health monitoring system based on edge computing. *Sensors, 25*(18), 5612. https://doi.org/10.3390/s25185612
- Wei, T. C., Narang, V., & Thean, A. “Electrical Characterization of FEOL Bridge Defects in Advanced Nanoscale Devices Using TCAD Simulations.” *International Symposium on Physical and Failure Analysis of Integrated Circuits,* 2020.
- Woo, S., Kim, J., Kim, J., & Kim, S. “Calibration of Accelerometer Using Fuzzy Inference System.” *IEEE Control, Automation and Systems Conference,* 2020.
- Wu, Z. Y., Cao, J., & Ding, L. “Structural State Assessment Based on Similarity Degree of Membership Cloud.” *International Conference on Fuzzy Systems and Knowledge Discovery,* 2020.
- Xiong, W., Cai, C., Zhang, R., Shi, H., & Xu, C. “Review of Hydraulic Bridge Failures: Historical Statistical Analysis, Failure Modes, and Prediction Methods.” *Journal of Bridge Engineering,* 2023.
- Yao, J., & Kaifeng, Z. “Evaluation Model of the Artist Based on Fuzzy Membership to Improve the Principal Component Analysis of Robust Kernel.” *IEEE International*

Conference on Big Data Security on Cloud / High Performance Smart Computing / Intelligent Data Security, 2020.

- Ye, L., & Su, S. "Optimum Experimental Design Applied to MEMS Accelerometer Calibration for 9-Parameter Auto-Calibration Model." *IEEE Engineering in Medicine and Biology Society Conference,* 2022.
- Yin, Z., & Li, Y. "Intelligent Decision Support System for Bridge Monitoring." *International Conference on Machine Vision and Human-Machine Interface,* 2020.
- Ying, H., Ding, Y., Li, S., & Shao, S. "Typical Takagi-Sugeno and Mamdani Fuzzy Systems as Universal Approximators: Necessary Conditions and Comparison." *IEEE International Conference on Fuzzy Systems,* 2020.
- Yu, Y., Xie, H., & Ou, J. "Vibration Monitoring Using Wireless Sensor Networks on Dongying Huanghe River Bridge." *Earth and Space 2020: Engineering, Science, Construction, and Operations in Challenging Environments,* ASCE, 2020.
- Zhang, Y. "Ambient Vibration Induced by Spatial Coupled Vibration of Vehicle-Track-Bridge System and Isolating Measures." *International Conference on Electrical Technology and Civil Engineering,* 2021.
- Zhou, X., Zou, K., & Wang, Y. "Fuzzy Variable Time Series Based on Fuzzy Membership Function and Econometrics." *International Symposium on Knowledge Acquisition and Modeling,* 2020.

Online/Internet Sources

- Borlaza, G. C., & Hernandez, C. G. "Philippines." *Encyclopedia Britannica,* 2021. Retrieved from https://www.britannica.com/place/Philippines
- Department of Public Works and Highways. *Bridge Inventory.* Retrieved from http://www.dpwh.gov.ph/DPWH/gis/
- Department of Public Works and Highways. *No Title.* Retrieved from http://www.dpwh.gov.ph/dpwh/2019%20DPWH%20Road%20and%20Bridge%20Inventory/Brid ge%20Data%202016/Bridge%20Data%202016/index.htm

- Department of Public Works and Highways. *Philippine National Bridge Network: Road and Bridge Information Application (RBIA),* 2020. https://www.dpwh.gov.ph/dpwh/gis/rbi
- Department of Public Works and Highways. *Summary of Existing National Bridges (2020 ATLAS).* https://www.dpwh.gov.ph/dpwh/references/atlas
- Department of Public Works and Highways, & Japan International Cooperation Agency. *The Project for Study on Improvement of Bridges through Disaster Mitigating Measures for Large Scale Earthquakes in the Republic of the Philippines.* Retrieved from https://openjicareport.jica.go.jp/pdf/12146015_01.pdf
- PHIVOLCS. *Seismicity Map: Establishment of Criteria Based on Risk Level of Existing Public School Building Along Marikina West Valley Fault.*
- https://www.phivolcs.dost.gov.ph/information-tools/the-phivolcs-faultfinder/
- PHIVOLCS. *Susceptibility Map of Metropolitan Manila.* Retrieved from https://gisweb.phivolcs.dost.gov.ph/phivolcs_hazardmaps/

APPENDICES

Appendix A

April 6, 2026
The Planning Department
Department of Public Works and Highways Region VII
Cebu City, Philippines

Dear Sir/Madam, Greetings!

We are Fourth-year BS Information Technology students from the University of Cebu – Banilad Campus. We are currently developing our capstone project entitled **"ViBridge: An IOT-Based Vibration and Structural Condition Monitoring System for Small Bridges."** The purpose of this letter is to formally request your valuable insights and professional perspective regarding bridge monitoring, planning, and maintenance practices.

As the Planning Department of the Department of Public Works and Highways Region VII, your knowledge and experience will greatly help us understand the current processes, challenges, and needs related to bridge infrastructure monitoring. Your input will guide the design of our system, which aims to support real-time vibration monitoring, anomaly detection, and data-driven decision-making for bridge safety and maintenance.

Please be assured that all information you provide will be used solely for academic purposes and will remain confidential.

Thank you for your time and consideration. Your support will play an important role in the successful development of our project.

Sincerely,

Paul Vincent T. Lepatan
Project Manager

Noted By:
Mr. Joemarie C. Amparo
Capstone Adviser
College of Computer Studies
University of Cebu – Banilad Campus

Mr. Eric P. Ortega
Chairperson
College of Computer Studies

Appendix B

Interview Guide

We are undergraduate students from the University of Cebu – Banilad Campus, currently pursuing a Bachelor of Science in Information Technology. As part of our academic requirements, we are conducting a research study titled: "ViBridge: An IoT-Based Bridge Vibration Monitoring System for Structural Health Assessment." In connection with this, we respectfully invite you to participate by taking a few minutes to complete the attached questionnaire/interview. Your honest responses will greatly contribute to the success of our study and help us gather essential data that will guide the development of our proposed system. Please be assured that all information you provide will be kept strictly confidential and will be used solely for academic and research purposes.

PART I: Profile of the Respondents

Gender:
☐ Male
☐ Female

Age: _____

Position / Role: _____

Organization / Agency: _____

How many years have you been involved in bridge inspection, maintenance, or structural monitoring?

☐ Less than 1 year
☐ 1–3 years
☐ 4–6 years
☐ More than 6 years

What is your area of work related to bridges?

☐ Bridge inspection
☐ Structural engineering
☐ Maintenance and repair
☐ Infrastructure management
☐ Other: _____

How often are you involved in bridge inspections?

☐ Regularly (scheduled inspections)
☐ Occasionally
☐ Rarely
☐ Never directly involved

PART II: Assessment of Bridge Inspection Practices

1. What methods are commonly used in your organization for bridge inspection? (Check all that apply)

   ☐ Visual inspection
   ☐ Manual measurement tools
   ☐ Structural monitoring devices
   ☐ Drone or camera-based inspection
   ☐ Other: _____

2. What are your experiences when conducting bridge inspections?

3. Which of the following structural conditions do you usually monitor during inspections? (Check all that apply)

   ☐ Cracks in structural components
   ☐ Corrosion of steel elements
   ☐ Deformation or displacement
   ☐ Vibration or structural movement
   ☐ Environmental effects (weather, flooding, etc.)
   ☐ None
   ☐ Others: _____

4. How do you currently monitor these bridge conditions?

   ☐ Visual inspection and experience
   ☐ Manual inspection tools
   ☐ Basic digital equipment
   ☐ Structural monitoring devices or sensors
   ☐ I do not monitor structural conditions directly

5. What problems do you commonly encounter in bridge inspection and monitoring? (Check all that apply)

   ☐ Limited inspection equipment
   ☐ Difficulty accessing some bridge areas
   ☐ Insufficient monitoring data
   ☐ Lack of trained personnel
   ☐ Delayed inspections
   ☐ Other: _____

6. How often do you encounter structural issues during bridge inspections?

   ☐ Very frequently
   ☐ Sometimes
   ☐ Rarely
   ☐ Never

---

7. How confident are you in identifying structural problems during bridge inspections?

   ☐ Very confident
   ☐ Somewhat confident
   ☐ Not very confident
   ☐ Not confident at all

---

8. How do you determine if a bridge is in good condition?

---

9. How do you identify signs that a bridge may require repair or maintenance?

---

10. Have you used any form of technology (sensors, monitoring systems, software tools, etc.) for bridge inspection or monitoring?

    ☐ Yes
    ☐ No

---

11. Which features would be most useful in a bridge monitoring system? (Check all that apply)

    ☐ Real-time structural monitoring
    ☐ Damage detection alerts
    ☐ Structural data recording and analysis
    ☐ Remote monitoring access
    ☐ Maintenance planning support
    ☐ Other: _____

12. Are you interested in using a system that automatically monitors bridge conditions?

    ☐ Yes
    ☐ No

13. What are your thoughts on using sensors to monitor bridge structural conditions?

    ☐ I'm interested in using sensors
    ☐ I'm not sure about using sensors
    ☐ I don't want to use sensors

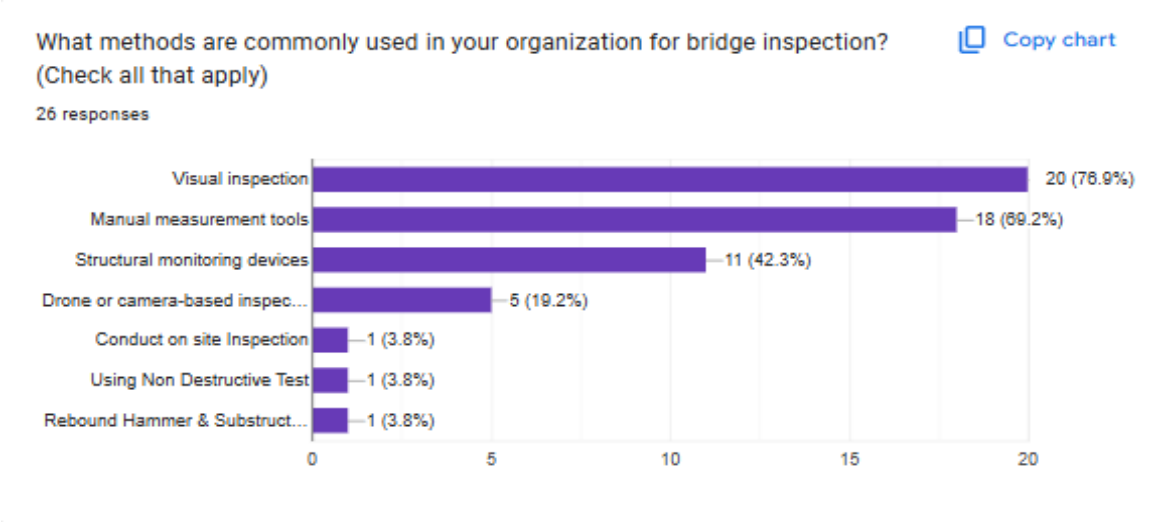
14. What are your thoughts on automating bridge monitoring (e.g., vibration, strain, structural movement)?

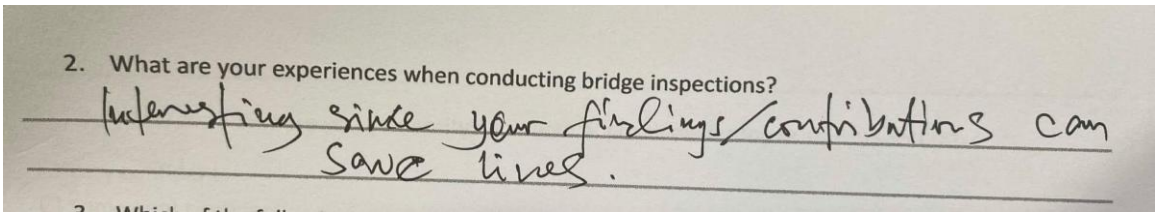
    ☐ I'm interested in automated monitoring
    ☐ I'm not sure about automation
    ☐ I don't want automated monitoring

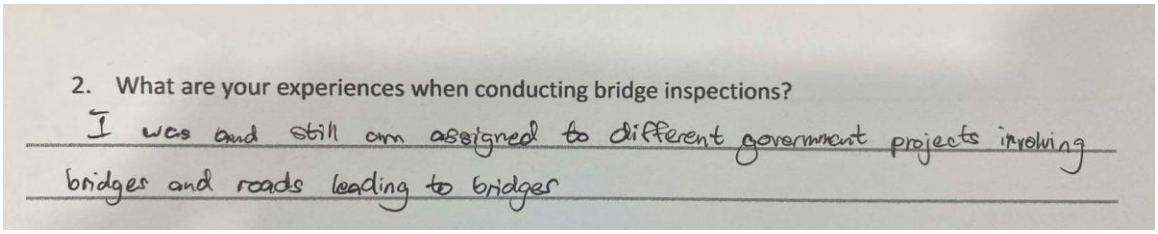
Appendix C

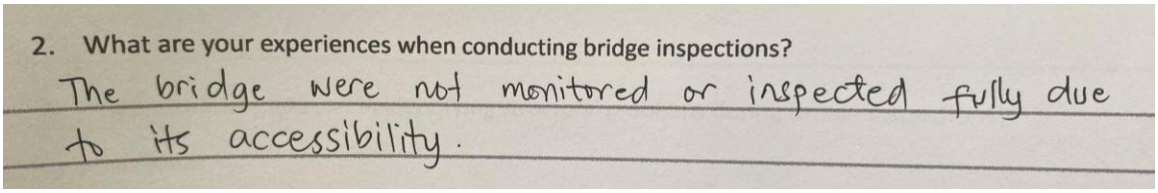
Results and Findings

The following findings are based on data acquired from the proponents' and respondents' interviews and surveys





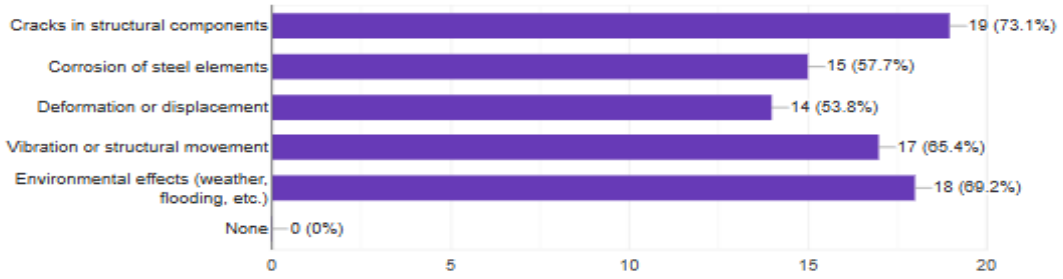




Which of the following structural conditions do you usually monitor during inspections? (Check all that apply)

Copy chart

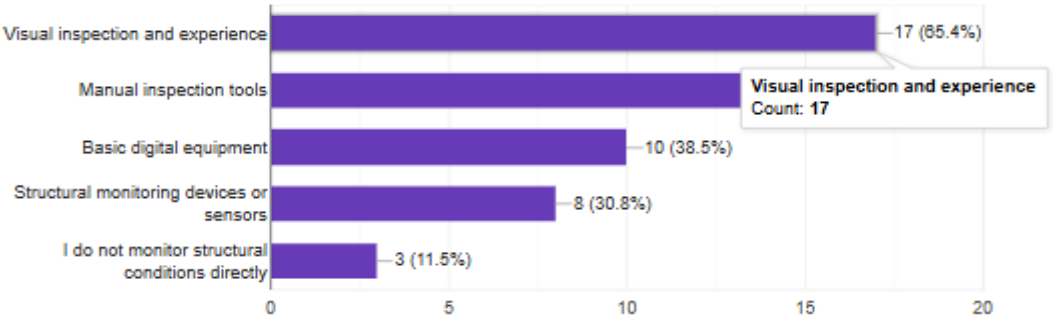
26 responses



How do you currently monitor these bridge conditions?

Copy chart

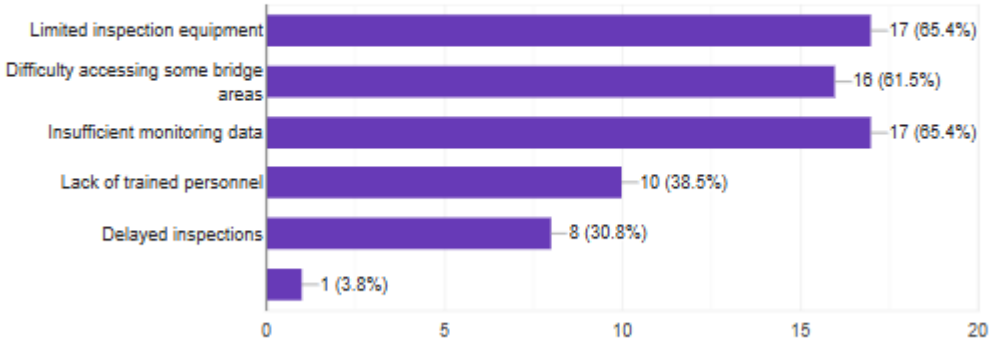
26 responses



What problems do you commonly encounter in bridge inspection and monitoring? (Check all that apply)

Copy chart

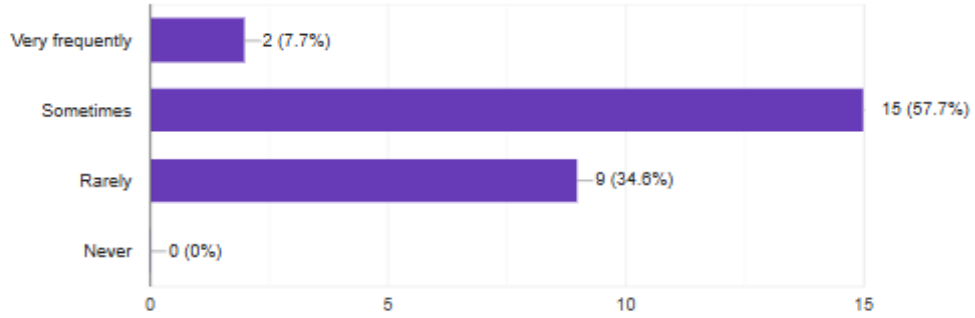
26 responses



How often do you encounter structural issues during bridge inspections?

Copy chart

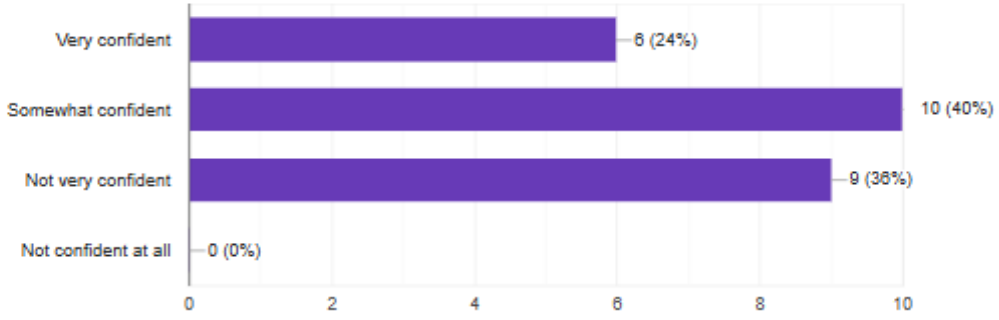
26 responses



How confident are you in identifying structural problems during bridge inspections?

Copy chart

25 responses



8. How do you determine if a bridge is in good condition?

THROUGH VISUAL INSPECTION AND EVALUATION OF KEY COMPONENTS SUCH AS DECK, SUPERSTRUCTURE & SUBSTRUCTURE.

9. How do you identify signs that a bridge may require repair or maintenance?

SIGNS INCLUDES VISIBLE CRACKS, SPALLING, EXPOSED REINFORCEM CORROSION, UNUSUAL DEFLECTION & SIGNS OF SCOUR OR EROSION.

8. How do you determine if a bridge is in good condition?

Bridge is in good condition if there are no structural Defects, No abnormal vibrations and No Structural deformation

9. How do you identify signs that a bridge may require repair or maintenance?

All defects observed, major or minor requires repair or maintenan Even Bridges in good condition must require maintenance

8. How do you determine if a bridge is in good condition?

No visible cracks or unnatural movements detected at any time

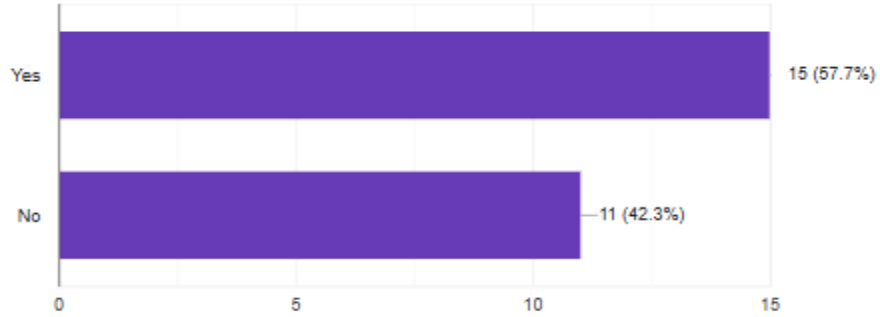
9. How do you identify signs that a bridge may require repair or maintenance?

When large cracks or unnatural movements of the bridge is noticeable which can affect the structural integrity of the bridge itself

Have you used any form of technology (sensors, monitoring systems, software tools, etc.) for bridge inspection or monitoring?

Copy chart

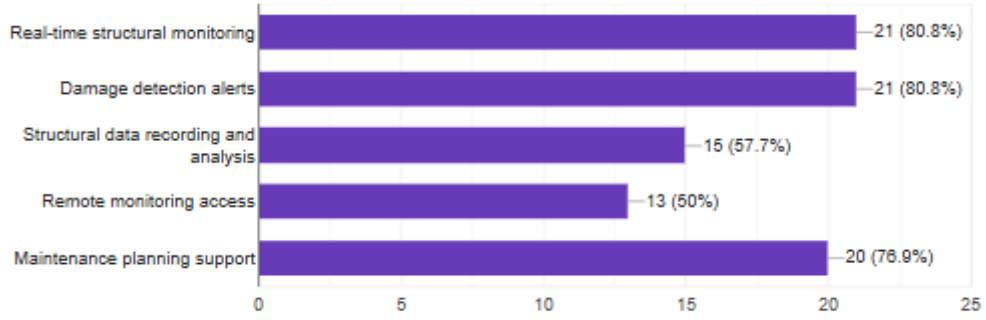
26 responses



Which features would be most useful in a bridge monitoring system? (Check all that apply)

Copy chart

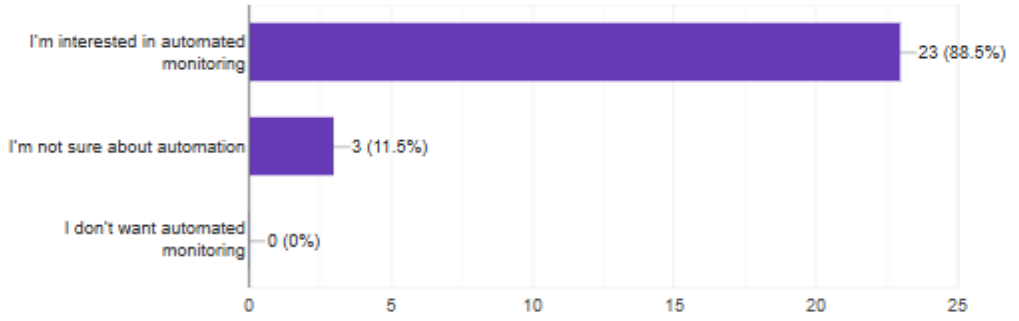
26 responses



What are your thoughts on automating bridge monitoring (e.g., vibration, strain, structural movement)?

Copy chart

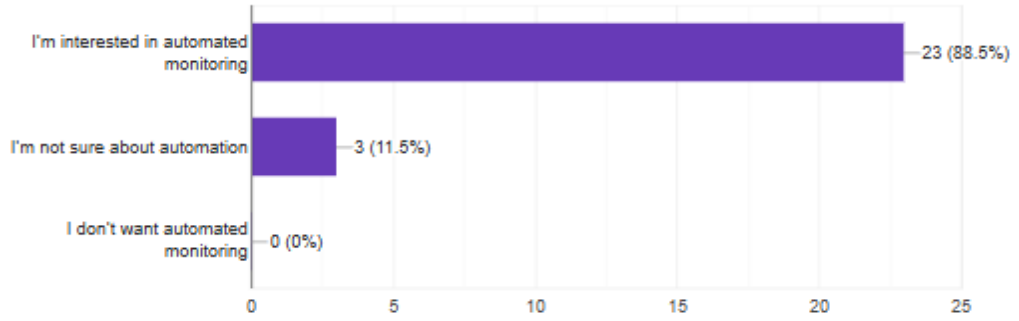
26 responses



What are your thoughts on automating bridge monitoring (e.g., vibration, strain, structural movement)?

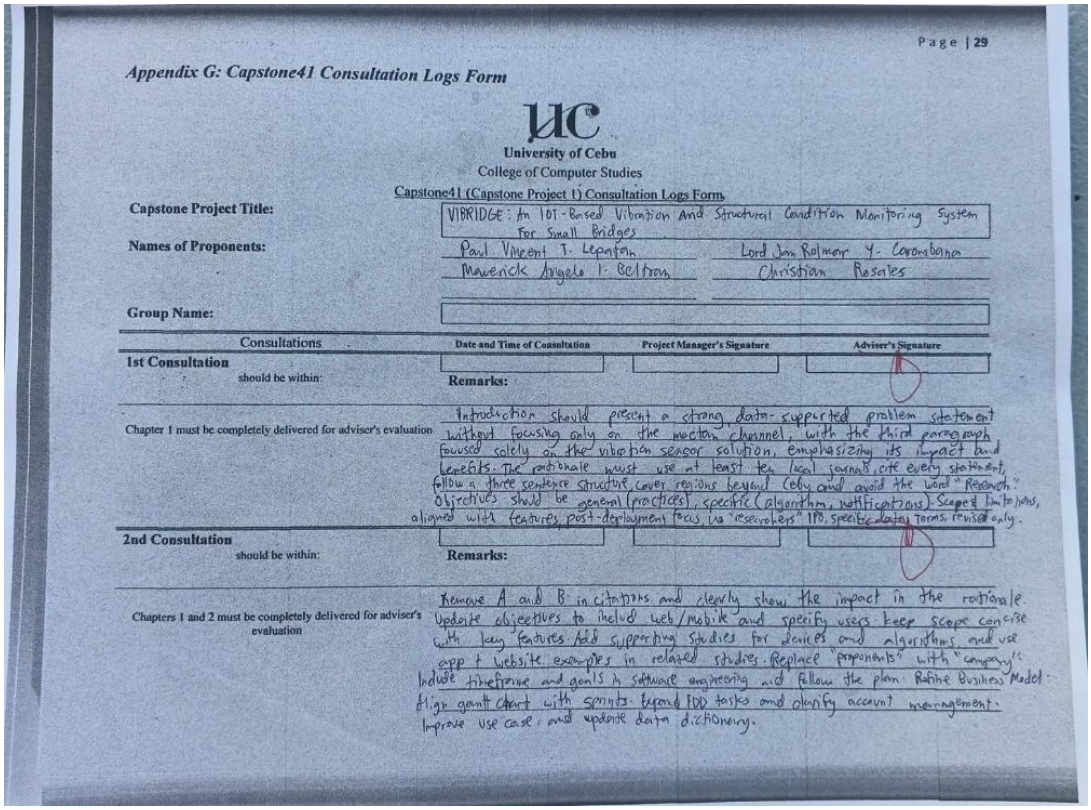
Copy chart

26 responses



Appendix D

Consultation Logs – Adviser



Page | 29

Appendix G: Capstone41 Consultation Logs Form

University of Cebu
College of Computer Studies
Capstone41 (Capstone Project 1) Consultation Logs Form

Capstone Project Title: VIBRIDGE: An IoT-Based Vibration And Structural Condition Monitoring System For Small Bridges

Names of Proponents: Paul Vincent T. Lepiten, Lord Jan Rolmer Y. Corombona, Maverick Angelo I. Beltran, Christian Rosales

Group Name:

Consultations	Date and Time of Consultation	Project Manager's Signature	Adviser's Signature
1st Consultation should be within:			

Remarks:

Chapter 1 must be completely delivered for adviser's evaluation

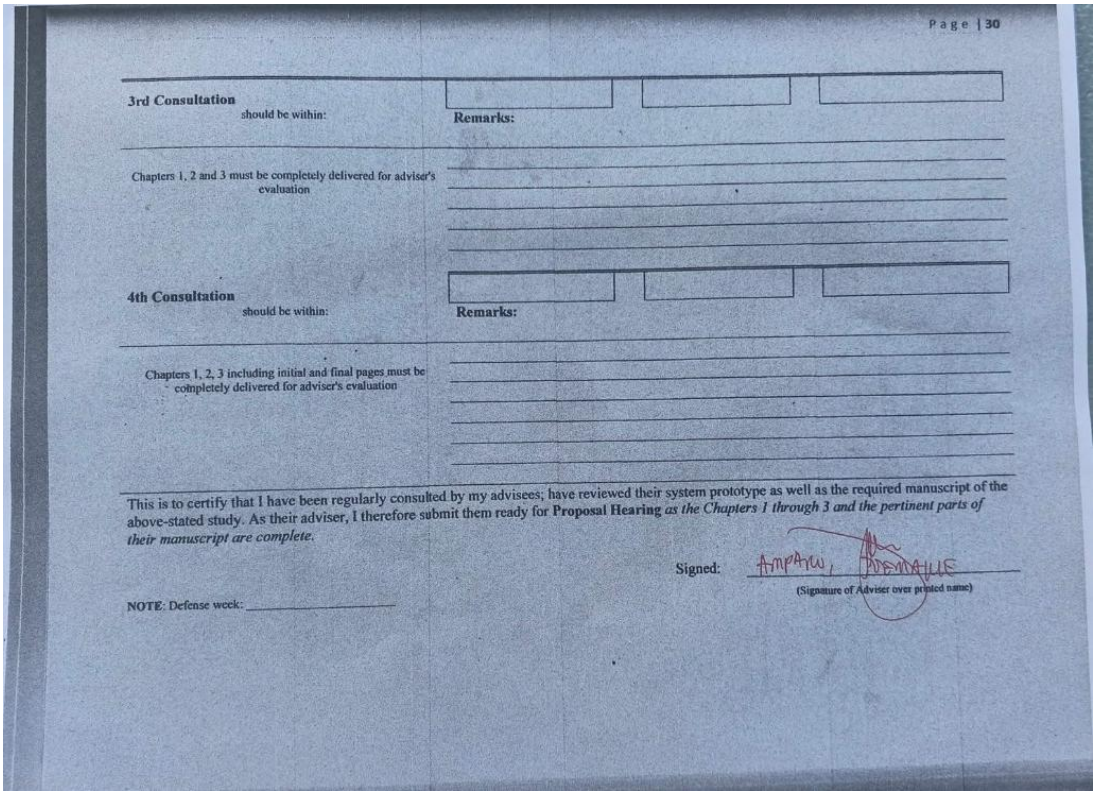
Introduction should present a strong, data-supported problem statement without focusing only on the vibration channel, with the third paragraph focused solely on the vibration sensor solution, emphasizing its impact and benefits. The rationale must use at least ten local journals, cite every statement, follow a three sentence structure, cover regions beyond Cebu and avoid the word "Research." Objectives should be general (practices), specific (algorithm, notifications). Scope & limitations, aligned with features, post-deployment focus via "researchers" IPO, specific data, terms revised only.

2nd Consultation should be within:			

Remarks:

Chapters 1 and 2 must be completely delivered for adviser's evaluation

Remove A and B in citations and clearly show the impact in the rationale. Update objectives to include web/mobile and specify users. Keep scope concise with key features. Add supporting studies for devices and algorithms, and use app & website examples in related studies. Replace "proponents" with "company." Include timeframe and goals in software engineering and follow the plan. Refine Business Model. Align gantt chart with sprints. Expand TDD tasks and clarify account management. Improve use case, and update data dictionary.



Page | 30

3rd Consultation should be within:			

Remarks:

Chapters 1, 2 and 3 must be completely delivered for adviser's evaluation

4th Consultation should be within:			

Remarks:

Chapters 1, 2, 3 including initial and final pages must be completely delivered for adviser's evaluation

This is to certify that I have been regularly consulted by my advisees; have reviewed their system prototype as well as the required manuscript of the above-stated study. As their adviser, I therefore submit them ready for **Proposal Hearing** *as the Chapters 1 through 3 and the pertinent parts of their manuscript are complete.*

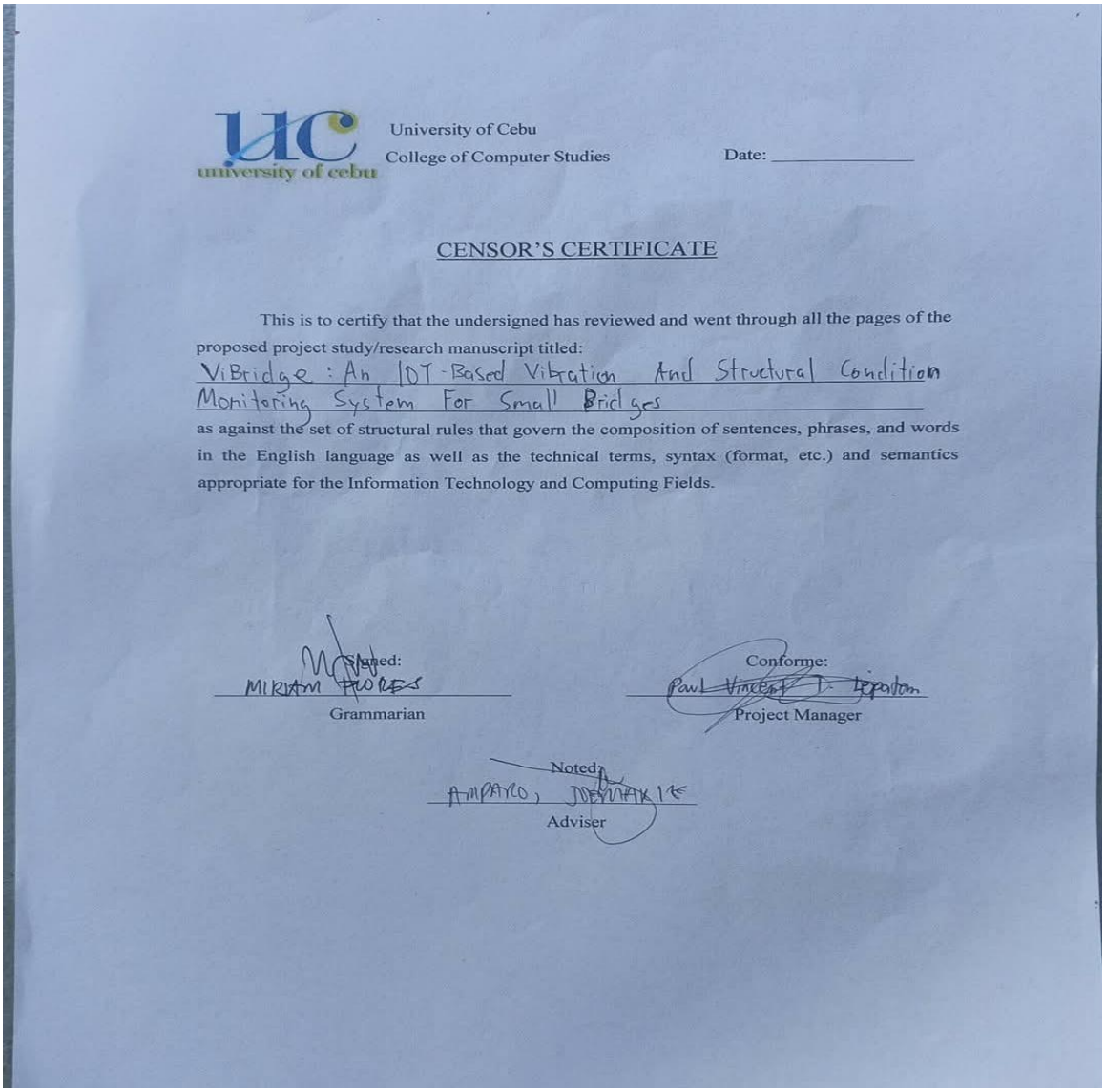
Signed: AMPARO, ADRIANNE LEE

(Signature of Adviser over printed name)

NOTE: Defense week: _____

Appendix E

Consultation Logs – Censor



University of Cebu
College of Computer Studies

Date: _____

CENSOR'S CERTIFICATE

This is to certify that the undersigned has reviewed and went through all the pages of the proposed project study/research manuscript titled:

ViBridge : An IOT-Based Vibration And Structural Condition Monitoring System For Small Bridges

as against the set of structural rules that govern the composition of sentences, phrases, and words in the English language as well as the technical terms, syntax (format, etc.) and semantics appropriate for the Information Technology and Computing Fields.

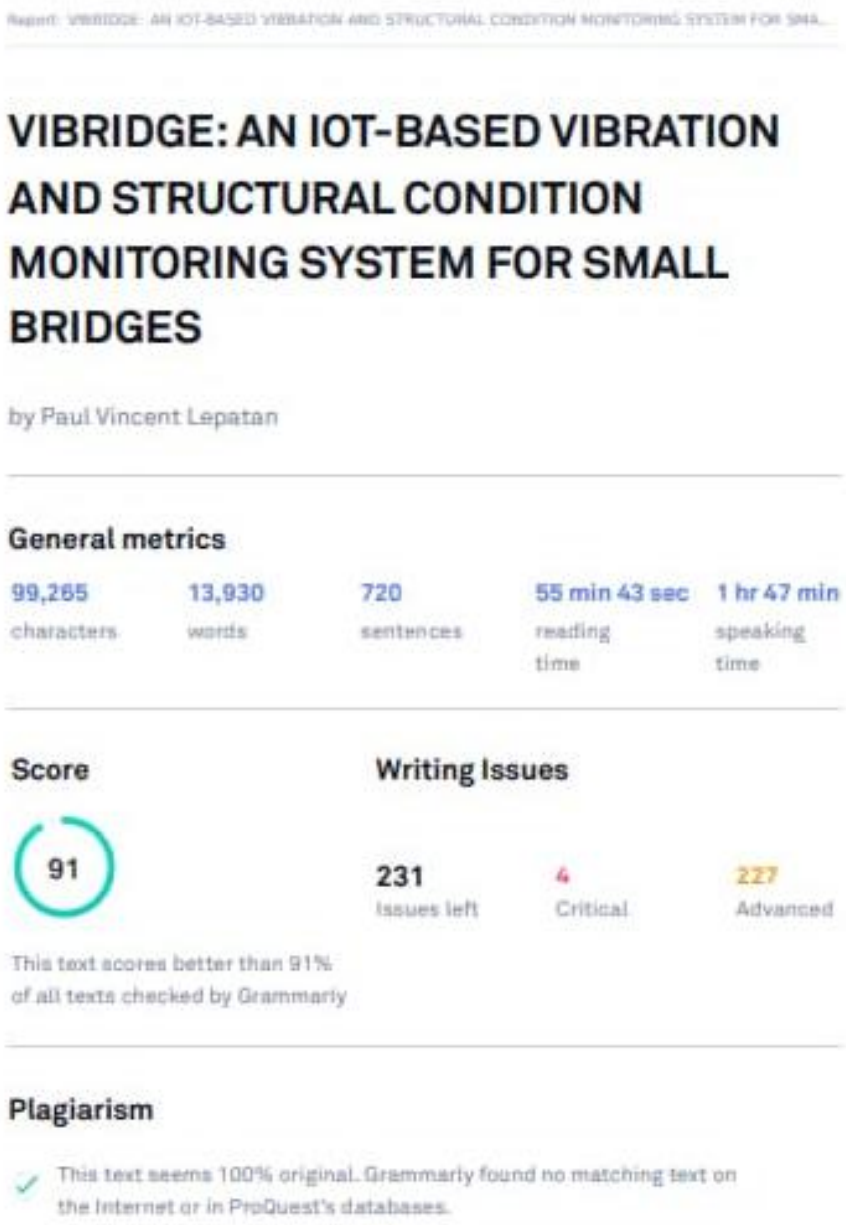
Signed:
MIRIAM FLORES
Grammarian

Conforme:
Paul Vincent T. Lepatan
Project Manager

Noted:
AMPARO, JOEMARK T.
Adviser

Appendix F

Grammarly Result



Report: VIBRIDGE: AN IOT-BASED VIBRATION AND STRUCTURAL CONDITION MONITORING SYSTEM FOR SMA...

VIBRIDGE: AN IOT-BASED VIBRATION AND STRUCTURAL CONDITION MONITORING SYSTEM FOR SMALL BRIDGES

by Paul Vincent Lepatan

General metrics

99,265	13,930	720	55 min 43 sec	1 hr 47 min
characters	words	sentences	reading time	speaking time

Score

91

This text scores better than 91% of all texts checked by Grammarly

Writing Issues

231	4	227
Issues left	Critical	Advanced

Plagiarism

✓ This text seems 100% original. Grammarly found no matching text on the Internet or in ProQuest's databases.

TEAM COMPOSITION



PAUL VINCENT T. LEPATAN

Address: Villa Donna Subdivision, Biasong, Talisay City
Contact No: 09159084615
Email: plepatan@gmail.com

Personal Information:

Date of Birth:	April 01, 2002
Place of Birth:	Talamban, Cebu
Age:	24
Civil Status:	Single
Sex:	Male
Religion:	Roman Catholic
Citizenship:	Filipino
Language Proficiency:	Cebuano, Filipino, and English

Educational Background:

Present Education:	University of Cebu - Banilad Campus Bachelor of Science in Information Technology Gov. M. Cuenco Ave, Cebu City, 6000 Cebu A.Y. 2022 - Present
Secondary Education:	University of Cebu Lapu-Lapu and Mandaue A.Y. 2016 – 2022

MAVERICK ANGELO I. BELTRAN



Address: Upper Casili Consolacion Cebu
Contact No: 09935471448
Email: bearkada01@gmail.com

Personal Information:

Date of Birth:	February 10, 2003
Place of Birth:	City of Balanga, Bataan
Age:	23
Civil Status:	Single
Sex:	Male
Religion:	Roman Catholic
Citizenship:	Filipino
Language Proficiency:	Cebuano, Filipino, and English

Educational Background:

Present Education:	University of Cebu - Banilad Campus Bachelor of Science in Information Technology Gov. M. Cuenco Ave, Cebu City, 6000 Cebu A.Y. 2022 - Present
Secondary Education:	Mount Olives School of the Phil. Inc. A.Y. 2016 – 2022

LORD JAN ROLMAR Y. CAROMBANA

Address: Sitio kalubihan, Tabok, Mandaue city
Contact No: 09647475289
Email: mamawkaayo211@gmail.com



Personal Information:

Date of Birth:	January 31, 2004
Place of Birth:	Mandaue City
Age:	22
Civil Status:	Single
Sex:	Male
Religion:	Roman Catholic
Citizenship:	Filipino
Language Proficiency:	Cebuano, Filipino, and English

Educational Background:

Present Education:	University of Cebu - Banilad Campus Bachelor of Science in Information Technology Gov. M. Cuenco Ave, Cebu City, 6000 Cebu A.Y. 2022 - Present
Secondary Education:	Badian National High School A.Y. 2016 – 2022

CHRISTIAN ROSALES

Address: Lataban, Liloan, Cebu
Contact No: 09924348693
Email: christianrosales11111@gmail.com



Personal Information:

Date of Birth:	October 14, 2003
Place of Birth:	Cebu City
Age:	22
Civil Status:	Single
Sex:	Male
Religion:	Roman Catholic
Citizenship:	Filipino
Language Proficiency:	Cebuano, Filipino, and English

Educational Background:

Present Education:	University of Cebu - Banilad Campus Bachelor of Science in Information Technology Gov. M. Cuenco Ave, Cebu City, 6000 Cebu A.Y. 2022 - Present
Secondary Education:	Divine Life Institute of Cebu 9XVM+9QP, Central Nautical Hwy, Liloan, 6002 Cebu A.Y. 2016 – 2022